

SEMESTER 3

CYBER PHYSICAL SYSTEMS

SEMESTER S3

MATHEMATICS FOR ELECTRICAL SCIENCE AND PHYSICAL SCIENCE – 3

(Common to B & C Groups)

Course Code	GYMAT301	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic knowledge in complex numbers.	Course Type	Theory

Course Objectives:

1. To introduce the concept and applications of Fourier transforms in various engineering fields.
2. To introduce the basic theory of functions of a complex variable, including residue integration and conformal transformation, and their applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fourier Integral, From Fourier series to Fourier Integral, Fourier Cosine and Sine integrals, Fourier Cosine and Sine Transform, Linearity, Transforms of Derivatives, Fourier Transform and its inverse, Linearity, Transforms of Derivative. (Text 1: Relevant topics from sections 11.7, 11.8, 11.9)	9
2	Complex Function, Limit, Continuity, Derivative, Analytic functions, Cauchy-Riemann Equations (without proof), Laplace's Equations, Harmonic functions, Finding harmonic conjugate, Conformal mapping, Mappings of $w = z^2$, $w = e^z$, $w = \frac{1}{z}$, $w = \sin z$. (Text 1: Relevant topics from sections 13.3, 13.4, 17.1, 17.2, 17.4)	9
3	Complex Integration: Line integrals in the complex plane (Definition & Basic properties), First evaluation method, Second evaluation method, Cauchy's integral theorem (without proof) on simply connected domain, Independence of path, Cauchy integral theorem on multiply connected domain (without proof), Cauchy Integral formula (without proof). (Text 1: Relevant topics from sections 14.1, 14.2, 14.3)	9

4	Taylor series and Maclaurin series, Laurent series (without proof), Singularities and Zeros – Isolated Singularity, Poles, Essential Singularities, Removable singularities, Zeros of Analytic functions – Poles and Zeros, Formulas for Residues, Residue theorem (without proof), Residue Integration- Integral of Rational Functions of $\cos\theta$ and $\sin\theta$. (Text 1: Relevant topics from sections 15.4, 16.1, 16.2, 16.3, 16.4)	9
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Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Determine the Fourier transforms of functions and apply them to solve problems arising in engineering.	K3
CO2	Understand the analyticity of complex functions and apply it in conformal mapping.	K3
CO3	Compute complex integrals using Cauchy's integral theorem and Cauchy's integral formula.	K3
CO4	Understand the series expansion of complex function about a singularity and apply residue theorem to compute real integrals.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	2	-	-	-	-	-	-	-	2
CO2	3	3	-	2	-	-	-	-	-	-	-	2
CO3	3	3	-	2	-	-	-	-	-	-	-	2
CO4	3	3	-	2	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Engineering Mathematics	Erwin Kreyszig	John Wiley & Sons	10 th edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Complex Analysis	Dennis G. Zill, Patrick D. Shanahan	Jones & Bartlett	3 rd edition, 2015
2	Higher Engineering Mathematics	B. V. Ramana	McGraw-Hill Education	39 th edition, 2023
3	Higher Engineering Mathematics	B.S. Grewal	Khanna Publishers	44 th edition, 2018
4	Fast Fourier Transform - Algorithms and Applications	K.R. Rao, Do Nyeon Kim, Jae Jeong Hwang	Springer	1 st edition, 2011

SEMESTER S3

MICROCONTROLLERS AND EMBEDDED SYSTEMS

Course Code	PCCPT302	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To develop simple programs for 8051 using assembly language programming
2. To Understand the interfacing of 8051 microcontroller with peripheral devices
3. To understand the architecture and design concept of embedded systems
4. To develop embedded systems based on Arduino.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	8051 microcontroller: Difference between microprocessor and microcontroller Architecture, 8051 pin diagram; Architecture, I/O Port structure, Register organization - special function registers, Memory organization, Instruction set, Addressing modes Simple Assembly language programs: Arithmetic (Addition, Subtraction, Multiplication & Division), Transfer a block of data from one internal memory location to another	11
2	8051 microcontroller Programming and Peripherals. Timers/Counters- Serial Communication, Interrupt structure-programming Interfacing of peripherals – LED (ALP and embedded C programming). OLED, LCD, ADC, DAC, sensors, simple Switch and keyboard interfacing, 7 segment LED (embedded C programming).	11

3	Embedded Systems Overview Introduction to Embedded Systems: Definition, Features, Simple Example of Embedded Systems, Applications of embedded systems- Consumer electronics, Robotics, Automobiles Embedded System Architecture: HW - Processor, Controller, SoC, Memory, Peripherals; SW - Application, Middleware, OS, Device Drivers, Tool chain- Assembler, Interpreter, Compiler, Linker, Loader, Debugger Embedded system design process: Requirement Analysis, Specification Development, HW & SW Co-Design and Development, Module Integration, Testing.	11
4	Embedded System Board Study (Arduino Uno) Arduino Uno Board: Board Study (Board level Block schematic) - Chip (Features only - Architecture not needed), GPIO, Memory, Programming Interface Programming: Arduino IDE, Sample Code (LED, Switch, DC motor, Stepper motor control), Temperature monitoring system using LM35 Temperature sensor & Seven Segment display Embedded Operating system basic concepts: Functional layers in a computer system OS terminology, Kernel Functions (Overview only), Types of Kernels (Monolithic kernel & Microkernel), Tasks/ Processes.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the internal architecture of 8051 Microcontroller.	K2
CO2	Develop simple programs for 8051 using assembly language programming.	K3
CO3	Understand the interfacing of 8051 microcontroller with peripheral devices using ALP/ Embedded C.	K2
CO4	Interpret the architecture and design concept of embedded systems.	K2
CO5	Develop embedded systems based on Arduino.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1									
CO2	3	2	3									
CO3	3	2	3		2							
CO4	3	2	2	2								
CO5	3	2	3		2							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The 8051 Microcontroller and Embedded Systems: Using Assembly and C	Muhammad Ali Mazidi	Pearson	2e, 2007
2	Embedded Systems: An Integrated Approach	Lyla B Das	Pearson	1e, 2012
3	Beginning Arduino	Michael McRoberts	Apress	1e, 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Embedded Systems Architecture, programming and Design	Raj Kamal,	Tata McGraw-Hill	3/e, 2013
2	Handbook of Modern Sensors: Physics, Designs, and Applications	Jacob Fraden	Springer	5/e, 2016
3	Embedded Systems Architecture, A Comprehensive Guide for Engineers and Programmers,	Tammy Noergaard	Newnes – Elsevier	2/e, 2012
4	Arduino: The complete guide to Arduino for beginners, including projects, tips, tricks and programming,	James Arthur,	Ingram	1/e, 2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_ee32/preview
2	https://onlinecourses.nptel.ac.in/noc21_ee32/preview
3	https://archive.nptel.ac.in/courses/106/105/106105193/
4	https://nptel.ac.in/courses/106105193

SEMESTER S3

FUNDAMENTALS OF CYBER PHYSICAL SYSTEMS

Course Code	PCCPT303	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GBEST104	Course Type	Theory

Course Objectives:

1. To understand the fundamental concepts and components of Cyber-Physical Systems (CPS).
2. To apply modelling techniques, simulation tools, and control strategies for CPS design and analysis.
3. To identify advanced control techniques and security measures in CPS designs.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to CPS : Overview, definition, and examples, Interdisciplinary nature and key components, Basics of embedded systems and real-time constraints, Modeling techniques for CPS (discrete-event, state-based), Simulation tools and environments	11
2	Sensors and Actuators : Types, principles of operation, and applications in CPS, Signal conditioning, interfacing, and control strategies. Communication Protocols and Networks : Wired and wireless communication standards, IoT (Internet of Things) protocols and architectures, Challenges and solutions in reliable communication for CPS	11
3	Control Systems in CPS : Fundamentals of control theory, Control systems for CPS applications, Real-time operating systems (RTOS) and scheduling algorithms. Advanced Control Techniques and Security in CPS : Advanced control strategies (adaptive control, model predictive control), Security challenges in CPS (authentication, encryption), Fault tolerance and reliability in CPS designs	11

4	CPS Applications in Industry : Automotive electronics, autonomous vehicles, smart grids, and energy management systems, Role of CPS in industrial automation and robotics Emerging Trends and Future Directions : Edge computing and its role in CPS, AI and machine learning applications in CPS, Ethical considerations and societal impacts of CPS technologies	11
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand fundamental concepts and components of CPS.	K2
CO2	Apply modelling techniques and simulation tools for CPS design.	K3
CO3	Utilize operation principles and control strategies of CPS sensors and actuators.	K3
CO4	Examine communication challenges and solutions in CPS.	K4
CO5	Develop CPS designs integrating advanced control and security measures.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									
CO2	2	2	3	2	3							
CO3	3	3		3	2							
CO4	3	3			2							
CO5	3	2	3		2							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cyber-Physical Systems	Raj Rajkumar, Dionisio De Niz, Mark Klein	Pearson Education	1/e, 2017
2	Principles of Cyber Physical Systems	Rajeev Alur	MIT Press	1/e, 2015
3	Introduction to Embedded Systems – A Cyber–Physical Systems Approach	E. A. Lee, Sanjit Seshia	MIT Press	2/e, 2017
4	Communication Networks for Smart Grids: Making Smart Grid Real	Kenneth C. Budka, Jayant G. Deshpande	Springer	1/e, 2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Real-Time Systems Design and Analysis: Tools for the Practitioner	Seppo J. Ovaska, Phillip A. Laplante	Wiley	4/e, 2013
2	Security and Privacy in Cyber-Physical Systems: Foundations, Principles, and Applications	Glenn A. Fink, Houbing Song, Sabina Jeschke	Wiley	1/e, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	
1 - 4	https://nptel.ac.in/courses/106105241

SEMESTER S3

ANALOG AND DIGITAL ELECTRONICS

Course Code	PBCPT304	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	PC-PBL

Course Objectives:

1. To develop different types of analog circuits using discrete electronic components.
2. To analyze the basics of Op-Amp circuits and its applications.
3. To understand the basic knowledge of logic circuits and enable students to apply it to design a digital system.
4. To understand basic techniques for the design of digital circuits and fundamental concepts used in the design of Combinational and Sequential Logic circuits.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	BJT Amplifiers: BJT Amplifiers, RC Coupled amplifier: Need of different components and design. Multistage amplifiers: Effect of cascading on gain and bandwidth. NMOSFET amplifiers: MOSFET circuits at DC, MOSFET as an amplifier, Biasing of discrete MOSFET amplifier. Oscillator Circuits: Effect of positive and negative feedback on gain, frequency response and distortion. The four basic feedback topologies - voltage gain, input and output impedance. Criterion for oscillation, Wien bridge oscillator: working and design. Hartley and Crystal oscillator (working principle and design equations).	9
2	Operational Amplifiers: Basics, Block diagram of an Op Amp, Op Amp parameters-Slew Rate, Band Width, CMRR, Gain, Offset Voltage and Offset Current, differential and common mode operation of Op Amps, Inverting & Non Inverting Amplifier, (Closed Loop), Inverting and Non inverting summing amplifier. Op-Amp based Phase Shift Oscillator.	9

3	Boolean Algebra: Binary logic functions, Boolean laws, truth tables, associative and distributive properties, De- Morgans theorems, realization of switching functions using logic gates. Classification of Logic Families. Combinational Logic: Canonical logic forms, sum of product & product of sums, Karnaugh maps, two and three variable Karnaugh maps, simplification of expressions	9
4	Combinational and Sequential Logic: Analysis & design of Combinational Logic: Introduction to combinational circuits, code conversions, decoder, encoder, priority encoder, binary adder, subtractor . Sequential Logic: Sequential circuits, flip-flops: JK, D & T, clocked and edge triggered flipflop, asynchronous and synchronous counters, registers, serial in serial out shift registers.	9

Project based learning (8Hrs)

- The objective of **Project based learning** to strengthen the understanding of student's fundamentals through effective application of theoretical concepts. Projects can help to boost student's skills and widen the horizon of their thinking.
- The ultimate aim of an engineering student is to resolve a problem by applying theoretical knowledge.
- In the beginning, students shall be given an awareness and/or guideline on how to implement and learn this course through a project and form groups in the first few weeks.
- Each group should identify a project that can be done to understand a part of the contents of this course and perform a literature survey.
- Choose a suitable platform for your application and the input output devices to be interfaced.
- Develop the block diagram and present before the jury within first month after the commencement of classes.
- Implement necessary hardware and software to develop the prototype.
- Prepare a report and present before the jury for final evaluation.

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16marks)	- Each question carries 6 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop basic amplifiers using BJT and MOSFET	K3
CO2	Construct feedback circuits using BJT.	K3
CO3	Understand the fundamental principles of operational amplifiers.	K2
CO4	Understand digital logic and Boolean algebra	K2
CO5	Analyse sequential and combinational circuits.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	2							
CO2	3	2	3	2	2							
CO3	3	2		2	2							
CO4	3	3		2	2							
CO5	3	2	2	2	2							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electronic Devices and Circuit Theory	Robert Boylestad and L Nashelsky	Pearson	11/e, 2015
2	Digital Circuits and Systems	D.V.Hall	Tata Mc Graw Hill	1/e, 1989
3	Digital Design	Mano M.M., Ciletti M.D	Pearson India	4/e, 2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Op- Amps and Linear Integrated Circuits	Ramakant A Gayakwad	Prentice Hall of India	4/e,2015
2	Analog and Digital Electronics	Sanjay Agarwal Sonveer Singh	TechSarPvt. Ltd	1/e,2020
3	Electronic Devices and Circuits	Anil K Maini, Varsha Agarwal	Wiley	2/e,2012
4	Linear Integrated Circuits	Roy D. C. and S. B. Jain	New Age International	3/e, 2010

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/117/103/117103063/
2	https://archive.nptel.ac.in/courses/117/101/117101106/
3	https://archive.nptel.ac.in/courses/117/103/117103064/
4	https://archive.nptel.ac.in/courses/117/106/117106086/

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation / Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

- 1. Project Planning and Proposal (5 Marks)**
 - Clarity and feasibility of the project plan
 - Research and background understanding
 - Defined objectives and methodology
- 2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)**
 - Individual contribution to the presentation
 - Effectiveness in answering questions and handling feedback
- 3. Involvement in the Project Work and Team Work (3 Marks)**
 - Active participation and individual contribution
 - Teamwork and collaboration
- 4. Execution and Implementation (10 Marks)**
 - Adherence to the project timeline and milestones
 - Application of theoretical knowledge and problem-solving
 - Final Result
- 5. Final Presentation (5 Marks)**
 - Quality and clarity of the overall presentation
 - Individual contribution to the presentation
 - Effectiveness in answering questions
- 6. Project Quality, Innovation, and Creativity (3 Marks)**
 - Overall quality and technical excellence of the project
 - Innovation and originality in the project
 - Creativity in solutions and approaches

SEMESTER S3

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Course Code	GNEST305	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Demonstrate a solid understanding of advanced linear algebra concepts, machine learning algorithms and statistical analysis techniques relevant to engineering applications, principles and algorithms.
2. Apply theoretical concepts to solve practical engineering problems, analyze data to extract meaningful insights, and implement appropriate mathematical and computational techniques for AI and data science applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to AI and Machine Learning: Basics of Machine Learning - types of Machine Learning systems-challenges in ML- Supervised learning model example- regression models- Classification model example- Logistic regression-unsupervised model example- K-means clustering. Artificial Neural Network- Perceptron- Universal Approximation Theorem (statement only)- Multi-Layer Perceptron- Deep Neural Network- demonstration of regression and classification problems using MLP.(Text-2)	11
2	Mathematical Foundations of AI and Data science: Role of linear algebra in Data representation and analysis – Matrix decomposition- Singular Value Decomposition (SVD)- Spectral decomposition- Dimensionality reduction technique-Principal Component Analysis (PCA). (Text-1)	11

3	Applied Probability and Statistics for AI and Data Science: Basics of probability-random variables and statistical measures - rules in probability-Bayes theorem and its applications- statistical estimation-Maximum Likelihood Estimator (MLE) - statistical summaries- Correlation analysis-linear correlation (direct problems only)- regression analysis- linear regression (using least square method) (Text book 4)	11
4	Basics of Data Science: Benefits of data science-use of statistics and Machine Learning in Data Science- data science process - applications of Machine Learning in Data Science- modelling process- demonstration of ML applications in data science- Big Data and Data Science. (For visualization the software tools like Tableau, PowerBI, R or Python can be used. For Machine Learning implementation, Python, MATLAB or R can be used.)(Text book-5)	11

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concept of machine learning algorithms including neural networks and supervised/unsupervised learning techniques for engineering applications.	K3
CO2	Apply advanced mathematical concepts such as matrix operations, singular values, and principal component analysis to analyze and solve engineering problems.	K3
CO3	Analyze and interpret data using statistical methods including descriptive statistics, correlation, and regression analysis to derive meaningful insights and make informed decisions.	K3
CO4	Integrate statistical approaches and machine learning techniques to ensure practically feasible solutions in engineering contexts.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3								
CO2	3	3	3	3								
CO3	3	3	3	3								
CO4	3	3	3	3								
CO5	3	3	3	3								

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Linear Algebra	Gilbert Strang	Wellesley-Cambridge Press	6/e, 2023
2	Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow	Aurélien Géron	O'Reilly Media, Inc.	2/e, 2022
3	Mathematics for machine learning	Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong	Cambridge University Press	1/e, 2020
4	Fundamentals of mathematical statistics	Gupta, S. C., and V. K. Kapoor	Sultan Chand & Sons	9/e, 2020
5	Introducing data science: big data, machine learning, and more, using Python tools	Cielen, Davy, and Arno Meysman	Simon and Schuster	1/e, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data science: concepts and practice	Kotu, Vijay, and Bala Deshpande	Morgan Kaufmann	2/e, 2018
2	Probability and Statistics for Data Science	Carlos Fernandez-Granda	Center for Data Science in NYU	1/e, 2017
3	Foundations of Data Science	Avrim Blum, John Hopcroft, and Ravi Kannan	Cambridge University Press	1/e, 2020
4	Statistics For Data Science	James D. Miller	Packt Publishing	1/e, 2019
5	Probability and Statistics - The Science of Uncertainty	Michael J. Evans and Jeffrey S. Rosenthal	University of Toronto	1/e, 2009
6	An Introduction to the Science of Statistics: From Theory to Implementation	Joseph C. Watkins	chrome-extension://efaidnbmnnnibpcajpcgclefindmkaj/https://www.math.arizo	Preliminary Edition.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/106/106106198/
2	https://archive.nptel.ac.in/courses/106/106/106106198/ https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/resources/lecture-29-singular-value-decomposition/
3	https://ocw.mit.edu/courses/18-650-statistics-for-applications-fall-2016/resources/lecture-19-video/
4	https://archive.nptel.ac.in/courses/106/106/106106198/

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism , Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution -Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places -accessibility and social impacts, Managing conflict , Collective bargaining, Confidentiality , Role of confidentiality in moral integrity, Codes of Ethics . Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education , employment and everyday life, History of women in Science & Technology,	6

	Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy	6

	implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
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**Course Assessment Method
(CIE: 50 marks , ESE: 50)**

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks				50

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges

and outcomes.

- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S3

ANALOG AND DIGITAL ELECTRONICS LAB

Course Code	PCCPL307	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Practical

Course Objectives:

1. To design and build simple Analog circuits using discrete components.
2. To understand the concept of Digital Logic Design through the implementation of logic circuits using ICs of basic logic gates & flip flops.

Expt. No.	Experiments
Part A - Analog Electronics (Minimum 5 experiments)	
1	Design and analysis of RC coupled amplifier, need for different components
2	Multistage Amplifiers-Effects of cascading on gain and bandwidth
3	MOSFET amplifier (CS) - frequency response characteristics
4	Feedback amplifiers (current series, voltage series) - gain and frequency response.
5	Low frequency oscillators –Wien bridge oscillator
6	Operational Amplifiers Basics-Measurement of Opamp parameters
7	Design Inverting & Non Inverting Amplifier-frequency response
8	Design Phase Shift Oscillator using Opamp
Part B - Digital Electronics (Minimum 5 experiments)	
9	Realization of functions using basic and universal gates (SOP and POS forms).
10	Design and realization of half /full adder and subtractor using basic gates and universal gates.
11	Design and implementation of code converters using logic gates(i) BCD to excess-3 code and vice versa (ii) Binary to gray and vice-versa
12	Design and implementation of multiplexer and demultiplexer using logic gates.
13	Construction and verification of 4 bit ripple counter.
14	Synchronous Counter: Realization of Mod-N counters.
15	Design and implementation of any of SISO/SIPO/PISO shift registers
16	Design and implementation of Ring counter and Johnson Counter

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.

Endorsement by External Examiner: The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop analog circuits using BJT/FETs and evaluate their performance characteristics	K3
CO2	Design and implement feedback amplifier and oscillator circuits	K6
CO3	Design and implement basic circuits using OPAMP IC	K6
CO4	Design and implement combinational logic circuits using logic gates.	K6
CO5	Design and implement sequential logic circuits using integrated circuits.	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3						3		2	
CO2	3	3	3						3		2	
CO3	3	3	3						3		2	
CO4	3	3	3						3		2	
CO5	3	3	3						3		2	

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Microelectronic Circuits	Sedra, Adel S., and Kenneth C. Smith.	Oxford University Press	8/e, 2015.
2	Electronic Devices and Circuit Theory	Boylestad, Robert L., and Louis Nashelsky	Pearson	11/e, 2016.
3	Digital Design	Mano, M. Morris, and Michael D. Ciletti	Pearson	6/e, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electronic Circuit Analysis and Design	Donald A. Neamen	McGraw-Hill Education	2/e, 2001
2	Design of Analog CMOS Integrated Circuits	Behzad Razavi	McGraw-Hill Education	1/e, 2000

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/108102095
2	https://youtu.be/9g9dowLjmCA
3	https://youtu.be/CeD2L6KbtVM
4	https://youtu.be/HcH0khFGwS8

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S3

INSTRUMENTATION LAB

Course Code	PCCPL308	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Practical

Course Objectives:

1. To Understand Instruments and Transducers.
2. To understand the concepts of Error Analysis and Calibration.
3. To understand the Principles of Measurement.

Expt. No.	Experiments
1	Study and calibration of LVDT transducer for displacement measurement
2	Calibration of Capacitive transducer for angular displacement
3	Design of temperature transmitter using RTD.
4	Calibration of strain gauge for temperature measurement.
5	Level Measurement using Capacitance Sensor
6	Bridge Amplifier for Temperature Measurement.
7	Calibration of thermistor for temperature measurement
8	Calibration of thermocouple for temperature measurement
9	Measurement of level using differential pressure transmitter.
10	Measurement of displacement using strain gauge based displacement transducer

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voice	Record	Total
10	15	10	10	5	50

Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.

Endorsement by External Examiner: The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify different measuring instruments for the measurement of various electrical and non-electrical parameters.	K3
CO2	Select various transducers for the measurement of physical quantities like temperature, pressure, distance and displacement.	K1
CO3	Analyze the factors contributing to the errors present in measuring instruments and calibrate them.	K4
CO4	Analyze the characteristics of capacitive transducers and temperature transducers.	K4
CO5	Understand the basics and working principle of pressure, temperature and flow measurement.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3						3		2	
CO2	3	3	3						3		2	
CO3	3	3	3						3		2	
CO4	3	3	3						3		2	
CO5	3	3	3						3		2	

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Course in Electrical and Electronic Measurements and Instrumentation	A.K Sawhney	Dhanpat Rai & Co.	12/e, 2015
2	Measurement of Instrumentation :Theory and Application	Alan S. Morris and Reza Langari	Butterworth-Heinemann	3/e, 2012
3	Principles of Measurement Systems	John P. Bentley	Pearson Education	4/e, 2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electronic Instrumentation and Measurements	David A.Bell	McGraw-Hill Education	2/e, 1990
2	Instrumentation Reference Book	Wait Boyes	Butterworth-Heinemann	4/e, 2010

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=qbKnW42ZM5c
2	https://archive.nptel.ac.in/courses/108/105/108105064/
3	https://onlinecourses.nptel.ac.in/noc24_bt32/preview
4	https://onlinecourses.nptel.ac.in/noc23_ee105/preview

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 4

CYBER PHYSICAL SYSTEMS

SEMESTER S4

MATHEMATICS FOR ELECTRICAL SCIENCE – 4

(Group B)

Course Code	GBMAT401	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic calculus	Course Type	Theory

Course Objectives:

1. To familiarize students with the foundations of probabilistic and statistical analysis mostly used in varied applications in engineering and science.
2. To expose the students to the basics of random processes essential for their subsequent study of analog and digital communication

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Random variables, Discrete random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Binomial distribution, Poisson distribution, Poisson distribution as a limit of the binomial distribution, Joint pmf of two discrete random variables, Marginal pmf, Independent random variables, Expected value of a function of two discrete variables. [Text 1: Relevant topics from sections 3.1 to 3.4, 3.6, 5.1, 5.2]	9
2	Continuous random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Uniform, Normal and Exponential distributions, Joint pdf of two Continuous random variables, Marginal pdf, Independent random variables, Expectation value of a function of two continuous variables.	9

	[Text 1: Relevant topics from sections 3.1, 4.1, 4.2, 4.3, 4.4, 5.1, 5.2]	
3	Confidence Intervals, Confidence Level, Confidence Intervals and One-side confidence intervals for a Population Mean for large and small samples (normal distribution and t-distribution), Hypotheses and Test Procedures, Type I and Type II error, z Tests for Hypotheses about a Population Mean (for large sample), t Test for Hypotheses about a Population Mean (for small sample), Tests concerning a population proportion for large and small samples. [Text 1: Relevant topics from 7.1, 7.2, 7.3, 8.1, 8.2, 8.3, 8.4]	9
4	Random process concept, classification of process, Methods of Description of Random process, Special classes, Average Values of Random Process, Stationarity- SSS, WSS, Autocorrelation functions and its properties, Ergodicity, Mean-Ergodic Process, Mean-Ergodic Theorem, Correlation Ergodic Process, Distribution Ergodic Process. [Text 2: Relevant topics from Chapter 6]	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	Each question carries 9 marks. Two questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the concept, properties and important models of discrete random variables and to apply in suitable random phenomena.	K3
CO2	Understand the concept, properties and important models of continuous random variables and to apply in suitable random phenomena.	K3
CO3	Estimate population parameters, assess their certainty with confidence intervals, and test hypotheses about population means and proportions using z-tests and the one-sample <i>t</i> -test.	K3
CO4	Analyze random processes by classifying them, describing their properties, utilizing autocorrelation functions, and understanding their applications in areas like signal processing and communication systems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	-	-	-	-	-	-	2
CO2	3	3	2	2	-	-	-	-	-	-	-	2
CO3	3	3	2	2	-	-	-	-	-	-	-	2
CO4	3	3	2	2	-	-	-	-	-	-	-	2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability and Statistics for Engineering and the Sciences	Devore J. L	Cengage Learning	9 th edition, 2016
2	Probability, Statistics and Random Processes	T Veerarajan	The McGraw-Hill	3 rd edition, 2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability, Random Variables and Stochastic Processes,	Papoulis, A. & Pillai, S.U.,	McGraw Hill.	4 th edition, 2002
2	Introduction to Probability and Statistics for Engineers and Scientists	Ross, S. M.	Academic Press	6 th edition, 2020
3	Probability and Random Processes	Palaniammal, S.	PHI Learning Private Limited	3 rd edition, 2015
4	Introduction to Probability	David F. Anderson, Timo, Benedek	Cambridge	1 st edition, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/117/105/117105085/
2	https://archive.nptel.ac.in/courses/117/105/117105085/
3	https://archive.nptel.ac.in/courses/117/105/117105085/
4	https://archive.nptel.ac.in/courses/117/105/117105085/

SEMESTER S4

SIGNAL PROCESSING FOR ROBOTICS

Course Code	PCCPT402	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	PCCPT402	CIE Marks	40
Credits	3-1-0-0	ESE Marks	60
Prerequisites (if any)	4	Exam Hours	2 Hrs. 30 Min.

Course Objectives:

1. To understand the fundamentals of signal processing.
2. To design and implement filtering and noise reduction techniques
3. To understand wavelet transform and its applications.
4. To apply signal processing techniques to sensor data.
5. To implement signal processing algorithms.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Signal Processing: Characterization & classification of signals, typical Signal Processing operations, example of typical signals. Time Domain Representation of Signals & Systems: Discrete Time Signals, Operations on Sequences, the sampling process, Quantization, Discrete-Time systems, Time Domain characterization of LTI Discrete-Time systems, Convolution and impulse response	11
2	Transform-Domain Representation of Signals: Discrete Fourier Transform (DFT), DFT properties, computation of the DFT of real sequences, Fast Fourier transform (FFT).	11
3	Filtering and Noise Reduction: Types of filters (low-pass, high-pass, band-pass, band-stop), Design and implementation of digital filters, Noise reduction techniques.	11

4	Wavelet Transform: Introduction to wavelets, Continuous wavelet transform (CWT), Discrete wavelet transform (DWT). Sensor Data Processing: Processing data from cameras, LIDAR, and IMUs, Sensor fusion techniques, Real-time data processing.	11
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Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	Each question carries 9 marks. Two questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Classify different types of signals and signal operations.	K2
CO2	Illustrate time-domain and transform-domain analysis of signals.	K2
CO3	Construct digital filters used in signal processing.	K3
CO4	Explain different wavelet transforms to analyze signals.	K2
CO5	Compare the processing of and data from multiple sensors.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2		2								
CO2	3	2	3	1								
CO3	3	2	3	1	2							
CO4	3	2		1								
CO5	3	2	3	1								

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Signal Processing: Principles, Algorithms, and Applications	John G. Proakis and Dimitris G. Manolakis	Pearson	4/e, 2007

2	Discrete-Time Signal Processing	Alan V. Oppenheim and Ronald W. Schafer	Pearson	3/e, 2010
3	Understanding Digital Signal Processing	Richard G. Lyons	Pearson	3/e, 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Wavelet Transforms and Time-Frequency Signal Analysis	Lokenath Debnath	Birkhauser Boston Inc	2/e, 2011
2	Robotics, Vision and Control: Fundamental Algorithms In MATLAB	Peter Corke	Springer	2/e, 2017
3	Wavelet Transforms and Time-Frequency Signal Analysis	Lokenath Debnath	Birkhauser Boston Inc	1/e, 2001

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/117102060
2	https://onlinecourses.nptel.ac.in/noc22_ee28/preview
3	https://nptel.ac.in/courses/108105055
4	https://nptel.ac.in/courses/112105249

SEMESTER S4

SOFT COMPUTING

Course Code	PCCPT403	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GBMAT101, GBMAT201, UCEST105, GBEST204, PBCPT304	Course Type	Theory

Course Objectives:

1. To explain various components of soft computing like fuzzy logic, neural networks and genetic algorithm.
2. To understand the fundamentals of Fuzzy sets, including properties, operations and membership functions.
3. To summarize the fuzzy inference systems and defuzzification methods.
4. To understand the principles of MOO and Neuro-Fuzzy hybrid systems, including classification and characteristics.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Soft Computing, Difference between Hard Computing & Soft Computing & Applications of Soft Computing, Artificial Neurons Vs Biological Neurons, Basic models of artificial neural networks, Activation Functions, McCulloch and Pitts Neuron, Hebb network, Perceptron networks – Learning rule, Training and testing algorithm	11
2	Adaptive Linear Neuron, Back propagation Network, Introduction to Fuzzy Set, Properties & operations on fuzzy sets, Fuzzy membership functions, Fuzzification, Methods of membership value assignments, Fuzzy relations, Operations on Fuzzy Relation, Fuzzy Propositions & Fuzzy Implications, Lamda cuts for fuzzy sets	11
3	Defuzzification methods, Fuzzy Inference Systems - Mamdani type, Fuzzy Inference Systems - Sugeno type, Fuzzy Logic Controller, Introduction to genetic algorithm, operators in genetic algorithm – coding, Selection, Cross over, Mutation, stopping condition for genetic algorithm	11

4	MOOP-Linear & Nonlinear, Convex & Non Convex, Principles of MOO- Illustrating Pareto Optimal Solutions, Objectives in MOOP, Dominance & Pareto-Optimality-Concept of Domination, Properties of Dominance Relation, Pareto Optimality, Procedure for finding a non dominated set, Optimality Conditions, Neuro Fuzzy hybrid system-Classification & characteristics, Genetic – neuro hybrid systems	11
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	Each question carries 9 marks. Two questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamental differences between hard computing and soft computing	K2
CO2	Understand the basic models of artificial neural networks, activation functions and principles of learning	K2
CO3	Summarize fuzzy set theory for performing operations such as fuzzification, defuzzification, and utilization fuzzy inference systems for decision-making processes.	K2
CO4	Identify genetic algorithms to solve optimization problems with multiple objectives (MOOP).	K3
CO5	Construct hybrid systems such as neuro-fuzzy and genetic-neuro hybrids in solving complex engineering problems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1		1							
CO2	3	2	1		1							
CO3	3	3	2		1							
CO4	3	3	2	1	2							
CO5	3	3	3	2	2							

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fuzzy Logic with Engineering Applications	Timothy J. Ross	John Wiley and Sons	3/e, 2010
2	Fuzzy Sets and Fuzzy Logic Theory and Applications	George J. Klir and Bo Yuan	Prentice Hall	1/3, 1995

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Applied Fuzzy Electronics	Ibrahim A. M.	Prentice Hall	2/e, 2013
2	Fuzzy Logic: Intelligence, Control and Information	J. Yen, R. Langari	Pearson Education	2/e, 1998

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1 - 4	https://nptel.ac.in/courses/127105006

SEMESTER S4

INTRODUCTION TO PROGRAMMING FOR CPS

Course Code	PBCPT404	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCCPT205	Course Type	PC-PBL

Course Objectives:

1. To understand the programming techniques for robotics and cyber-physical systems using Python & MATLAB.
2. To understand the concepts of robot kinematics, dynamics, sensor integration, and hardware interfacing.
3. To build computational models, robot behaviours, enabling them to design, test, optimize robotic systems in a virtual environment using simulation techniques and tools.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Python Programming Variables, data types, and basic operations, Control Structures and Functions in Python Conditionals, loops, functions, modules, and packages	9
2	Introduction to MATLAB Environment, MATLAB syntax, variables, and arrays, MATLAB Scripts and Functions, Writing scripts, creating functions, and plotting	9
3	Robot Kinematics and Dynamics, Understanding robot movement and dynamics, Sensor Integration in Robotics, Reading sensor data, interfacing with hardware	9
4	Simulation Techniques for Robotics, Building computational models, simulating robot behaviors	9

Project based learning (8 Hrs)

- The objective of **Project based learning** to strengthen the understanding of student's fundamentals through effective application of theoretical concepts. Projects can help to boost student's skills and widen the horizon of their thinking.
- The ultimate aim of an engineering student is to resolve a problem by applying theoretical knowledge.
- In the beginning, students shall be given an awareness and/or guideline on how to implement and learn this course through a project and form groups in the first few weeks.
- Each group should identify a project that can be done to understand a part of the contents of this course and perform a literature survey.
- Choose a suitable platform for your application and the input output devices to be interfaced.
- Develop the block diagram and present before the jury within first month after the commencement of classes.
- Implement necessary hardware and software to develop the prototype.
- Prepare a report and present before the jury for final evaluation.

Course Assessment Method **(CIE: 60 marks , ESE: 40 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop algorithms for CPS control using Python and MATLAB.	K3
CO2	Utilize sensors and actuators in CPS.	K3
CO3	Examine data from CPS sensors using Python and MATLAB.	K4
CO4	Analyze robotic systems and cyber-physical interactions.	K4
CO5	Evaluate computational models and simulate robot behaviours	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1		3							
CO2	3	2	2		3							
CO3	3	2	2	2	3							
CO4	3	2	3		3	1			1	1		
CO5	3	2	3	2	3				1	1		

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Python Crash Course	Eric Matthes	No Starch Press	3/e, 2023
2	MATLAB for Engineers	Holly Moore	Pearson Education	6/e, 2022

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Learning Robotics Using Python	Lentin Joseph	Packt Pub Ltd	1/e, 2015
2	Numerical Computing with MATLAB	Cleve B. Moler	Society for Industrial and Applied Mathematics	1/e, 2004

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/106106145
2	https://nptel.ac.in/courses/111102137
3	https://in.mathworks.com/discovery/robot-programming.html
4	https://in.mathworks.com/discovery/robot-programming.html

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

Project Assessment and Evaluation criteria (30 Marks)

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S4

DIGITAL SYSTEM DESIGN

Course Code	PECPT411	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To develop an in-depth understanding of clocked synchronous and asynchronous sequential networks.
2. To summarize the architecture and applications of programmable logic devices.
3. To compare various modeling techniques using Verilog HDL.
4. To construct combinational and sequential circuits using Verilog HDL

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Analysis of clocked Synchronous Sequential Networks(CSSN): Modelling of CSSN – State assignment and reduction, Design of CSSN Analysis of Asynchronous Sequential Circuits (ASC): Flow table reduction- Races in ASC,State assignment problem and the transition table- Design of AS	9
2	Programmable Devices: Architecture of Programmable Array Logic and PLA, Architecture description of Field Programmable Gate Array and Complex Programmable Logic Devices, Case studies of robotic application using FPGA/CPLD.	9
3	Gajski's 'Y' Chart, Behavioral Modeling, Data flow modeling, Structural modeling Hardware Description Language: Structure of Verilog module, Operators, Data Types, Styles of Description. Verilog Data flow	9

	description: Highlights of Data flow description, Structure of Data flow description.	
4	Verilog Behavioral description: Structure, Variable Assignment Statement, Sequential Statements, Loop Statements, Verilog Behavioral Description of Multiplexers (2:1, 4:1, 8:1). Verilog Structural description: Highlights of Structural description, Organization of structural description, Structural description of ripple carry adder.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	Each question carries 9 marks. Two questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Construct clocked synchronous sequential networks.	K3
CO2	Model asynchronous sequential circuits.	K3
CO3	Develop digital systems on FPGAs and CPLDs, and apply them in robotic applications	K3
CO4	Demonstrate the structure, types and data flow of Verilog code for combinational and sequential circuits.	K2
CO5	Compare digital systems using behavioural and structural Verilog descriptions.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									
CO2	3	2	3									
CO3	3	2	3		3							
CO4	3	2	2		2							
CO5	3	2	3		2							

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Principles & Design	Donald G Givone	Tata McGraw Hill	1/e, 2003
2	HDL Programming VHDL and Verilog.	Nazeih M Botros,	Dreamtech press	3/e, 2009

3	Introduction to Digital Design Using Digilent FPGA Boards,	Richard E. Haskell, Darrin M. Hanna	LBE Books- LLC Understanding	3/e, 2011
Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Digital Systems	M. Ercegovac, T. Lang and L.J. Moreno	Wiley	1/e, 2000
2	Digital Design	John F Wakerly	Pearson Education	3/e, 2002
3	Verilog HDL: A Guide to Digital Design and Synthesis	Samir Palnitkar	Pearson Education	2/e, 2003
4	Advanced Digital Design with the Verilog HDL	Michael D Ciletti	Pearson education	2/e, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/X7M3rUxUpOc
2	https://youtu.be/Y8FvvzcocT4
3	https://youtu.be/BoIOLczVulQ
4	https://nptel.ac.in/courses/117106114

SEMESTER S4

KINEMATICS OF MACHINERY

Course Code	PECPT412	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GBEST213	Course Type	Theory

Course Objectives:

1. To Understand the fundamentals of kinematics and dynamics of machinery
2. To summarize techniques for studying motion of machines and their components.
3. To Illustrate the synthesis, analysis of cam mechanisms and explain the characteristics of involute action in gear trains.
4. To explain the kinematic analysis of spatial chains and robot kinematics.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to mechanisms: kinematic pairs, kinematic diagrams, classification of kinematic chains, kinematic inversions, and equivalent linkages. Kinematic analysis of planar mechanisms: mobility analysis and range of movement, Grashof criterion and inversions, displacement analysis, relative instantaneous centers, Aronhold-Kennedy theorem, velocity and acceleration analysis.	9
2	Dimensional synthesis of planar mechanisms: three position synthesis for function generation, path generation and rigid body guidance, dead center problems, branch and order defects.	9
3	Cams: synthesis of translating flat-face, translating roller and oscillating roller follower cams. Gears: fundamental law of gearing, characteristics of involutive action, analysis of gear trains.	9
4	Spatial kinematic chains and robot kinematics: kinematic analysis of spatial chains, Denavit- Hartenberg parameters, robot kinematics.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	Each question carries 9 marks. Two questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the classification and characteristics of kinematic chains and mechanisms	K2
CO2	Understand the mobility and range of movement of planar mechanisms	K2
CO3	Apply planar mechanisms for three-position synthesis, path generation, and to address dead center problems.	K3
CO4	Apply knowledge of cam profiles and motion characteristics.	K3
CO5	Understand fundamental laws of gearing, characteristics of involute action, and analyze gear trains for various applications.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									
CO2	3	2	2		2							
CO3	3	2	2		2							
CO4	3	2	2		1							
CO5	3	2	2		2							

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Theory of Applied Robotics	Reza N. Jazar	Springer	2/e, 2010
2	Modern Robotics	Kevin M. Lynch, Frank C. Park	Cambridge University Press	1/e, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Theory Of Machines	R S Khurmi, J K Gupta	S Chand	4/e, 2019
2	Mechatronics : Principles, Concepts And Applications	Nitaigour Mahalik	McGraw Hill Education	2/e, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-4	https://nptel.ac.in/courses/112105236

SEMESTER S4

SECURITY OF CYBER-PHYSICAL SYSTEMS

Course Code	PECPT413	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	none	Course Type	Theory

Course Objectives:

1. To understand various types of cyber-attacks.
2. To understand the defensive techniques against these attacks.
3. To illustrate cyber security standards applicable to CPS.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Definition and examples of CPS, Integration of computing, networking, and physical elements in CPS. Unique vulnerabilities and attack surfaces in CPS. CIA triad (Confidentiality, Integrity, Availability) in CPS, Security goals and principles tailored to CPS. Identifying threats and assessing risks specific to CPS, Risk mitigation strategies for CPS environments.	9
2	Principles of defense-in-depth in CPS, Security by design approaches for CPS architectures. Implementing authentication mechanisms in CPS, Role-based access control (RBAC) and other access control models in CPS. Symmetric and asymmetric encryption in CPS, Key management considerations for secure communication in CPS. Secure coding guidelines for CPS, Testing methodologies (unit testing, integration testing) for secure CPS applications.	9

3	MQTT, CoAP, and other protocols used in CPS, Securing data in transit and at rest in CPS environments. Implementing VLANs and subnetting for network security in CPS, Validating network segmentation strategies in CPS. Signature-based vs. anomaly-based detection in CPS, Deploying IDPS solutions tailored to CPS environments. Importance of logging and monitoring for detecting security incidents in CPS, Tools and techniques for effective monitoring in CPS.	9
4	Developing incident response plans (IRPs) specific to CPS, Incident handling and escalation procedures in CPS environments. Conducting penetration testing and vulnerability assessments in CPS, Importance of responsible disclosure policies in CPS security. Integrating security into the development lifecycle of CPS projects, Continuous improvement and adaptation of CPS security practices. Overview of cybersecurity standards applicable to CPS, Compliance considerations in CPS deployments.	9

Course Assessment Method

(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	Each question carries 9 marks. Two questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify and classify unique vulnerabilities in Cyber-Physical Systems (CPS).	K4
CO2	Develop defence strategies tailored to CPS environments.	K3
CO3	Analyze and select appropriate network security measures for CPS deployments.	K4
CO4	Develop and simulate incident response plans (IRPs) specific to Cyber-Physical Systems.	K3
CO5	Analyze compliance requirements and cyber security standards applicable to CPS.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2									
CO2	3	3	2									
CO3	3	3	2									
CO4	3	3	2									
CO5	3	3	2									

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Security and Privacy in Cyber-Physical Systems: Foundations, Principles, and Applications	Houbing Song, Glenn A. Fink, Sabina Jeschke	Wiley-IEEE Press	1/e, 2017
2	Securing Cyber-Physical Systems	Al-Sakib Khan Pathan	CRC Press	1/e, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Security of Cyber-Physical Systems	Jeeva Jose, Vijo Mathews	Khanna Publishing	1/e, 2022
2	Cyber-Physical Systems: From Theory to Practice	Danda B. Rawat, Joel J.P.C. Rodrigues, Ivan Stojmenovic	CRC Press	1/e, 2020

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1 - 4	https://nptel.ac.in/courses/106104467

SEMESTER S4

FUNDAMENTALS OF HUMAN COMPUTER INTERACTION

Course Code	PECPT414	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To explain the steps of experimental design, usability and experimental testing, and evaluation of human computer interaction systems.
2. To explain the capabilities of both humans and computers from the viewpoint of human information processing.
3. To illustrate an understanding of guidelines, principles, and theories influencing human computer interaction.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Importance of user Interface – definition, importance of good design. Benefits of good design. A brief history of Screen design. The graphical user interface – popularity of graphics, the concept of direct manipulation, graphical system, Characteristics, Web user – Interface popularity, characteristics- Principles of user interface.	9
2	Design process: Human interaction with computers, importance of human characteristics human consideration, Human interaction speeds, understanding business junctions. Screen Designing: Design goals – Screen planning and purpose, organizing screen elements, ordering of screen data and content – screen navigation and flow – Visually pleasing composition – amount of information – focus and emphasis – presentation information simply and meaningfully – information retrieval on web – statistical graphics – Technological consideration in interface design.	9

3	Windows: New and Navigation schemes selection of window, selection of devices based and screen based controls. Components – text and messages, Icons and increases – Multimedia, colors, uses problems, choosing colors. Software tools: Specification methods, interface – Building Tools, Interaction Devices – Keyboard and function keys – pointing devices – speech recognition digitization and generation – image and video displays – drivers.	9
4	HCI in the software process: The software life cycle Usability engineering, Iterative design and prototyping Design Focus: Prototyping in practice Design rationale Design rules Principles to support usability Standards Golden rules and heuristics HCI patterns Evaluation techniques, Goals of evaluation, Evaluation through expert analysis, Evaluation through user participation, Choosing an evaluation method. Universal design, Universal design principles Multi-modal interaction	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	Each question carries 9 marks. Two questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the capabilities of both humans and computers from the viewpoint of human information processing.	K2
CO2	Illustrate typical human-computer interaction (HCI) models, styles, and various historic HCI paradigms.	K2
CO3	Apply an interactive design process and universal design principles to designing HCI systems	K3
CO4	Explain the future user interface designer to tools, techniques, and ideas for interface design.	K2
CO5	Apply HCI design principles, standards and guidelines.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1					1		1	
CO2	2	2	1	1					1		1	
CO3	2	2	3	1					1		1	
CO4	2	2	3	1					1		1	
CO5	2	2	3	1					1		1	

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human Computer Interface	I. A. Dhotre, Soumitra Das	Technical Publications	3/e, 2023
2	Human-Computer Interaction: User-Centric Computing for Design	Samit Bhattacharya	McGraw Hill Education	1e/, 2019
3	Human Computer Interaction	P.KrishanSankar , N.P.Shankaranarayana	ARS Publications	1/e, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	User Interface Design	Soren Lauesen	Pearson Education	1/e, 2005
2	Human – Computer Interaction	Alan Dix, Janet Fincay, GreGoryd, Abowd, Russell Bealg	Pearson Education	3/e, 2004
3	The essential guide to user interface design	Wilbert O Galitz	Wiley DreamaTech	2/e, 1947

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/106/106106177/
2	https://archive.nptel.ac.in/courses/106/103/106103237/
3	https://nptel.ac.in/courses/106103237
4	https://archive.nptel.ac.in/courses/106/106/106106177/

SEMESTER S4

INTERNET OF THINGS (IOT)

Course Code	PECPT415	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the fundamentals of IoT, its vision, and strategic research directions.
2. To analyze various connectivity technologies and their application in sensor networks and M2M communications
3. To apply practical skills in Arduino / Raspberry Pi by integrating sensors and actuators to design and implement functional projects..
4. To analyze the interoperability in IoT systems and software-defined networking.
5. To understand data handling, analytics, and the role of cloud and fog computing in IoT.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to IoT: The Internet of Things Today, Time for Convergence, Towards the IoT Universe, Internet of Things Vision, IoT Strategic Research and Innovation Directions, IoT Applications, Future Internet Technologies, Review of sensing and actuation, Basics of IoT Networking	9
2	Connectivity Technologies, Sensor Networks, UAV networks. Machine-to-Machine Communications: M2M to IoT – A Basic Perspective– Introduction, Some Definitions, M2M Value Chains, IoT Value chains, LoRa Technology in M2M Communications:- Introduction to LoRa and LoRaWAN, LoRa Protocols and Standards, Applications of LoRa in M2M and IoT	9

3	Interoperability in IoT, Introduction to Arduino Programming, Integration of Sensors and Actuators with Arduino. Introduction to Raspberry Pi, Implementation of IoT with Raspberry Pi, Software Defined Networking, Software Defined IoT Networking	9
4	Data Handling and Analytics, Cloud Computing, and Fog Computing, Introduction to Industrial IoT. Case Studies: Smart Cities and Smart Homes, Connected Vehicles, Smart Grid	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Students should evaluate and analyze a real-world problem, assess the proposed solutions, conclude which solution is most appropriate for the problem, and then implement the chosen solution using executable code or simulate it using relevant software or supported hardware platforms.

Criteria for evaluation:

1. Problem Definition (K4 - 4 points)

- a. Clearly defines the real-world problem.
- b. Examine and identifies relevant contextual factors (constraints, resources, objectives).

2. Problem Analysis (K4 - 4 points)

- a. Break-down and presents a well-reasoned solution approach.
- b. Compare and justify the proposed solutions with evidence and logical reasoning.

3. Evaluate (K5 - 4 points)

- a. Thoroughly evaluate the proposed solutions.
- b. Compares trade-offs, advantages, and disadvantages.
- c. Considers feasibility, scalability, and practical implications.

4. Implementation (K5 - 4 points)

- a. Select the most feasible solution by implementing the proposed solutions.
- b. Successfully translates the chosen solution into an executable code or simulation using relevant software suit or supported hardware platform.
- c. Demonstrates proficiency in simulation / emulation practices (readability, efficiency, error handling).

5. **Conclusion (K4- 2 points, K5 – 2 points)**

- a. Summarizes findings and insights. State which solution is most appropriate for the problem. **(K4)**
- b. Reflects critical thinking and informed decision-making. **(K5)**

Scoring:

1. **Accomplished (4 points):** Exceptional analysis, clear implementation, and depth of understanding.
2. **Competent (3 points):** Solid performance with minor areas for improvement.
3. **Developing (2 points):** Adequate effort but lacks depth or clarity.
4. **Minimal (1 point):** Incomplete or significantly flawed.

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	Each question carries 9 marks. Two questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamental concepts of the Internet of Things (IoT).	K2
CO2	Demonstrate the integration of sensors and actuators with Arduino / Raspberry Pi to create basic IoT prototypes and systems.	K3
CO3	Analyze the role of Machine-to-Machine (M2M) communications in enhancing IoT applications and interoperability.	K4
CO4	Evaluate the use of data handling, analytics, cloud computing, and fog computing in the implementation of Industrial IoT.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1									
CO2	3	3	3	1	2		1					
CO3	3	3	3	2	2	1	1					1
CO4	3	3	3	2	2	1	1					1

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Internet of Things: A Hands-On Approach	Arshdeep Bahga and Vijay Madisetti	VPT (Universal Publishing)	1/e, 2014
2	The Internet of Things: Enabling Technologies, Platforms, and Use Cases	Pethuru Raj and Anupama C. Raman	CRC Press	1/e , 2017
3	Internet of Things (A Hands-on-Approach)	Raj Kamal	McGraw Hill Education	2/e, 2017

Reference Books				
Sl No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Internet of Things: How Smart TVs, Smart Cars, Smart Homes, and Smart Cities Are Changing the World.	Michael Miller	Pearson Education	1/e, 2015
2	Internet of Things	RMD Sundaram, Shriram K Vasudevan, and Abhishek S Nagarajan	John Wiley & Sons.	2/e, 2019
3	https://loradevelopers.semtech.com/uploads/documents/files/LoRa_and_LoRaWAN-A_Tech_Overview-Downloadable.pdf			

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1 - 4	https://archive.nptel.ac.in/courses/106/105/106105166/

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism, Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places-accessibility and social impacts, Managing conflict, Collective bargaining, Confidentiality, Role of confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education, employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in	6

	connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental	6

	policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
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**Course Assessment Method
(CIE: 50 marks , ESE: 50)**

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks				50

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business

plan.

- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S4

COMPUTER AIDED MODELING AND SIMULATION OF MECHANICAL SYSTEMS

Course Code	PCCPL407	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Practical

Course Objectives:

1. To utilize CAD software for basic part modelling and design.
2. To utilize MATLAB and relevant toolboxes (e.g., Robotics System Toolbox) for simulating robotic systems.

Expt. No.	Experiments
	Part A- MODELING Introduction to 3D modeling-basic part modeling, surface design, assembly design, Generation of 2D drawings from 3D Models, Sheet Metal Design, Kinematic simulation and 3D printing, Analysis of simple structures. Note: For Modelling/Analysis anyone of the 3D modelling software can be used
1	Part Modelling & drafting of Simple Machine parts Modelling of Hat, Angle, Angle with stiffener, C Brackets, sheet metal parts etc
2	Wireframe and Surface design of parts Surface modelling of outer cover of machine parts /electrical/electronic appliances/mouse etc
3	Assembly of machine elements Assembly of universal joints, couplings, plates with nuts-and-bolts etc
4	3D Printing of objects 3D printing of simple parts, gears, buckles etc
5	Kinematic simulation of four bar and slider crank Mechanisms
6	Static stress analysis of simple structures Static stress Analysis of cantilever, simply supported beam etc

	Part B- Simulation Lab(Minimum FIVE experiments) Tool: MATLAB
7	Simulation Environment Setup Set up MATLAB environment for simulating robotic systems and visualize workspace.
8	Forward Kinematics Simulation Implement forward kinematics for a 2-DOF robotic manipulator in MATLAB.
9	To interface Push button/Digital sensor (IR/LDR) with Arduino/Raspberry Pi and write a program to turn ON LED when push button is pressed or at sensor detection Inverse Kinematics Simulation Develop MATLAB solver for inverse kinematics of a 2-DOF robotic manipulator.
10	To interface DHT11 sensor with Arduino/Raspberry Pi and write a program to print temperature and humidity readings. Dynamics Simulation Model dynamics of a three-link robotic arm and simulate responses in MATLAB..
11	Sensor Simulation Integrate and simulate camera sensor for a mobile robot in MATLAB.
12	Path Planning and Motion Simulation Implement A* algorithm in MATLAB for collision-free path planning of a mobile robot.

Course Assessment Method

(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Utilize CAD software tools for 3D modelling and drafting of parts and Assembly of mechanical parts	K3
CO2	Evaluate the prototypes of designed models using 3D printing	K5
CO3	Analyse the simple parts for its structural strength and mechanisms for its kinematics.	K4
CO4	Understand the use of MATLAB and relevant toolboxes (e.g., Robotics System Toolbox) for simulating robotic systems.	K2
CO5	Analyze robot kinematics and dynamics, Sensor Integration and Fusion, Path Planning and Motion Control through MATLAB simulations	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2	2				3		2	
CO2	3	3	3	2	2				3		2	
CO3	3	3	3	2	2				3		2	
CO4	3	3	3	2	2				3		2	
CO5	3	3	3	2	2				3		2	

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Textbooks/Reference				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Design with SOLIDWORKS 2024: A Step-by-Step Project Based Approach	David C. Planchard	SDC Publications	1/e, 2023.
2	https://www.mathworks.com/help/robotics/			
3	https://petercorke.com/toolboxes/robotics-toolbox/			

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.

- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER: S4

MICROCONTROLLERS AND IOT LABORATORY

Course Code	PCCPL408	CIE Marks	50
Teaching Hours/Week (L: T:P:R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To apply and analyze Embedded C/Assembly Language Programming techniques for modern micro controllers.
2. To develop the ability to interface microcontrollers through Embedded C or Assembly Language Programming.
3. To apply concepts related to IoT system using Arduino/Raspberry Pi/open platform.

Expt. No.	Experiments
	<u>Part A- Microcontroller Lab (Minimum 5 experiments)</u> Conduct ANY FIVE of the following experiments on an ARM Cortex-M3/M4 evaluation board using ARM assembly/Embedded 'C' & Keil μ Vision tool/compiler or using 8051 Simulators or other similar tools.
1	Display “Hello World” message using Internal UART
2	Interface and Control a DC Motor
3	Interface a Stepper motor and rotate it in clockwise and anti-clockwise direction
4	Interface a DAC and generate Triangular and Square waveforms.
5	Interface a 4x4 keyboard and display the key code on an LCD.
6	Using the Internal PWM module of ARM controller generate PWM and vary its duty cycle.
7	Demonstrate the use of an external interrupt to toggle an LED On/Off.
8	Display the Hex digits 0 to F on a 7-segment LED interface, with an appropriate delay in between.
9	Interface a simple Switch and display its status through Relay, Buzzer and LED

	<u>Part B- IoT Lab(Minimum 5 experiments)</u>
10	Familiarization with Arduino/Raspberry Pi and perform necessary software installation
11	To interface LED/Buzzer with Arduino/Raspberry Pi and write a program to turn ON LED for 1 sec after every 2 second.
12	To interface Push button/Digital sensor (IR/LDR) with Arduino/Raspberry Pi and write a program to turn ON LED when push button is pressed or at sensor detection.
13	To interface DHT11 sensor with Arduino/Raspberry Pi and write a program to print temperature and humidity readings.
14	To interface Bluetooth with Arduino/Raspberry Pi and write a program to send sensor data to smartphone using Bluetooth..
15	To interface Bluetooth with Arduino/Raspberry Pi and write a program to turn LED ON/OFF when 1'/'0' is received from smartphone using Bluetooth
16	Write a program on Arduino/Raspberry Pi to upload temperature and humidity data to Thing speak cloud
17	Write a program on Arduino/Raspberry Pi to retrieve temperature and humidity data from thing speak cloud.

Course Assessment Method

(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply basic programming concepts to develop and debug programs utilizing ARM Cortex-M3/M4 and 8051 microcontrollers.	K3
CO2	Utilize internal modules like UART, PWM, and timers in ARM microcontrollers.	K3
CO3	Design and build a system to Interface and control motors using microcontrollers.	K6
CO4	Create and analyze waveforms through the use of DAC interfaces.	K4
CO5	Develop programs IoT applications using Arduino/Raspberry P.	K3
CO6	Design of solutions to control various sensors and modules with Arduino/Raspberry Pi for IoT applications.	K6
CO7	Develop wireless communication using Bluetooth in IoT applications.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2	2				3		2	
CO2	3	3	3	2	2				3		2	
CO3	3	3	3	2	2				3		2	
CO4	3	3	3	2	2				3		2	
CO5	3	3	3	2	2				3		2	
CO6	3	3	3	2	2				3		2	
CO7	3	3	3	2	2				3		2	

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The 8051 Microcontroller and Embedded Systems Using Assembly and C	Muhammad Ali Mazidi, Janice Gillispie Mazidi, Rolin McKinlay	Pearson Education	2/e, 2007
2	Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language & C	Yifeng Zhu	E-Man Press LLC	3/e, 2017
3	ARM Assembly Language: Fundamentals and Techniques	William Hohl and Christopher Hinds	CRC Press	2/e, 2015
4	Arduino Cookbook	by Michael Margolis	O'Reilly Media	3/e, 2020
5	Exploring Arduino: Tools and Techniques for Engineering Wizardry	Jeremy Blum	Wiley	2/e, 2019

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted.

SEMESTER 5

CYBER PHYSICAL SYSTEMS

SEMESTER S5

CONTROL SYSTEMS

Course Code	PCCPT501	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-1-0-0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	—	Course Type	Theory

Course Objectives:

1. To understand the fundamental principles of control systems, including the modeling, analysis, and design of both linear and nonlinear systems.
2. To illustrate time domain analysis of first and second-order systems, including transient and steady-state responses.
3. To construct frequency domain analysis using Bode plots and build P, PI, and PID controllers.
4. To understand state space analysis and represent systems in various canonical forms.
5. To summarize nonlinear systems using describing functions and classify singular points.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Control System - Definition, Open loop vs closed loop control systems- components of atypical control system- Necessity of a control system in a Robot, bird's eye view of typical actuators in robot control systems- hydraulic, pneumatic and electric actuators- over view of basic types-DC motors for speed control, DC and AC servo motors for position control, Brushless DC motors for speed control of quadcopters and linear actuation mechanisms. Basic idea of feedbacks in robotic systems-sensors-eg. Linear and rotary encoders. Linear time invariant Systems- Transfer function, Necessity of knowing the transfer function, Modelling -Mechanical and Electromechanical systems –block diagram representation -block diagram reduction, characteristic equation, signal flow graph, overview Mason's gain formula	11

2	Time domain analysis of control systems: Transient and steady state responses - time domain specifications - first and second order systems - step responses of first and second order systems, steady state error analysis - static error coefficient of type 0,1, 2 systems –Dynamic error coefficients. Concept of stability: Time response for various pole locations - stability of feedback system -Routh's stability criterion. Root locus - General rules for constructing Root loci – stability from root loci - effect of addition of poles and zeros.	11
3	Frequency domain analysis: Frequency domain specifications- Analysis based on Bode plot -Log magnitude vs. phase plot. Design of P, PI and PID controller using Ziegler-Nichols tuning method, Overview-Necessity of using Lag, lead and lag-lead compensators in Control Systems. State space analysis of systems: Introduction to state concept - state equation of linear continuous time systems, matrix representation of state equations. Phase variable and canonical forms of state representation- controllable, observable, diagonal and Jordan canonical forms- solution of time invariant autonomous systems	11
4	Nonlinear systems: Introduction - characteristics of nonlinear systems. Types of nonlinearities. Determination of describing function of nonlinearities (relay, dead zone and saturation only) - application of describing function for stability analysis of autonomous system with single nonlinearity. Singular points – Classification of singular points. Definition of stability- asymptotic stability and instability, overview of Lyapunov methods to stability of linear and nonlinear, continuous time systems. Case study-kinematic modelling of a differential drive robot and controlling the same to move to a point	11

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Compare open-loop and closed-loop control systems and identify their components.	K2
CO2	Develop time domain analysis of first and second-order systems, including transient and steady-state responses.	K3
CO3	Analyze frequency domain using Bode plots and build P, PI, and PID controllers.	K3
CO4	Understand state space analysis and represent systems in various canonical forms.	K2
CO5	Explain nonlinear systems using describing functions and classify singular points.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										
CO2	3	2	3	2								
CO3	3	2	3	2								
CO4	3	2	2	2								
CO5	3	2	3	1								

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), : No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Modern Control Engineering	Katsuhiko Ogata	Pearson Education India	5/e, 2009
2	Control Systems Principles and Design	M. Gopal	McGraw Hill Education (India)	4/e, 2019
3	Control Systems Engineering	I. J. Nagrath and M. Gopal	New Age International Pvt Ltd	6/e, 2018
4	Control Systems.	A. Anand Kumar	PHI	2/e, 2018
5	Robotics, Vision and Control: Fundamental Algorithms in MATLAB	Peter Corke	Springer	2/e, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Robotics- Fundamental Concepts and Analysis	Ashitava Ghosal	Oxford University press	1/e, 2009
2	Control Systems Engineering	Nise N. S	Wiley Eastern	6/e, 2010
3	Modern Control Systems	Dorf R. C. and R. H. Bishop	Pearson Education	6/e, 2011
4	Nonlinear Systems, Prentice	Hassan K Khalil	Hall International (UK)	3/e, 2002

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/107/106/107106081/
2	https://onlinecourses.nptel.ac.in/noc20_ee90/preview
3	https://onlinecourses.nptel.ac.in/noc17_ee12/preview
4	https://nptel.ac.in/courses/107106081/

SEMESTER S5

INTRODUCTION TO WIRELESS COMMUNICATION

Course Code	PCCPT502	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	—	Course Type	Theory

Course Objectives:

1. To understand the basic concepts of cellular system and Trend in Cellular radio and personal communication.
2. To compare various wireless communication systems and networks.
3. To explain the benefits of diversity techniques in improving system performance.
4. To Compare and contrast different spectrum sharing technologies and multiple access techniques in various scenarios.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Wireless Communication System: Evolution of mobile communications, Mobile Radio System around the world, Types of Wireless communication System, Comparison of Common wireless system, Trend in Cellular radio and personal communication. Second generation Cellular Networks, Third Generation (3G) Wireless Networks, Wireless Local Loop (WLL), Wireless Local Area Network (WLAN), Bluetooth and Personal Area Networks. Introduction to Wi-Fi, WiMAX, ZigBee Networks	11

2	The Cellular Concept- System Design Fundamentals: Cellular system, Hexagonal geometry cell and concept of frequency reuse, Channel Assignment Strategies Distance to frequency reuse ratio, Channel & co-channel interference reduction factor, S/I ratio consideration and calculation for Minimum Co-channel and adjacent interference, Handoff Strategies, Umbrella Cell Concept, Trunking and Grade of Service, Improving Coverage & Capacity in Cellular System-cell splitting, Cell sectorization, Repeaters, Micro cell zone concept, Channel antenna system design considerations.	11
3	Mobile Radio Propagation Model, Small Scale Fading and diversity: Large scale path loss, Free Space Propagation loss equation, Path- loss of NLOS and LOS systems, Reflection, Ray ground reflection model, Diffraction, Scattering, Link budget design, Max. Distance Coverage formula, Empirical formula for path loss, Indoor and outdoor propagation models, Small scale multipath propagation, Statistical for models multipath fading channels and diversity techniques in brief.	11
4	Multiple Access Techniques: Introduction, Comparisons of multiple Access Strategies TDMA,CDMA, FDMA, OFDM , CSMA Protocols. Wireless Systems: GSM system architecture, Radio interface, Protocols, Localization and calling, Handover, Authentication and security in GSM, GSM speech coding, Concept of spread spectrum, GPRS system architecture. WI-FI and IEEE 802.11 WIRELESS LAN STANDARDS: IEEE 802 Architecture, IEEE 802.11 Architecture and Services, 802.11Medium Access Control, 802.11 Physical Layer, Other IEEE 802.11 Standards, Wi-Fi Protected Access.	11

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basics of wireless communication and propagation of radio signals.	K2
CO2	Understand basic principles behind radio resource management techniques such as power control, channel allocation and handoffs.	K2
CO3	Summarize basic principles of Mobile Radio Propagation Model, Small Scale Fading and diversity.	K2
CO4	Compare multiple access techniques and various wireless communication systems and networks.	K2
CO5	Understand the design consideration and architecture for different Wireless Systems like GSM, CDMA, GPRS, IEEE 802.11.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									
CO2	3	2	2			1						
CO3	3	3	2									
CO4	3	3	3		2							
CO5	3	3	3		2	1						1

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), : No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Wireless Communications: Principles and Practice	Theodore S. Rappaport	Prentice Hall.	2/e, 2002
2	Wireless Communication Systems	William C.Y.Lee	McGraw-Hill	4/e, 2006
3	Wireless Communications and Networks	William Stallings	Pearson	2/e, 2005

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mobile and Wireless Communications	R. Prasad	Artech House	3/e, 2010
2	Wireless Communication Fundamentals	A. Molisch	Wiley	2/e, 2011
3	Wireless Communications: Algorithms, Techniques, and Protocols	C. S. R. Murthy and B. S. Manoj	CRC Press	2/e, 2013
4	Wireless Communication Systems: Advanced Techniques for 5G and Beyond	Y. Chen and X. Zhang	Springer	1/e, 2020

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/117/102/117102062/
2	https://archive.nptel.ac.in/courses/108/106/106106167/
3	https://nptel.ac.in/courses/106106167
4	https://onlinecourses.nptel.ac.in/noc22_ee85/preview

SEMESTER S5

ELECTRIC DRIVES AND CONTROL

Course Code	PCCPT503	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	—	Course Type	Theory

Course Objectives:

1. To understand Basic elements-types of electric drives and Drive motor characteristics
2. To understand Starting methods and Conventional and solid state speed control of D.C & A.C Drives

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic elements-types of electric drives-factors influencing electric drives-heating and cooling curves-loading conditions and classes of duty-Selection of power rating for drive motors with regard to thermal overloading and load variation factors	9
2	Mechanical characteristics- speed- torque characteristics of various types of load and drive motors -braking of electrical motors-dc motors: shunt, series, compound motors-single phase and three phase induction motors	9
3	Types of d.c motor starters-typical control circuits for shunt and series motors-Star-delta starters, Auto-transformer starters, Soft starters for induction motors, three phase squirrel and slip ring induction motors	9
4	Speed control of DC series and shunt motors-Armature and field control, ward-leonard control system-using controlled rectifiers and DC choppers – applications Speed control of three phase induction motor-Voltage control, voltage/frequency control, slip power recovery scheme-using inverters and AC voltage regulators-applications	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 = 24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To understand basic elements, types of electric drives, and factors influencing their selection.	K2
CO2	To recall speed-torque characteristics of various types of loads and drive motors.	K1
CO3	To understand starting methods for DC motors and three-phase induction motors.	K2
CO4	To understand speed control techniques for DC series and shunt motors.	K2
CO5	To apply speed control methods for three-phase induction motors.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	2								
CO2	3	3	2	2								
CO3	2	2	3	3								
CO4	3	2	3	3								
CO5	2	1	2	2								

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), : No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A first course on Electric drives	Pillai S.K	Wiley Eastern Limited	1/e, 1998
2	Power electronics	M.D. Singh, K.B. Khanchandani	Tata McGraw-Hill	1/e, 1998
3	Fundamentals of Electrical Drives	Dubey G.K	Narosa Publishing House	2/e, 2002

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Art and science and utilization of electrical energy	H.Partab	Dhanpat Rai and sons	3/e, 1994
2	Electric Drives: Concepts and Application	Vedam Subramanyam	McGraw-Hill	1/e, 2001

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1 - 4	https://archive.nptel.ac.in/courses/108/104/108104140/

SEMESTER S5

FUNDAMENTALS OF REAL TIME DIGITAL SIGNAL PROCESSING

Course Code	PBCPT504	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCCPT402	Course Type	PC-PBL

Course Objectives:

1. To understand the basic elements and concepts of real-time signal processing systems.
2. To understand DSP architecture design
3. To understand the principles of FIR and IIR digital filter design and the implementation of these filters using MATLAB.
4. To illustrate the methods for generating and detecting digital signals, including sine waves, chirp signals, and noise generators, as well as DTMF generation and detection.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to real time Signal Processing: Basic elements of Real-Time Signal Processing Systems, Analog Interface- Sampling, Quantization and encoding, Smoothing filters, data converters, DSP hardware- DSP hardware options, Digital signal processors, Fixed point and floating point processors, Real time constraints, DSP System Design - Algorithm development, Selection of DSP hardware, software development and tools	9
2	Introduction to Digital Signal Processor: Introduction to a popular DSP from Texas Instruments, CPU Architecture, CPU Data Paths and Control, Timers, Interrupts, Internal Data/ Program Memory, External Memory Interface, pipelining, Number system (Fixed and floating point format), Fixed point Instruction Set and Addressing Modes, TMS 320C67XX CPU Simple programming examples using C and assembly.	9
3	Filter Design: FIR & IIR Digital Filter Design, filter Design programs using MATLAB, Fourier Transform: DFT, FFT programs using MATLAB	9

4	Digital signal generation and detection: Sine wave generators- Lookup table method, linear Chirp signal, Noise generators- Linear Congruential Sequence generator, Pseudo random binary sequence generator, White colour and Gaussian noise, DTMF Generation and Detection-DTMF Generator and Detector.	9
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Project based learning (8 hrs)

- The objective of Project based learning to strengthen the understanding of student's fundamentals through effective application of theoretical concepts. Projects can help to boost student's skills and widen the horizon of their thinking.
- The ultimate aim of an engineering student is to resolve a problem by applying theoretical knowledge.
- In the beginning, students shall be given an awareness and/or guideline on how to implement and learn this course through a project and form groups in the first few weeks.
- Each group should identify a project that can be done to understand a part of the contents of this course and perform a literature survey.
- Choose a suitable platform for your application and the input output devices to be interfaced.
- Develop the block diagram and present before the jury within first month after the commencement of classes.
- Implement necessary hardware and software to develop the prototype.
- Prepare a report and present before the jury for final evaluation.

Course Assessment Method (CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	- Each question carries 6 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic elements and functioning of real-time signal processing systems, including the analog interface and DSP hardware options	K2
CO2	Analyse the architecture and operational principles of a popular DSP, including CPU data paths, memory interfaces, and pipelining.	K4
CO3	Develop FIR and IIR digital filters and implement these designs using MATLAB.	K3
CO4	Analyse the methods for generating and detecting various digital signals, including sine waves, chirp signals, and noise, as well as DTMF signals.	K4
CO5	Develop and implement real-time DSP algorithms using appropriate hardware and software tools, demonstrating the ability to solve practical DSP problems in various applications.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3			2					2	1	2	
CO2	3	3							2	1	2	
CO3	3	3	2						2	1	2	
CO4	3	3	2	3					2	1	2	
CO5	3	3			2				2	1	2	

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), : No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Real-Time Digital Signal Processing Fundamentals, Implementations and Applications	Sen M. Kuo, Bob H. Lee, Wenshun Tian	John Wiley & Sons, Ltd	3/e, 2013
2	Digital Signal Processing and Application with C6713 and C6416 DSK	Rulph Chassaing	A Wiley-Interscience Publication	1/e, 2004
3	Digital Signal Processing Implementation using the TMS320C6000 DSP Platform	Naim Dahnoun	Prentice Hall	1/e, 2000

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Signal Processing - A Student Guide	T.J. Terrel and Lik-Kwan Shark	Macmillan Press Ltd	1/e, 1996
2	Digital Signal Processing: A System Design Approach	David J Defatta J, Lucas Joseph G & Hodkiss William S	John Wiley	1/e, 1988

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1 - 4	https://archive.nptel.ac.in/courses/108/108/108108185/

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S5

SENSOR AND ACTUATOR INTEGRATION

Course Code	PECPT521	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCCPT205	Course Type	Theory

Course Objectives:

1. To understand the knowledge of interfacing sensors and actuators with embedded controllers using various circuits and communication protocols.
2. To understand advanced techniques for interfacing resistive and capacitive sensors, focusing on linearity improvement and error reduction.
3. To understand the principles of miniaturization technology for smart sensors and actuators and their fabrication processes.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Overview of embedded system; Importance of sensors, actuators and interfacing circuits in embedded control system Embedded Sensors and Actuators: Various types of important sensors, actuators and their working principles: e.g, thermal, mechanical, electrical, magnetic, optical, chemical, smart material and meta material based.	9
2	Interfacing Aspects of Sensors and Actuators to Embedded Controller and their Communication Protocols: Signal conditioning circuits; Various Op-Amp based interfacing circuit implementation: Amplifier, Filter, ADC, DAC etc.; Various Serial Communication protocols for interfacing. Advancement in Interfacing Schemes of Resistive Sensors for Linearity Improvement and Error Reduction: Resistive sensor examples; Non-idealities in basic interfacing circuits; Linearization techniques; Error reduction schemes due to environmental effects and remote communication. Advanced Techniques for Direct Interfacing of Resistive Sensors with	9

	Embedded controller: Embedded controller based excitation system; Direct interfacing schemes of various resistive sensors topologies (e.g., single, differential and bridge type) to microcontrollers; Interfacing scheme for sensor array.	
3	Advanced Techniques for Direct Interfacing of Capacitive Sensors with Embedded Controller: Capacitive sensor examples; Interfacing scheme for different capacitive sensor configurations; Direct interfacing schemes. Advancement in Design of Interfacing Circuits for Lossy Capacitive Sensors: Lossy Capacitive sensor characteristics; Various advanced interfacing schemes for lossy capacitive sensor.	9
4	Miniaturization Technology for Smart Sensors and Actuators: Background of miniaturization; Miniaturized device fabrication process technology for Smart sensors and actuators. Miniaturized Sensors, Actuators and their Interfacing Electronics: Various types of important MEMS sensors and actuators: Design and operation; Interfacing Electronics for MEMS Devices; System-on-Chip integration; Applications.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the working principles of various sensors and actuators used in embedded systems.	K2
CO2	Develop signal conditioning circuits for interfacing sensors with embedded controllers.	K3
CO3	Develop advanced interfacing techniques for resistive and capacitive sensors to improve linearity and reduce errors.	K3
CO4	Illustrate an understanding of miniaturization technology and the fabrication process for smart sensors and actuators.	K2
CO5	Understand various types of MEMS sensors and actuators.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2								
CO2	3	3	2	2								
CO3	3	3	2	2								
CO4	3	3	2	2								
CO5	3	3	2									

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), : No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Sensors, Actuators, and their Interfaces	Nathan Ida	SciTech Publishing,	1/e, 2014
2	Advanced Interfacing Techniques for Sensors	B. George, J. Roy, V. Jagadeesh Kumar, S. C. Mukhopadhyay	Springer	1/e, 2017
3	Analog Interfacing to Embedded Microprocessor Systems	Stuart R. Ball	Elsevier	1/e, 2004

4	Sensors and Signal Conditioning	John G. Webster and Ramón Pallás-Areny	John Wiley & Sons	2/e, 2000
5	Fundamentals of Microfabrication and Nanotechnology	Marc Madou	CRC press	3/e, 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Smart Sensors and MEMS	S. Nihtianov, A. Luque	Elsevier	1/e, 2014
2	Instrument Engineers Handbook	Bela G Liptak	CRC press	4/e, 2003
3	Understanding Automotive Electronics: An Engineering Perspective	William B. Ribbens	Elsevier	8/e, 2017
4	Advanced Digital Design with the Verilog HDL	Michael D Ciletti	Pearson education	2/e, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1 - 4	https://youtu.be/kTLoQskfPx0

SEMESTER S5

COMPUTER COMMUNICATION

Course Code	PECPT522	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To summarize about various computer transmission modes, network models and network switching.
2. To apply knowledge of OSI layers to real-world networking scenarios.
3. To compare and Contrast the OSI model with other network models (eg: TCP/IP).
4. To understand the concepts, protocols, and security measures of computer networks.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to computer communication: Transmission modes - serial and parallel transmission, asynchronous, synchronous, simplex, half duplex, full duplex communication. Switching: circuit switching and packet switching Networks: Network criteria, physical structures, network models, categories of networks, Interconnection of Networks: Internetwork Network models: Layered tasks, OSI model, Layers in OSI model, TCP/IP protocol suite.	9
2	Physical Layer: Guided and unguided transmission media (Coaxial cable, UTP,STP, Fiber optic cable) Data Link Layer: Framing, Flow control (stop and wait , sliding window flow control) Error control, Error detection(check sum, CRC), Bit stuffing,HDLC Media access control: Ethernet (802.3), CSMA/CD, Logical link control, Wireless LAN (802.11), CSMA/CA.	9

3	Network Layer Logical addressing : IPv4 & IPV6 Address Resolution protocols (ARP, RARP) Subnetting, Classless Routing(CIDR), ICMP, IGMP, DHCP Virtual LAN, Networking devices (Hubs, Bridges & Switches) Transport Layer –UDP, TCP Congestion Control & Quality of Service – Data traffic, Congestion, Congestion Control, QoS and Flow Characteristics.	9
4	Application Layer – DNS, Remote Logging (Telnet), SMTP, FTP, WWW, HTTP, POP3, MIME, SNMP Security at Application Layer (E-MAIL, PGP and S/MIME), Security at Transport Layer (SSL and TLS), Security at Network Layer (IPSec). Routing : Routing and Forwarding, Static routing and Dynamic routing Routing Algorithms: Distance vector routing algorithm, Link state routing (Dijkstra's algorithm), Routing Protocols: Routing Information protocol (RIP), Open Shortest Path First (OSPF), Border Gateway Protocol (BGP), MPLS.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the Different types of network topologies and protocols.	K2
CO2	Summarize different layers of the OSI model and TCP/IP with their functions.	K2
CO3	Identify the basic protocols of computer networks, and how they can be used to assist in network design and implementation	K3
CO4	Understand OSI reference model and it's significance in computer communication.	K2
CO5	Explain the concept of subnetting and Routing mechanisms.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									
CO2	3	2	2									
CO3	3	3	2		1							
CO4	3	3	3		1							
CO5	3	3	3		1							

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), : No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Communication A Top -Down Approach	James F.Kurose and Keith W.Ross	Addison-Wesly	7/e, 2017
2	Computer Networks: A Systems Approach	Larry L.Peterson and BruceS.Davie	McGraw-Hill	5/e, 2016
3	Data Communications and Networking	Behrouz A. Forouzan	McGraw-Hill	5/e, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Communication Systems	R.J Evans	Wiley	1/e, 2017
2	Computer Communication: Principles and Protocols	R.S Salaria	I.K International Publishing House	2/e, 2018
3	Computer Communication and Networks	A.K Singh	SK kataria & sons	1/e, 2018
4	Computer Networks and Protocols: A Practical Approach	N.C. Pabrai	McGraw-Hill	1/e, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/105/106105183/
2	https://nptel.ac.in/courses/106105183
3	https://onlinecourses.nptel.ac.in/noc22_ee61/preview
4	https://onlinecourses.nptel.ac.in/noc22_cs19/preview

SEMESTER S5
CRYPTOGRAPHY FOR CPS

Course Code	PECPT523	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PECPT 413	Course Type	Theory

Course Objectives:

1. To understand the fundamental concepts of Classical and modern Cryptosystem.
2. To compare various private and public key Cryptosystem for encryption, key exchange and authentication algorithms.
3. To understand the different applications of cryptography.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Integer Arithmetic - Divisibility - Greatest Common Divisor - Extended Euclidean Algorithm- Linear Diophantine equation, Modular Arithmetic - Modulo Operator - Congruence - Addition and multiplicative inverse, Algebraic structures - Field – Finite fields of the form GF(p) and GF(2n) - Polynomial arithmetic, Prime Numbers – Fermat’s and Euler’s Theorem - Prime Factorization by Trial Division Method - Chinese Remainder Theorem. Security Goals - Security Services - Confidentiality -Integrity - Authentication- Nonrepudiation- Access control, Security Mechanisms – Encipherment- Data Integrity – Signature - Authentication Exchange - Traffic Padding- Routing Control – Notarization Access Control	9

2	Introduction to Cryptography -Classification of Cryptosystems - Cryptanalytic attacks - CipherProperties – Confusion - Diffusion. Classical Cryptosystems - Substitution Techniques - Monoalphabetic Cipher – Caesar Cipher- AffineCipher, Polyalphabetic Ciphers - Autokey Cipher- Playfair Cipher-Hill Cipher-VigenereCipher- One Time Pad Cipher, Stream and Block Ciphers, Modern Secret Key Ciphers - Substitution Box-Permutation Box-Product Ciphers Data Encryption Standard - DES - Structure of DES - DES Attacks - 2-DES - 3-DES, Advanced Encryption Standard - AES - Structure-Analysis, Cryptographic Hash Functions - Properties - Secure Hash Algorithm - SHA-512 Logic - SHA-512 Round Function, Message Authentication Code-Hash-based Message Authentication Code - HMAC -Cipher-based Message Authentication Code - CMAC	9
3	Public Key Cryptosystems -PKC - Types of PKC - Trapdoor - One-way functions, RSA Cryptosystem -Key Generation – Encryption- Decryption, ElGamal Cryptosystem – Key Generation – Encryption-Decryption, Diffie-Hellman Key Exchange Protocol- Man in the Middle attack on Diffie-Hellman Protocol. Digital Signature- Signing - Verification, Digital signature forgery-Existential forgery- Selective forgery - Universal forgery, RSA Digital Signature Scheme - ElGamal Signature Scheme.	9
4	Symmetric Key Distribution using Symmetric and Asymmetric Encryption - Distribution of public keys -Public announcement- Public available directory- Public-key authority- Public key Certificates. Electronic Mail Security -Pretty Good Privacy- PGP message format - Transmission and Reception of PGP Messages, IP Security Overview - IP Authentication Header - Encapsulating Security Payload - Distributed Denial of Service attacks, Secure Electronic Transaction – Payment Processing - Dual Signature, Firewalls - Firewall Design Principles	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concepts of number theory in designing crypto systems	K3
CO2	Explain various network security aspects, cryptanalytic attacks and classical cryptosystems	K2
CO3	Illustrate various symmetric key cryptosystems, hash and message authentication functions.	K2
CO4	Apply the principles of asymmetric key cryptosystems and digital signature.	K3
CO5	Illustrate various protocols to ensure Email Security and Network Security.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	2								
CO2	3	2	3									
CO3	3	3	3	2	2							
CO4	3	3	3	2	2							
CO5	3	3	3		2							

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), : No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cryptography and Network security: Principles and Practice	Stallings W	Pearson Education Asia	7/e, 2017
2	Cryptography and Network Security	Behrouz A Forouzan	Tata McGraw Hill Education Pvt Ltd Publication	2/e, 2007
3	Cryptography and Network Security	Atul Kahate	Tata McGraw Hill	4/e, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Network Security and Cryptography	Bernard Menezes	Cengage Learning India	1/e, 2011
2	Security in Computing	Charles P. Pleegeer, Shari Lawrence Pleegeer	Pearson Education Asia	5/e, 2015

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1 - 4	https://nptel.ac.in/courses/106105162

SEMESTER S5

INTERFACE DESIGN FOR CPS

Course Code	PECPT524	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PECPT 414	Course Type	Theory

Course Objectives:

1. To understand various types of input and output devices used in CPS, including sensors, actuators, displays, and feedback mechanisms, and their integration into system designs.
2. To understand the challenges and considerations involved in integrating input and output devices into CPS architectures, ensuring seamless operation and effective user interaction.
3. To illustrate methods for evaluating CPS interfaces, including usability testing, validation techniques, and assessing interface effectiveness in real-world scenarios.
4. To explore emerging technologies and future trends in CPS interface design, preparing students to adapt to advancements in the field.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of Interface Design for CPS: Introduction to Interface Design - Overview of interface design principles, Importance of interfaces in Cyber-Physical Systems Human-Computer Interaction (HCI) Basics - Foundations of HCI and its relevance to CPS, Cognitive psychology principles in interface design User-centered Design Process - Understanding user needs and requirements, Methods for gathering and analyzing user data in CPS contexts, Interface Design Methodologies - Design thinking and iterative design processes, Prototyping techniques for CPS interfaces	9

2	Designing Input Devices for CPS: Types of Input Devices - Sensors, actuators, and their roles in CPS, Integration challenges and considerations for input devices Interaction Techniques in CPS - Gestural interfaces: design principles and implementation in CPS, Voice interfaces: technologies, usability considerations, and applications Multi-Modal Interfaces - Integration of multiple input modalities in CPS, Case studies of effective multi-modal interface designs Integrating Input Devices into CPS Systems - Methods and strategies for integrating input devices into CPS architectures, Data acquisition, processing, and communication protocols	9
3	Designing Output Devices for CPS: Types of Output Devices - Displays: types, technologies, and applications in CPS, Feedback mechanisms: haptic feedback, auditory feedback, and their design considerations Interface Models and Information Visualization - Information visualization techniques for CPS interfaces, Designing effective visual and auditory outputs for real-time data in CPS Usability and Accessibility in CPS Interfaces - Usability principles specific to CPS environments, Ensuring accessibility for diverse user groups in CPS interface design Integrating Output Devices into CPS Systems - Strategies for integrating output devices to provide meaningful feedback in CPS, Synchronization and coordination of output devices with system responses	9
4	Evaluation, Testing, and Future Trends in CPS Interfaces Evaluation Methods for CPS Interfaces - Methods for evaluating interface effectiveness in CPS, Usability testing techniques and metrics specific to CPS environments Testing Protocols and Validation - Protocols for testing CPS interfaces in real-world scenarios, Validation techniques for ensuring interface reliability and performance Future Trends in CPS Interface Design - Emerging technologies and their impact on CPS interface design, Predictions and discussions on the future directions of CPS interfaces	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply fundamental principles and methodologies of interface design to CPS environments	K3
CO2	Analyze various types of input devices used in CPS, and understand their integration challenges	K4
CO3	Analyze various types of output devices used in CPS, and understand their integration challenges	K4
CO4	Examine CPS interfaces through usability testing, validation methods, and assessing interface effectiveness in real-world scenarios.	K4
CO5	To understand Evaluation Methods for CPS Interfaces	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	2								
CO2	3	2	3	2	1							
CO3	3	2	3	2	1							
CO4	3	2	3	2	1							
CO5	3	2	2	2								

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), : No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Interaction Design: Beyond Human-Computer Interaction	Jenny Preece, Helen Sharp, Yvonne Rogers	Wiley	6/e, 2023
2	Human Computer Interaction	Dr. Samit Bhattacharya	McGraw Hill	1/e, 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Designing Interfaces: Patterns for Effective Interaction Design	Jenifer Tidwell	O'Reilly	3/e, 2020
2	Designing for Interaction: Creating Smart Applications and Clever Devices	Dan Saffer	New Riders	2/e, 2010

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1 - 4	https://nptel.ac.in/courses/106103237

SEMESTER S5

CLOUD COMPUTING

Course Code	PECPT525	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To identify the significance of implementing virtualization techniques.
2. To interpret the various cloud computing models and services
3. To compare the various public cloud platforms and software environments.
4. To apply appropriate cloud programming methods to solve big data problems. appreciate the need of security mechanisms in cloud
5. To illustrate the use of various cloud services available online.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamental Cloud Computing: Traditional computing- Limitations. Overview of Computing Paradigms-Grid Computing, Cluster Computing, Distributed Computing, Utility Computing, Cloud Computing. NIST reference Model-Basic terminology and concepts. Cloud characteristics, benefits and challenges, Roles and Boundaries. Cloud delivery (service) models-Infrastructure-as-a-Service(IaaS),Platform-as-aService(PaaS),Software-as-a-Service (SaaS), XaaS (Anything-as-a-service)-Cloud deployment models- Public cloud, Community cloud, Private cloud, Hybrid cloud.	9
2	Introduction to virtualization -Virtualizing physical computing resources, Virtual Machines (Machine virtualization), non-virtualized v/s virtualized machine environments. Types of VMs-process VM v/s system VM, Emulation, interpretation and binary translation. Hardware-levelvirtualization-Hypervisors/VMM.Typesof Hypervisors. Full	9

	Virtualization, Para-Virtualization, Hardware-assisted virtualization, OS level virtualization. Basics of Network Virtualization, Storage Virtualization and Desktop Virtualization, Pros and cons of virtualization. Case Study- Xen: Para-virtualization, VMware: full virtualization.	
3	Cloud-Enabling Technologies, Private cloud platforms and Security: Broadband networks and internet architecture- Internet Service Providers (ISPs), Data centre technology, Web technology, Multitenant technology, Service technology. Resource provisioning techniques-static and dynamic provisioning. Open-source software platforms for private cloud-OpenStack, CloudStack, Basics of Eucalyptus, Open Nebula, Nimbus. Basic terms and concepts in security: Threat agents, Cloud security threats/risks, Trust. Operating system security-Virtual machine security-Security of virtualization- Security Risks Posed by Shared Images, Security Risks Posed by Management OS.	9
4	Popular Cloud Platforms: Amazon Web Services(AWS):- AWS ecosystem- Computing services, Amazon machine images, Elastic Compute Cloud (EC2), Advanced compute services. Storage services- Simple Storage System (Amazon S3), Elastic Block Store (Amazon EBS), Database Services, Amazon CDN Services and Communication services. Google Cloud Platform:- IaaS Offerings: Compute Engine (GCE), Cloud Storage, PaaS Offerings: Google App Engine (GAE), Storage services, Application services, Compute services, Database Services, SaaS Offerings: Gmail, Docs, Google Drive. Microsoft Azure: Azure Platform Architecture, Hyper-V, Azure Virtual Machine, Compute services, Storage services	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Students should evaluate and analyze a real-world problem, assess the proposed solutions, conclude which solution is most appropriate for the problem, and then implement the chosen solution using executable code or simulate it using relevant software or supported hardware platforms.

Criteria for evaluation:

- 1. Problem Definition (K4 - 4 points)**
 - a. Clearly defines the real-world problem.
 - b. Examine and identifies relevant contextual factors (constraints, resources, objectives).
- 2. Problem Analysis (K4 - 4 points)**
 - a. Break-down and presents a well-reasoned solution approach.
 - b. Compare and justify the proposed solutions with evidence and logical reasoning.
- 3. Evaluate (K5 - 4 points)**
 - a. Thoroughly evaluate the proposed solutions.
 - b. Compares trade-offs, advantages, and disadvantages.
 - c. Considers feasibility, scalability, and practical implications.
- 4. Implementation (K5 - 4 points)**
 - a. Select the most feasible solution by implementing the proposed solutions.
 - b. Successfully translates the chosen solution into an executable code or simulation using relevant software suit or supported hardware platform.
 - c. Demonstrates proficiency in simulation / emulation practices (readability, efficiency, error handling).
- 5. Conclusion (K4- 2 points, K5 – 2 points)**
 - a. Summarizes findings and insights. State which solution is most appropriate for the problem. **(K4)**
 - b. Reflects critical thinking and informed decision-making. **(K5)**

Scoring:

- 1. Accomplished (4 points):** Exceptional analysis, clear implementation, and depth of understanding.
- 2. Competent (3 points):** Solid performance with minor areas for improvement.
- 3. Developing (2 points):** Adequate effort but lacks depth or clarity.
- 4. Minimal (1 point):** Incomplete or significantly flawed.

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the various cloud computing models and services.	K2
CO2	Illustrate the significance of implementing virtualization techniques.	K2
CO3	Explain different cloud enabling technologies and compare private cloud platforms.	K2
CO4	Analyse the security mechanisms in cloud.	K4
CO5	Evaluate the different popular cloud computing platforms.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	1		2							
CO2	3	2	2		3							
CO3	3	3	1		3							
CO4	3	2	2		3							
CO5	3	3	1		3							

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), : No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cloud Computing Concepts, Technology & Architecture	Thomas, E., Zaigham M., Ricardo P	Prentice Hall	1/e, 2013
2	Cloud computing	Bhowmik, S	Cambridge University Press.	1/e, 2017
3	Cloud computing: Principles and paradigms	Buyya, R., Broberg, J., & Goscinski, A. M	John Wiley & Sons	1/e, 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cloud computing: theory and practice	Marinescu, D. C	Morgan Kaufmann	1/e, 2017
2	Cloud Computing: Implementation – Management – and Security	John W Rittinghouse and James F Ransome	CRC Press	1/e, 2010
3	Cloud Computing: Web-Based Applications That Change the Way You Work and Collaborate Online	Michael Miller	Pearson Education	1/e, 2009
4	Cloud Computing – A Practical Approach	Toby Velte, Anthony Velte and Robert Elsenpeter	TMH	1/e, 2009

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/105/106105167/
2	https://www.shiksha.com/online-courses/cloud-computing-by-nptel-course-nptel18
3	https://onlinecourses.nptel.ac.in/noc21_cs14/preview
4	https://www.classcentral.com/course/swayam-cloud-computing-and-distributed-systems-17544

SEMESTER S5

DSP LAB

Course Code	PCCPL507	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0: 0: 3: 0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCCPT402	Course Type	Practical

Course Objectives:

1. To create discrete time signals and test the sampling theorem.
2. To Formulate the DFT for a discrete signal and verification of its properties using Python/ Scilab/MATLAB.
3. To design and implement algorithms for solving difference equations, computing convolution and correlation and verifying their properties using numerical methods and simulations.
4. To design and test the filtering operations and compare with the theoretical values.
5. To implement the DSP computations on DSP hardware and verify the result

Details of Experiment

(Perform at least 10 Experiments from the list below, in which a minimum of 3 experiments with DSP kits are included.)

Expt. No	Experiment
1	Simulation of Signals: Simulate the following signals using Python/ Scilab/MATLAB. 1. Unit impulse signal 2. Unit pulse signal 3. Unit ramp signal 4. Bipolar pulse 5. Triangular signal
2	Verification of sampling theorem (use interpolation function) using Python/Scilab/MATLAB.
3	Verify the properties of DFT (like Linearity and Parseval 's theorem, etc.) using Python/ Scilab/MATLAB.
4	Compute the DFT of square pulse and Sinc function using Python/ Scilab/MATLAB.
5	Find the DFT / IDFT of a given DT signal using Python/ Scilab/MATLAB.
6	Find Linear convolution using DFT(Overlap-add and Overlap-Save methods) in Python/ Scilab/ MATLAB.
7	Find Auto and cross correlation of two sequences and verification of their properties using Python/ Scilab/MATLAB.

8	Compute the N point DFT of a given sequence and plot magnitude and phase spectrum (using DFT equation and verify it by built-in routine) using Python/ Scilab/MATLAB.
9	Design and implement a Lowpass and High pass FIR filter to meet the desired specifications (using different window techniques) and test the filter with an audio file. Plot the spectrum of audio signal before and after filtering using Python/ Scilab/MATLAB.
10	Design and implement a digital IIR filter (Low pass and High pass) to meet given specifications and test with an audio file. Plot the spectrum of audio signal before and after filtering using Python/ Scilab/MATLAB.
11	Obtain the Linear convolution of two sequences using DSP kit
12	Obtain the Circular convolution of two sequences using DSP kit
13	Compute the N-point DFT of a given sequence using DSP kit
14	Determine the Impulse response of first order and second order system using DSP kit
15	Generate a sine wave and standard test signals using DSP kit
16	Sample an analog signal and study the aliasing effect using DSP kit

Course Assessment Method (CIE: 50 Marks, ESE 50 Marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work, experiments, Viva and Timely completion of Lab Reports / Record. (Continuous Assessment)	Internal Exam	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

Mandatory requirements for ESE:

- Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- Endorsement by External Examiner: The external examiner shall endorse the record.

Course Outcomes (COs)

At the end of the course the student will be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the concepts of analog to digital conversion of signals and frequency domain sampling of signals.	K2
CO2	Model the discrete time signals and systems and verify its properties and results.	K3
CO3	Solve discrete computations using DSP processor and verify the results.	K6
CO4	Design the digital filters using a simulation tool and analyze the re-sponse of the filter for an audio signal.	K6
CO5	Develop programs using Matlab/Scilab/Octave to illustrate DSP concepts.	K6

K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	2				2			
CO2	3	3	3	3	2				2			
CO3	3	2	3	3	3				2			
CO4	2	2	2	3	3				2			
CO5	2	1	3	2	3				2			

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), : No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Signal Processing using MATLAB	Vmay K Ingle, John G Proakis	Cengage India Private Limited	4/e, 2017
2	Think DSP: Digital Signal Processing using Python	Allen B. Downey	O'Reilly Media	1/e, 2019
3	DSP Applications Using C and the TMS320C6x DSK (Topics in Digital Signal Processing)	Rulph Chassaing	Wiley	1/e, 2000

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Signal processing	S. K. Mitra	Tata Mc-Graw Hill	2/e, 2004
2	Digital Signal Processing	Alan V Oppenheim	Pearson	2/e, 1999

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Digital Signal Processing - Course (nptel.ac.in)
2	NPTEL:: Electrical Engineering - NOC:Digital Signal Processing and its Applications

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S5
CYBER PHYSICAL SYSTEMS LAB

Course Code	PCCPL508	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	–	Course Type	Practical

Course Objectives:

1. To identify Embedded components such as sensors and actuators
2. To design and test the calibration of sensors or controllers using microcontrollers like Arduino/Raspberry Pi or any compatible embedded platforms.
3. To model and develop simulations for CPS with real time data using MATLAB/SIMULINK or Gazebo tools.

Details of Experiment

(Perform at least 10 Experiments from the list below)

Expt. No	Experiment
1	Study about embedded components such as sensors and actuators.
2	Signal Integration, Differentiation, Logarithmic computation activities to understand the computational logic and the physical end result
3	Signal Addition, Subtraction and Multiplication computations activities to understand the computational logic and the physical end result
4	Implementation and calibration of humidity sensor with microcontrollers
5	Implementation and calibration of pressure sensor with microcontrollers
6	Implementation and calibration of temperature sensor with microcontrollers
7	Implementation and validation of RPM controller with microcontrollers
8	Implementation and validation of Illumination controller with microcontrollers
9	Implementation and validation of water level controller with microcontrollers
10	Implementation and validation of temperature controller with microcontrollers

11	Implementation and validation of displacement controller with microcontrollers
12	Create a program that displays data from the sensor in regular intervals in a compact format.
13	From the available sensors and actuators and find or create code that will display the sensed data on the PC.

Course Assessment Method
(CIE: 50 Marks, ESE 50 Marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work, experiments, Viva and Timely completion of Lab Reports / Record. (Continuous Assessment)	Internal Exam	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

Mandatory requirements for ESE:

- Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- Endorsement by External Examiner: The external examiner shall endorse the record.

Course Outcomes (COs)

At the end of the course the student will be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To identify Embedded components such as sensors and actuators	K3
CO2	To develop sensors and controllers using Arduino/Raspberry Pi or any compatible embedded platforms.	K6
CO3	Model the CPS and compose simulations with real world data using simulation tools like MATLAB/Simulink or Gazebo.	K6

K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2			3				2			
CO2	3	2	3	2	3				2			
CO3	3	2	3	2	3				2			

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), : No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Handbook of Modern Sensors: Physics, Designs, and Applications	Jacob Fraden	Springer	5/e, 2016
2	Embedded Systems: Real-Time Interfacing to Arm Cortex-M Microcontrollers	Jonathan W. Valvano	CreateSpace Independent Publishing Platform	2/e, 2012
3	Designing Embedded Systems with PIC Microcontrollers: Principles and Applications	Tim Wilmshurst	Newnes	1/e, 2006
4	MATLAB and Simulink for Engineers	Agam Kumar Tyagi	Oxford University Press	1/e, 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Sensors and Actuators: Engineering System Instrumentation	Clarence W. de Silva	CRC Press	1/e, 2007
2	The Art of Electronics	Paul Horowitz, Winfield Hill	Cambridge University Press	3/e, 2015
3	Introduction to Embedded Systems: Using Microcontrollers and the MSP430	Manuel Jiménez, Rogelio Palomera, Isidoro Couvertier	Springer	1/e, 2013

Video Links (NPTEL, SWAYAM...)	
Sl. No.	Link ID
1	https://youtu.be/cx6z0mkVF9E?si=BYC3Yqek0Q9_doSZ
2	https://youtu.be/CFL3E0or2HU?si=AFaoMBli_QEukHKx
3	https://youtu.be/IuEOMyGuuIg?si=bI8uYkJedMBVvKfk

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 6

CYBER PHYSICAL SYSTEMS

SEMESTER S6

SPECIAL PURPOSE MACHINES FOR AUTOMATION

Course Code	PCCPT601	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the basic principles of special purpose machines (SPMs) in automation.
2. To explore the design, operation, and control of various SPMs.
3. To gain knowledge about the integration of SPMs into automated systems.
4. To study the advancements and emerging trends in SPM technology and to solve industrial problems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Special Purpose Machines: Overview of Special Purpose Machines-Definition, scope, and role in automation. Types of Special Purpose Machines-Classification based on applications (e.g., assembly machines, inspection machines, testing machines). Components and Subsystems- Mechanical, electrical, and control components. Comparison with General-Purpose Machines-Advantages and limitations.	11
2	Design Principles for Special Purpose Machines: Mechanical Design Considerations-Structural design, kinematics, and dynamics. Control Systems-PLCs, sensors, actuators, and feedback systems. Integration of Components- Combining mechanical and electrical systems for automation. Design Methodologies-Approaches to designing efficient and reliable SPMs.	11

3	Automation Technologies and Implementation: Automation Technologies-Overview of technologies used in SPMs (e.g., robotics, vision systems, AI). Programming and Control- Writing and implementing control algorithms, interfacing with PLCs. Safety and Standards - Safety considerations and industry standards for automation systems. Troubleshooting Techniques- Common issues and solutions	11
4	Theoretical Aspects of Troubleshooting and Future Trends: Theoretical Troubleshooting Techniques-Concepts and strategies for diagnosing and resolving issues. Simulation and Testing Theories-Theoretical understanding of simulation tools and methodologies. Future Trends and Developments - Emerging technologies and future directions in SPMs and automation.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand and Explain the concept and significance of special purpose machines in automation and differentiate them from general-purpose machines.	K2
CO2	Design basic special purpose machines using mechanical design principles and automation technologies.	K4
CO3	Apply control systems and automation technologies in SPMs.	K3
CO4	Analyze and understand troubleshooting techniques and future trends in SPMs	K4
CO5	Compare the performance and efficiency of SPMs.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO11	PO 12
CO1	3	2	1	1								
CO2	3	3	3	2	1							
CO3	2	3	3	2	2							
CO4	2	3	2	3	2							1
CO5	3	3	3	2	3							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Automation, Production Systems, and Computer-Integrated Manufacturing	Mikell P. Groover	Pearson	4/e, 2015
2	Programmable Logic Controllers: Principles and Applications	John W. Webb and Ronald A. Reis	Prentice Hall	6/e, 2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Special Purpose Production Machines	B.W Charman	Harper Collins Distribution Services	1/e, 1977
2	Introduction to Robotics: Mechanics and Control	John J. Craig	Pearson	3/e, 2004

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/112102011
2	https://onlinecourses.nptel.ac.in/noc23_ee140/preview
3	https://archive.nptel.ac.in/courses/112/103/112103293/
4	https://archive.nptel.ac.in/courses/112/102/112102011/

SEMESTER S6
COMMUNICATION FOR CPS

Course Code	PCCPT602	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCCPT502	Course Type	Theory

Course Objectives:

1. To design and develop communication systems for CPS applications.
2. To evaluate and optimize CPS communication systems.
3. To implement secure and efficient communication protocols.
4. To apply CPS communication technologies to various domains and applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to communication for CPS: Overview of CPS and communication requirements, Types of communication in CPS (eg: sensor-sensor, sensor-actuator, actuator-actuator), Communication protocols and standards for CPS. Network topologies and architecture for CPS (TCP/IP, UDP, HTTP), Network performance matrices (eg: Throughput, latency, jitter)	9
2	Wireless communication for CPS: Wireless communication technologies (Wi-Fi, Bluetooth, Zigbee) Wireless network protocols and standards (802.11,802.15.4), Wireless Communication challenges (interference, fading), Wired communication for CPS: Wired communication technologies (Ethernet, USB, CAN), Wired network topologies and standards (TCP/IP, UDP, Modbus)	9

3	Data compression and Encryption for CPS: Data compression techniques (Huffman coding, LZ77), Encryption techniques (AES, ESA), Data compression and Encryption algorithms for CPS) Communication protocols for CPS applications: IoT communication protocols (MQTT, CoAP, LWM2M), Industrial automatic communication protocols (Modbus, Profibus, Profinet), Healthcare communication protocols (HL7, DICOM)	9
4	Performance Evaluation and Optimization: Performance matrices for CPS communication (throughput, latency, reliability) performance Evaluation tools (simulation, testing), Optimization techniques for CPS communication (buffer sizing, packet scheduling), Security and Privacy in CPS communication: Threats and vulnerabilities in CPS communication, Security measures for CPS communication (encryption and authentication), privacy concerns and solutions in CPS communication.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Compare different communication techniques to CPS communication.	K2
CO2	Identify the role of communication in ensuring reliability, safety and efficiency in CPS.	K3
CO3	Understand the data compression and expansion techniques to CPS communication.	K2
CO4	Analyze and mitigate security threats and vulnerabilities in CPS communication.	K4
CO5	Examine the performance of CPS communication systems.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2								
CO2	3	3	2	2								
CO3	3	3	2	2	2							
CO4	3	3	3	3	2							
CO5	3	2	2	3	2							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Communication protocols for cyber physical systems	Jiafu Wan, Lie shu, Xiaohua	Springer	2019
2	Communication for Cyber Physical Systems	Wilbert O. Galitz	Wiley-Blackwell	2017
3	Cyber physical systems- Design, Implementation and Applications.	Houbing song, Danda B Rawat and Ravi B Shashtha	CRC press	2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Communication for IoT and CPS	Quasay F.Hassan and Atta badii	Springer	2020
2	CPS communication Architectures and Protocols	Jose.L Martinez, Fernando Auat	IGI global	2020
3	Cyber physical Systems -modeling simulation and Analysis	Arunabha sen and Sudip misra	Wiley	2020
4	Communications for Control in cyber physical systems	Husheng Li	MK	2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/8wBQMbOntNc?si=F4NAVT9XBHekNCcK
2	https://youtu.be/W2EjVB6jLzU?si=4T4XCfmIgJPdOzrI
3	https://youtu.be/XCjjvhpw6GQ?si=ysz1Tug45MYkCI71
4	https://youtu.be/p7kYStiASLo?si=MjJMrljqgqif78ez

SEMESTER S6
ADVANCED MICROCONTROLLERS

Course Code	PECPT631	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCCPT302	Course Type	Theory

Course Objectives:

1. To understand RISC and ARM Design Philosophy
2. To gain in-depth knowledge of the ARM core data flow model.
3. To explore THUMB and ARM Programming
4. To provide a detailed understanding of exception handling in ARM processors

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	ARM Embedded Systems and ARM Processor Fundamentals: The RISC design philosophy, ARM design philosophy, embedded system hardware- AMBA bus protocol, embedded system software- applications. ARM core data flow model, Registers, CPSR-Processor modes, Banked registers. Pipeline- Characteristics.	9
2	ARM Instruction Set: Fundamentals of ARM instructions, Barrel shifter, Classification and explanation of instructions with examples-Data processing, Branch, Load-store, SWI and Program Status Register instruction.	9
3	Introduction to THUMB and ARM Programming: Introduction to THUMB, Differences between ARM and THUMB, Register usage in Thumb, ARM Thumb Interworking. General Structure of ARM assembly module, Assembler directives- AREA, ENTRY, END, SPACE, DCD, DCB, DCW, DCI, DCQ, EQU, EXPORT, ALIGN, CODE16, CODE32, DATA. Simple ALP programs on Arithmetic & logical operations, Factorial, string operation, sorting, searching, and Scan	9

4	Exception and Interrupt handling schemes: Exception handling- ARM processor exceptions and modes, vector table, exception priorities, link register offsets. Interrupts- assigning interrupts, interrupt latency, IRQ and FIQ exceptions with example- code for enabling and disabling IRQ and FIQ exceptions, Comparison between exception and interrupts. Interrupt handling schemes- nested interrupt handler, non-nested interrupt handler. Basic interrupt stack design.	9
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Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the RISC design philosophy and ARM processor architecture.	K2
CO2	Understand and explain the ARM core data flow model, including registers, CPSR, processor modes, and pipeline characteristics.	K2
CO3	Develop and debug assembly language programs using ARM and THUMB instructions.	K3
CO4	Explain the concepts of exception and interrupt handling in ARM processors and implement related schemes in assembly language.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2					2			
CO2	3	3	2	2					2			
CO3	3	3	2	2	2				2			
CO4	3	3	2	2	2				2			

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	ARM System Developer's guide	Andrew N. SLOSS	ELSEVIER Publications, ISBN 978-81-8147-646-3	2016
2	ARM Assembly Language	William Hohl,	CRC Press, ISBN:978-81-89643-04-1	2e, 2014
3	ARM System-on-chip Architecture	Steve Furber	Pearson Education ISBN 978-81-317-0840-8	2e, 2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	http://www.ocfreaks.com/lpc2148-gpio-programming-tutorial/			
2	http://www.ocfreaks.com/lpc214x-pll-tutorial-for-cpu-and-peripheral-clock/			
3	http://www.ocfreaks.com/lpc2148-pwm-programming-tutorial/			

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/108105102
2	https://nptel.ac.in/courses/108105102

SEMESTER S6

NETWORKED CONTROL SYSTEMS

Course Code	PECPT632	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3: 0: 0: 0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	–	Course Type	Theory

Course Objectives:

1. To understand the principles of networked control systems and their advantages over traditional control systems.
2. To study the impact of network-induced delays and packet losses on control system performance.
3. To analyze different control strategies to mitigate the adverse effects of network issues.
4. To design and implement networked control systems for real-world applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Networked Control Systems: Fundamentals of networked control systems, Advantages and challenges, Applications of networked control systems, Network protocols and communication architectures	9
2	Network-induced Issues: Network delays and their modelling, Packet losses and their impact, Sampling and data quantization.	9
3	Control Strategies for Networked Systems: Predictive control. Robust control techniques, Event-triggered and self-triggered control, Compensation strategies for delays and losses.	9
4	Design and Implementation: System design methodologies, Simulation tools and environments, Real-time implementation, Performance analysis and optimization.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamental concepts of networked control systems	K2
CO2	Analyze the effects of network delays and packet losses on control performance	K4
CO3	Apply various control strategies to address network-induced issues	K3
CO4	Model networked control systems for specific applications	K3
CO5	Examine the performance of networked control systems using different metrics	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		2	2								
CO2	3	3	2	2								
CO3	3	3	2	2	2							
CO4	3	3	2	3	2							
CO5	3	3	3	2	2							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Optimal Control: Linear Quadratic Methods	Anderson, B. D. O., & Moore, J. B.	Dover Publications	1/e, 1990
2	Automatic Control Systems	Kuo, B. C.	Wiley	8/e, 2002

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Feedback Control of Dynamic System	Franklin, G. F., Powell, J. D., & Emami-Naeini, A.	Pearson	6e, 2009
2	Modern Control Systems	Dorf, R. C., & Bishop, R. H	Pearson	13e, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/107/106/107106081/

SEMESTER S6

SECURE COMMUNICATION PROTOCOLS

Course Code	PECPT633	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	–	Course Type	Theory

Course Objectives:

1. Introduction to overview of network security issues, cryptographic algorithms, authentication techniques
2. To understand various Protocols and Transport Layer Security (TLS).
3. To study various attacks and threats in systems and security in new generation systems
4. To understand various Cyber risk assessment methods

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction (overview of network security issues, cryptographic algorithms, authentication techniques) - Layer 2/3 security - Authentication systems, Key establishment protocols, Kerberos- Secure communication at the data link and network layers (IPSEC and IKE) - Secure communication at the transport and application layers (SSL/TLS, email security, PGP) - Vulnerabilities of Internet protocols	9
2	Secure Sockets Layer (SSL) - SSL Protocol Stack - SSL Record Protocol Operation IP Security (IPsec) - IP Security (IPsec) architecture - Benefits of IPsec Hypertext Transfer Protocol Secure (HTTPS) Kerberos - Drawbacks and limitations - Kerberos Vulnerabilities Transport Layer Security (TLS).	9

3	Denial of service (DoS) attacks and defences - Firewalls, IP spoofing prevention - Routing protocols security and router security - Domain name server (DNS) security - Traffic monitoring, Intrusion detection, Honeypots - Wireless networks security - Spam, Phishing, and Pharming - Malware propagation and containment, Botnets - Anonymity and privacy on the Web. Internet Security Protocols and Standards - TCP Attacks, DNS Vulnerabilities, SSL/TLS, DDoS - Next Generation System Designs and Challenges Cyber-Physical System Overview and Security - Internet of Things and Smart Grid Security - Data & Infrastructure Security in Cloud/Edge Computing.	9
4	Blockchain and Decentralized Applications- Hash cash and other Consensus Protocols Blockchain Security- Smart Contracts - Scalability and Privacy challenges, Security Economics and Risk Modelling - Cyber-Risk Assessment - Threat Information Sharing - Cyber-insurance	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyse the security issues and authentication algorithms.	K4
CO2	Express the various protocols in different layers.	K2
CO3	Explain the use of different attacks and security mechanism in current networking scenario.	K2
CO4	Analyse the security issues in IoT and smart grid applications.	K4
CO5	Examine the security methods using blockchain. An Ability to analyse the cyber risk assessment.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	3	2	2							
CO2	3	2	2	2	2							1
CO3	3	2	2	2	2							1
CO4	3	2	2	2	2							1
CO5	2	3	2	2	2							1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Networks: Protocols, Standards and Interface	Uyless D. Black	Prentice Hall	2e, 2015
2	Computer Networking - A Top-Down Approach	James F Kurose and Keith W. Ross,	Addison-Wesley	5e, 2010.
3	Computer Networks: A Systems Approach	L. Peterson and B. Davie	Elsevier Inc	5e, 2011.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Network Security Essentials: Applications and Standards	William Stallins	Pearson Publication	2016
2	Security and Privacy Management, Techniques, and Protocols	Yassine Maleh	Morocco	2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/pV11L1jrbFE?si=T5xaQ5Tv_y5IgVM9
2	http://acl.digimat.in/nptel/courses/video/106106129/L42.html
3	https://youtu.be/8wBQMbOntNc?si=j7prZCIG03Qfn_nX
4	https://youtu.be/mzPoUjQC4WU?si=EIDGJC5GFz8roZ7g

SEMESTER S6

AUGMENTED AND VIRTUAL REALITY

Course Code	PECPT634	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3: 0: 0: 0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	–	Course Type	Theory

Course Objectives:

1. To learn the fundamental Computer Vision, Computer Graphics and Human-Computer interaction Techniques related to VR/AR
2. To review the Geometric Modeling Techniques
3. To discuss and Examine VR/AR Technologies
4. To use of various types of Hardware and Software in Virtual Reality systems and apply Virtual/Augmented Reality to varieties of Applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Virtual Reality (VR): Virtual Reality and Virtual Environment, Computer graphics, Real time computer graphics, Flight Simulation, Virtual environment requirement, benefits of virtual reality, Historical development of VR, Scientific Landmark	9
2	Computer Graphics and Geometric Modelling: The Virtual world space, positioning the virtual observer, the perspective projection, human vision, stereo perspective projection, Color theory, Conversion From 2D to 3D, 3D space curves, 3D boundary representation, Simple 3D modelling, 3D clipping, Illumination models, Reflection models, Shading algorithms, Geometrical Transformations: Introduction, Frames of reference,	9

3	Augmented Reality (AR): Taxonomy, Technology and Features of Augmented Reality, AR Vs VR, Challenges with AR, AR systems and functionality, Augmented Reality Methods, Visualization Techniques for Augmented Reality, Enhancing interactivity in AR Environments, Evaluating AR systems	9
4	Development Tools and Frameworks: Human factors - Introduction, the eye, the ear, the somatic senses. Hardware - Introduction, sensor hardware, Head-coupled displays, Acoustic hardware, Integrated VR systems. Software - Introduction, Modelling virtual world, Physical simulation, VR toolkits, Introduction to VRML AR / VR Applications: Introduction, Engineering, Entertainment, Science, Training, Game Development	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand fundamental Computer Vision, Computer Graphics and Human Computer Interaction Techniques related to VR/AR	K2
CO2	Understand Geometric Modeling Techniques	K2
CO3	Analyze and evaluate VR/AR Technologies	K4
CO4	Develop Virtual/Augmented Reality Applications	K5
CO5	Apply various types of Hardware and Software in VR systems	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									
CO2	3	2	2									
CO3	3	2	2	2	2							
CO4	3	2	2	2	2							
CO5	3	2	2									

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Virtual Reality Technology	Burdea, G. C. and P. Coffet	Wiley-IEEE Press	2/e, 2006
2	Augmented Reality: Principles & Practice,	Schmalstieg, D., Höllerer, T	Pearson	2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The VR Book: Human-Centered Design for Virtual Reality	Jason Jerald, Morgan &	Claypool Publishers	1/e, 2015
2	Virtual Reality Technology	Grigore C. Burdea and Philippe Coiffet	Wiley-Interscience	2/e, 2003

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.coursera.org/learn/vr-and-ar
2	https://www.youtube.com/playlist?list=PLyqSpQzTE6M_IpDCkR2Ry29taJVXak0e8

SEMESTER S6

MACHINE LEARNING FOR CPS

Course Code	PECPT635	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBCPT 404	Course Type	Theory

Course Objectives:

1. To understand the fundamental concepts of Machine Learning (ML) and its relevance to Cyber-Physical Systems (CPS).
2. To gain proficiency in implementing supervised, unsupervised, and reinforcement learning algorithms.
3. To explore the real-time applications of ML in optimizing and controlling CPS.
4. To learn advanced ML techniques and understand the ethical, security, and privacy considerations in CPS.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Machine Learning Basics of Machine Learning: Definition and significance, Types of Machine Learning: Supervised, Unsupervised, Reinforcement Learning, Common algorithms and their applications, The role of Machine Learning in CPS	9
2	Supervised and Unsupervised Learning for CPS Supervised Learning: Classification algorithms - Decision Trees, Support Vector Machines, Regression algorithms- Linear Regression, Polynomial Regression, Evaluation metrics- accuracy, precision, recall Unsupervised Learning: Clustering algorithms (K-means, Hierarchical Clustering), Dimensionality reduction techniques (PCA, t-SNE)	9

3	Reinforcement Learning and Real-Time Applications in CPS Reinforcement Learning (RL): Fundamentals of RL, Key concepts: states, actions, rewards, policy, Q-learning and Deep Q-networks Real-Time Applications: ML in real-time decision making, Optimization of control systems	9
4	Advanced Topics and Future Trends in ML for CPS Advanced ML Techniques: Deep Learning and Neural Networks, Transfer Learning, Explainable AI Future Trends and Challenges: Edge Computing and IoT integration, Security and privacy concerns in CPS, Ethical considerations in ML applications	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Students should evaluate and analyze a real-world problem, assess the proposed solutions, conclude which solution is most appropriate for the problem, and then implement the chosen solution using executable code or simulate it using relevant software or supported hardware platforms.

Criteria for evaluation:

1. Problem Definition (K4 - 4 points)

- a. *Clearly defines the real-world problem.*
- b. *Examine and identifies relevant contextual factors (constraints, resources, objectives).*

2. Problem Analysis (K4 - 4 points)

- a. *Break-down and presents a well-reasoned solution approach.*
- b. *Compare and justify the proposed solutions with evidence and logical reasoning.*

3. Evaluate (K5 - 4 points)

- a. *Thoroughly evaluate the proposed solutions.*
- b. *Compares trade-offs, advantages, and disadvantages.*
- c. *Considers feasibility, scalability, and practical implications.*

4. Implementation (K5 - 4 points)

- a. Select the most feasible solution by implementing the proposed solutions.
- b. Successfully translates the chosen solution into an executable code or simulation using relevant software suit or supported hardware platform.
- c. Demonstrates proficiency in simulation / emulation practices (readability, efficiency, error handling).

5. Conclusion (K4- 2 points, K5 – 2 points)

- a. Summarizes findings and insights. State which solution is most appropriate for the problem. **(K4)**
- b. Reflects critical thinking and informed decision-making. **(K5)**

Scoring:

1. **Accomplished (4 points):** Exceptional analysis, clear implementation, and depth of understanding.
2. **Competent (3 points):** Solid performance with minor areas for improvement.
3. **Developing (2 points):** Adequate effort but lacks depth or clarity.
4. **Minimal (1 point):** Incomplete or significantly flawed.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. • Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the core concepts and applications of machine learning in the context of CPS.	K2
CO2	Implement and evaluate supervised and unsupervised learning algorithms	K3
CO3	Evaluate reinforcement learning solutions for real-time control and optimization	K5
CO4	Analyze advanced ML techniques and their implications on the security, privacy, and ethics of CPS.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2						2			
CO2	3	2	3	2	2				2		2	
CO3	3	2	3	2	2				2		2	
CO4	3	2		2		2	2	2	2	2		

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Machine Learning: A Probabilistic Perspective	Kevin P. Murphy	MIT Press	1/e, 2012
2	Pattern Recognition and Machine Learning	Christopher M. Bishop	Springer	1/e, 2006
3	Reinforcement Learning: An Introduction	Richard S. Sutton and Andrew G. Barto	MIT Press	2/e, 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cyber-Physical Systems: Foundations, Principles and Applications	Houbing Song, Danda B. Rawat, Sabina Jeschke, Christian Brecher	Morgan Kaufmann	1/e, 2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.coursera.org/learn/machine-learning
2	https://www.coursera.org/specializations/deep-learning
3	https://www.youtube.com/watch?v=2pWv7GOvuf0&list=PLqYmG7hTraZBiG_XpjnPrSNw-1XQaM_gB

SEMESTER S6
ADVANCED CYBER PHYSICAL SYSTEMS

Course Code	PBCPT604	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCCPT303	Course Type	PC-PBL

Course Objectives:

1. To understand and apply advanced control strategies for CPS.
2. To analyze and implement security measures specific to CPS environments.
3. To evaluate real-time systems within the context of CPS.
4. To investigate and propose solutions for complex CPS challenges in various domains.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Advanced Control Strategies: Nonlinear Control Systems - Introduction to nonlinear dynamics and control, Lyapunov methods and stability analysis. Optimal and Robust Control - Overview of optimal control theory, Robust control techniques and their applications. Adaptive and Learning-Based Control - Adaptive control mechanisms, Reinforcement learning in CPS.	9
2	CPS Security and Privacy Security Challenges in CPS - Common vulnerabilities and attack vectors, Security protocols and encryption techniques. Privacy in CPS - Privacy concerns and regulations, Privacy-preserving techniques in data collection and processing. Intrusion Detection and Response - Techniques for detecting and mitigating attacks, Incident response strategies.	9

3	Real-Time Systems and Scheduling Fundamentals of Real-Time Systems - Real-time system requirements and classifications, Scheduling algorithms: Rate Monotonic Scheduling (RMS), Earliest Deadline First (EDF). Real-Time Communication Protocols - Communication protocols in real-time CPS, Network-on-Chip (NoC) and Time-Triggered Protocols. Performance Analysis and Optimization - Techniques for performance analysis of real-time systems, Optimization strategies for resource management	9
4	Interdisciplinary Applications of CPS CPS in Healthcare - Applications and challenges in healthcare CPS CPS in Smart Cities - Smart infrastructure and urban planning. Emerging Trends and Research Overview of cutting-edge research in CPS, IoT integration, AI and CPS, edge computing.	9

Project based learning (8 Hrs)

- The objective of **Project based learning** to strengthen the understanding of student's fundamentals through effective application of theoretical concepts. Projects can help to boost student's skills and widen the horizon of their thinking.
- The ultimate aim of an engineering student is to resolve a problem by applying theoretical knowledge.
- In the beginning, students shall be given an awareness and/or guideline on how to implement and learn this course through a project and form groups in the first few weeks.
- Each group should identify a project that can be done to understand a part of the contents of this course and perform a literature survey.
- Choose a suitable platform for your application and the input output devices to be interfaced.
- Develop the block diagram and present before the jury within first month after the commencement of classes.
- Implement necessary hardware and software to develop the prototype.
- Prepare a report and present before the jury for final evaluation.

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	- Each question carries 6 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply advanced control strategies for CPS.	K3
CO2	Identify and analyze common security vulnerabilities in CPS	K4
CO3	Analyze the performance of real-time systems	K4
CO4	Create CPS solutions for specific interdisciplinary applications, such as healthcare and smart cities, demonstrating the ability to integrate advanced CPS concepts with practical requirements.	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	3	2				2	1	2	
CO2	3	3	2	3	3				2	1	2	
CO3	3	3	2	3	3				2	1	2	
CO4	3	3	3	3	3				2	1	2	

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Modern Control Engineering	K. Ogata	Pearson Education India	5/e
2	Network Security Essentials: Applications and Standards	William Stallings	Pearson	6/e
3	Real-Time Systems: Design Principles for Distributed Embedded Applications	Philip Koopman	Springer	3/e
4	Cyber-Physical Systems: From Theory to Practice	Danda B. Rawat, Joel J.P.C. Rodrigues, Ivan Stojmenovic	CRC Press	1/e

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Real-Time Systems: Scheduling, Analysis, and Verification	Albert M.K. Cheng	Wiley	2010
2	Cybersecurity and Privacy in Cyber Physical Systems	Yassine Maleh, Mohammad Shojafar, Ashraf Darwish, Abdelkrim Haqiq	CRC Press	1/e
3	Real-Time Systems: Design Principles for Distributed Embedded Applications	Hermann Kopetz, Wilfried Steiner	Springer	3/e

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/108103007
2	https://nptel.ac.in/courses/106104467
3	https://nptel.ac.in/courses/106105036
4	https://nptel.ac.in/courses/106105241

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S6

**INTRODUCTION TO PROGRAMMING FOR
CYBER PHYSICAL SYSTEMS**

Course Code	OECPT 611	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	UCEST105, GBEST204	Course Type	Theory

Course Objectives:

1. To introduce the students to programming techniques for robotics and cyber-physical systems using Python, MATLAB

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Python Programming Variables, data types, and basic operations, Control Structures and Functions in Python Conditionals, loops, functions, modules, and packages	9
2	Introduction to MATLAB Environment, MATLAB syntax, variables, and arrays, MATLAB Scripts and Functions, Writing scripts, creating functions, and plotting	9
3	Robot Kinematics and Dynamics, Understanding robot movement and dynamics, Sensor Integration in Robotics, Reading sensor data, interfacing with hardware	9
4	Simulation Techniques for Robotics, Building computational models, simulating robot behaviors	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Implement algorithms for CPS control using Python and MATLAB.	K3
CO2	Integrate sensors and actuators in CPS.	K3
CO3	Analyze and visualize data from CPS sensors using Python and MATLAB.	K3
CO4	Simulate robotic systems and cyber-physical interactions.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1		3							
CO2	3	2	1		3							
CO3	3	2	1		3							
CO4	3	2	3		3							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Python Crash Course	Eric Matthes	No Starch Press	3/e, 2023
2	MATLAB for Engineers	Holly Moore	Pearson Education	6/e, 2022

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Learning Robotics Using Python	Lentin Joseph	Packt Pub Ltd	2015
2	Numerical Computing with MATLAB	Cleve B. Moler	Society for Industrial and Applied Mathematics	2004

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/106106145
2	https://nptel.ac.in/courses/111102137
3	https://in.mathworks.com/discovery/robot-programming.html
4	https://in.mathworks.com/discovery/robot-programming.html

SEMESTER S6

IOT AND APPLICATIONS

Course Code	OECPT612	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3: 0: 0: 0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	–	Course Type	Theory

Course Objectives:

1. To understand the fundamentals of IoT, its vision, and strategic research directions.
2. To explore various connectivity technologies and their application in sensor networks and M2M communications
3. To gain practical experience with Arduino and Raspberry Pi, integrating sensors and actuators.
4. To learn about interoperability in IoT systems and software-defined networking.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to IoT: The Internet of Things Today, Time for Convergence, Towards the IoT Universe, Internet of Things Vision, IoT Strategic Research and Innovation Directions, IoT Applications, Future Internet Technologies, Review of sensing and actuation, Basics of IoT Networking.	9
2	Connectivity Technologies, Sensor Networks, UAV networks. Machine-to-Machine Communications: M2M to IoT – A Basic Perspective– Introduction, Some Definitions, M2M Value Chains, IoT Value chains, LoRa Technology in M2M Communications:- Introduction to LoRa and LoRaWAN, LoRa Protocols and Standards, Applications of LoRa in M2M and IoT	9

3	Interoperability in IoT, Introduction to Arduino Programming, Integration of Sensors and Actuators with Arduino. Introduction to Raspberry Pi, Implementation of IoT with Raspberry Pi, Software Defined Networking, Software Defined IoT Networking	9
4	Smart and Connected Cities-An IoT Strategy for Smarter Cities-Vertical IoT Needs for Smarter Cities, Global vs. Siloed Strategies-Smart City IoT Architecture-Street Layer, City Layer, Data Center Layer, Services Layer-Smart City Security Architecture - Smart City Use-Case Examples – Street lighting, smart parking, smart traffic and air pollution monitoring	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the current state and future prospects of IoT technologies.	K2
CO2	Explain various connectivity technologies and sensor network frameworks.	K2
CO3	Develop basic IoT applications using Arduino and Raspberry Pi platforms.	K3
CO4	Apply IoT concepts to real-world scenarios	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										
CO2	3	2	3									
CO3	3	2	3	2	2							
CO4	3	2		2	2							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Internet of Things: A Hands-On Approach" by is the 1st Edition	Arshdeep Bahga and Vijay Madisetti	VPT (Universal Publishing)	1e, 2014
2	The Internet of Things: Enabling Technologies, Platforms, and Use Cases	Pethuru Raj and Anupama C. Raman	CRC Press	1e, 2017
3	Internet of Things (A Hands-on-Approach)	Raj Kamal	McGraw Hill Education	2e, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Internet of Things: How Smart TVs, Smart Cars, Smart Homes, and Smart Cities Are Changing the World.	Michael Miller	Pearson Education	2015
2	Internet of Things	RMD Sundaram, Shriram K Vasudevan, and Abhishek S Nagarajan	John Wiley & Sons.	2019
3	https://lora-developers.semtech.com/uploads/documents/files/LoRa_and_LoRaWAN-A_Tech_Overview-Downloadable.pdf			

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/WUYAjsxnwjU4 , https://youtu.be/nE1C4ghfvac
2	https://youtu.be/WUYAjsxnwjU4
3	https://youtu.be/WUYAjsxnwjU4
4	https://youtu.be/WUYAjsxnwjU4

SEMESTER S6

NETWORKING AND COMMUNICATION PROTOCOLS

Course Code	OECPT613	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	–	Course Type	Theory

Course Objectives:

1. To introduce the fundamental various types of computer networks.
2. To demonstrate the TCP/IP and OSI models with merits and demerits.
3. To explore the various layers of OSI Model.
4. To introduce UDP and TCP Models.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Network hardware, Network software, OSI, TCP/IP Reference models, Example Networks: ARPANET, Internet. Physical Layer: Guided Transmission media: twisted pairs, coaxial cable, fiber optics, Wireless transmission. Data link layer: Design issues, framing, Error detection and correction.	9
2	Elementary data link protocols: Simplex protocol, A simplex stop and wait protocol for an error-free channel, A simplex stop and wait protocol for noisy channel. Sliding Window protocols: A one-bit sliding window protocol, A protocol using Go-Back-N, A protocol using Selective Repeat, Example data link protocols. Medium Access sub layer: The channel allocation problem, Multiple access protocols: ALOHA, Carrier sense multiple access protocols, collision free protocols. Wireless LANs, Data link layer switching.	9

3	Network Layer: Design issues, Routing algorithms: shortest path routing, Flooding, Hierarchical routing, Broadcast, Multicast, distance vector routing, Congestion Control Algorithms, Quality of Service Internetworking, The Network layer in the internet.	9
4	Transport Layer: Transport Services, Elements of Transport protocols, Connection management, TCP and UDP protocols. Application Layer –Domain name system, SNMP, Electronic Mail; the World WEB, HTTP, Streaming audio and video.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe communication protocols and layered network architectures	K1
CO2	Explain convention computer system interfacing standards and peer to peer data link	K2
CO3	Design basis network systems.	K2
CO4	Analyze data communication technology.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									
CO2	2	3	3		1					1		2
CO3	3	1	2		1					1		2
CO4	1	2	2		1					1		2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data Communications and Networking with TCP-IP protocol suite	Behrouz A. Forouzan.	MH	6e, 2022
2	Computer Networks	Andrew S Tanenbaum, David. j. Wetherall	PHI	5e, 2010
3	Data & Computer Communications	William Stallings	Pearson Education	9e, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	An Engineering Approach to Computer Networks-	S. Keshav	Pearson Education	2/e
2	Data Communications and Networking	Behrouz A. Forouzan.	TMH	3/e

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/lnU-Zw3NEEQ?si=aeVvjQLfrZNbIG1c
2	https://youtu.be/29Qdz0FmvmQ?si=CTT0LZ5HpcmLBkml
3	https://youtu.be/b6f9vh3cd6w?si=2TFAJFIE3tBcVwUf
4	https://youtu.be/8-3CSAkscYU?si=TVLkSN-0HXLjfer2

SEMESTER S6

INTRODUCTION TO HUMAN MACHINE INTERACTION

Course Code	OECP614	CIE Marks	40
Teaching Hours/Week (L:T:P:R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To identify User Interface (UI) design principles.
2. To analysis of effective user friendly interfaces
3. To apply Interactive Design process in real world applications
4. To understand various design technologies to meet user requirements.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Foundations Of HMI: The Human: History of User Interface Designing, I/O channels, Hardware, Software and Operating environments, The Psychopathology of everyday Things, Psychology of everyday actions, Reasoning and problem solving. The computer: Devices, Memory, processing and networks. Interaction: Models, frameworks, Ergonomics, styles, elements, interactivity, Paradigms.	9
2	Design & Software Process: Mistakes performed while designing a computer system, Human interaction with computers, importance of human characteristics human consideration, Human interaction speeds. Interactive Design basics, process, scenarios, navigation, Iteration and prototyping. HMI in software process: software life cycle, usability engineering, Prototyping in practice, design rationale. Design rules: principles, standards, guidelines, rules. Recognize the goals, Goal directed design process. Evaluation Techniques: Universal Design.	9

3	The graphical User Interface: Popularity of graphics, the concept of direct manipulation, graphical systems, Characteristics. Web user Interface: Interface popularity, characteristics. The merging of graphical Business systems and the Web. Principles of user interface design.	9
4	Mobile Ecosystem: Platforms, Application frameworks: Types of Mobile Applications: Widgets, Applications, Games, Mobile Information Architecture, Mobile 2.0, Mobile Design: Elements of Mobile Design, Tools.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamental principles of HMI, Interaction such as Models, frameworks, Ergonomics etc	K2
CO2	Understand the design concepts, HMI in software process and Evaluation Techniques.	K2
CO3	Analyse various design technologies to meet user requirements.	K4
CO4	Identify User Interface (UI) design principles, characteristics.	K2
CO5	Identify mobile interaction design and its usage in day – to – day activities	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3				2							2
CO2	3	2	3		3							2
CO3	3	3	3		3							2
CO4	3	3	3		3							2
CO5	3	3	3		3	3						2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human Computer Interaction	Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale	Pearson Education	3e, 2004
2	Human Computer Interaction: An Empirical Study of User Interface Design	Kent.L.Norman	CRC Press	2e, 2017
3	Galitz's Human Machine Interaction	Dhananjay R.Kalbande	Wiley	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human Computer Interaction-A critical approach	Joanna Mc Grenere	MIT press	1e, 2017
2	The design of everyday things	Don Norman	Basic Books	2013
3	Human machine interface technology -Advancements and Applications	RaviChandar Janapati	CRC press	2023
4	Human Computer Interface	I.A.Dhotre, Soumitra Das	Technical publications	2023

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/uFYuHHglC6U?si=mOJOdYdypc7jb9if
2	https://youtu.be/HSSFVBWFiLg?si=lrW5wbTCJREn0z1M
3	https://nptel.ac.in/courses/106103115
4	https://onlinecourses.nptel.ac.in/noc19_cs86/preview

SEMESTER S6

ADVANCED CYBER PHYSICAL SYSTEMS LAB

Course Code	PCCPL607	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCCPL508	Course Type	Practical

Course Objectives:

1. To introduce basics of cyber-physical system and Industrial revolution 4.0 concepts.
2. To develop an exposition of the challenges in implementing a cyber-physical system.
3. To analyze the functional behaviour of CPS based on standard modeling formalisms.
4. To design CPS requirements based on operating system and hardware architecture constraints.
5. To understand the concepts involved in Cyber Physical Systems Security.

Details of Experiment

(Perform at least 10 Experiments from the list below, in which a minimum of 4 experiments from Exp. No. 1-6 and 6 experiments from Exp. No. 7-14 are included)

Expt. No	Experiment
1	Use simulation tools like MATLAB/ Simulink or Gazebo to model and simulate CPS. Validate the simulations with real-world data.
2	Login to devicehub.net and create a project then create a virtual device. Add the corresponding sensor and actuator to the virtual device.
3	Study the MQTT protocol. Examine the components of the protocol.
4	Create a connection from an MQTT capable device/software with an MQTT broker then send and receive data using it.
5	To configuring the gateways and exchange the data to local database using Dream factory.
6	To configuring the gateways and upload the data to cloud server using Dream factory or Ubidots.
7	Assemble a computer controlled planar robot with 2 rotary joints and fixed link lengths. The end effector should move in the 2D plane according to the set values of joint angles.

8	For the computer controlled 2D planar robot, when the desired destination position (X,Y) is given, move the end effector to this point and display the computed values of joint angles. Demonstrate importance of singularities.
9	Graphically simulate the forward kinematics of the following 3 joint manipulators Cartesian robot b) Cylindrical robot c) Spherical robot d) Articulated robot
10	Using suitable programming language, find the homogenous transformation matrix for a set of sequential joint movements (translation and rotation) if the parameters of joint movements are specified with respect to the fixed frame.
11	Repeat the above experiment, if the parameters of joint movements are specified relative to the moving frames.
12	Using a DC motor and suitable measuring instruments, plot the torque versus current, torque versus angular velocity and power versus angular velocity.
13	Using ultrasonic sensors, stepper/servo motors and infrared sensors, develop a line follower robot that will follow a trajectory, avoiding obstacles on its path.
14	Using DC servo / stepper motors, implement a pick and place SCARA robot arm in 3D plane.

Course Assessment Method

(CIE: 50 Marks, ESE 50 Marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work, experiments, Viva and Timely completion of Lab Reports / Record. (Continuous Assessment)	Internal Exam	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

Mandatory requirements for ESE:

- Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- Endorsement by External Examiner: The external examiner shall endorse the record.

Course Outcomes (COs)

At the end of the course the student will be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	An ability to expose the student to real world problems in CPS and Industrial revolution 4.0 best practices.	K3
CO2	Identify the limitations of some computational models.	K1
CO3	Apply the theoretical knowledge the design of compilers.	K3
CO4	Student can Analyze and verify the correctness of CPS implementations against system requirements and timing constraints.	K4
CO5	Categorize the essential modeling formalisms of Cyber-Physical Systems (CPS)	K4

K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	2	2	3				2	1		1
CO2	3	1	2	2	3				2	1		1
CO3	3	1	2	2	3				2	1		1
CO4	3	1	2	2	3				2	1		1
CO5	3	1	2	2	3				2	1		1

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), : No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Embedded Systems Design with 8051 Microcontrollers: Hardware and Software	Zdravko Karakehayov,	CRC Press	1/e, 1999.
2	Introduction to Embedded Systems: Using Microcontrollers and the MSP430	Manuel Jiménez, Rogelio Palomera, and Isidoro Couvertier	Springer	1/e,, 2013

Video Links (NPTEL, SWAYAM...)	
Sl. No.	Link ID
1	https://youtu.be/8wBQMbOntNc?si=dm9shnF0DJQPfQ31
2	https://youtu.be/p7kYStiASLo?si=xL8Pq7OpmCO9xZp8
3	https://youtu.be/urUBLmXFKl0?si=3Skks7UMk19Bu0Nj

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 7

CYBER PHYSICAL SYSTEMS

SEMESTER S7

REAL TIME OPERATING SYSTEMS

Course Code	PECPT741	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3: 0: 0: 0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	-	Course Type	Theory

Course Objectives:

1. To understand the fundamental concepts and techniques of Operating Systems.
2. To study fundamental concepts, design principles and hardware considerations
3. To explore multithreading concepts, processes, threads, and various scheduling structures.
4. To Gain practical skills in designing and implementing operating system components to optimize system performance and reliability.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Operating System: Computer Hardware Organization, BIOS and Boot Process, Multithreading concepts, Processes, Threads, Scheduling, Structures - Simple Batch, Multi-programmed, Time-shared, Personal Computer, Parallel, Distributed Systems, Real-Time Systems, System components, Operating System services.	9
2	Basics of Real Time Concepts: Terminology: RTOS concepts and definitions, real-time design issues, examples, Hardware Considerations: logic states, CPU, memory, I/O, Architectures, RTOS building blocks, Real-Time Kernel	9
3	Process Management: Concepts, scheduling, IPC, RPC, CPU Scheduling, scheduling criteria, scheduling algorithms Threads: Multi-threading models, threading issues, thread libraries, synchronization Mutex: creating, deleting, prioritizing mutex, mutex internals	9

4	Inter-process Communication: Messages, Buffers, mailboxes, queues, semaphores, deadlock, priority inversion Pipes memory management: - Process stack management, run-time buffer size, swapping, overlays, block/page management, replacement algorithms, real-time garbage collection	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamental concepts of computer hardware organization, BIOS, and the boot process.	K2
CO2	Analyze memory management techniques: process stack management, swapping, overlays, and block/page management.	K4
CO3	Develop efficient inter-process communication mechanisms to facilitate data exchange.	K3
CO4	Analyse solutions to common challenges such as deadlock and priority inversion in concurrent systems.	K4
CO5	Apply knowledge of system components and services to optimize system performance in practical scenarios.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1									
CO2	3	3	2	2								
CO3	3	3	3	3	2							
CO4	3	3	3	3	2							
CO5	3	3	2	3	3	2						

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	MicroC/OS-II: The Real –Time Kernel	J. J Labrosse	Newnes	1/e, 2002
2	Real-time systems	Jane W. S. Liu	Prentice Hall	1/e, 2000
3	Operating System Principles	Abraham Silberchatz, Peter B. Galvin, Greg Gagne	John Wiley	7/e, 2005

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Programming in the UNIX® Environment	W. Richard Stevens	Pearson Education India	2/e, 2011
2	Real-Time System Design and Analysis	Philips A. Laplante	John Wley& Sons	3/e, 2004
3	Linux for Embedded and Real-Time Applications	Doug Abbott	Newnes	2/e, 2011
4	Operating Systems – Internals and Design Principles	Stallings	Pearson Education/PHI	5/e, 2005

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Introduction to Operating Systems - Course (nptel.ac.in)
2	NPTEL :: Computer Science and Engineering - NOC:Real Time Operating System
3	Real Time Operating System - Course (nptel.ac.in)
4	NPTEL :: Computer Science and Engineering - NOC:Real Time Operating System

SEMESTER S7

DISTRIBUTED SYSTEMS AND MIDDLEWARE

Course Code	PECPT742	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the fundamental concepts of distributed systems
2. To learn about various middleware technologies and their applications
3. To develop the skills to design and implement distributed applications.
4. To explore case studies and real-world applications of distributed systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Distributed Systems: Definition and characteristics, Examples and applications, Challenges in distributed systems. Communication in Distributed Systems: Interprocess communication: Remote Procedure Call (RPC), Remote Method Invocation (RMI), Message-oriented communication.	9
2	Middleware Technologies: Overview of middleware, Types of middleware (e.g., message-oriented, object-oriented, service-oriented), Middleware platforms and frameworks. Synchronization and Coordination: Logical clocks, Mutual exclusion, Election algorithms.	9

3	Distributed Middleware: Middleware architecture, Types of middleware, Case studies (e.g., CORBA, DCOM). Fault Tolerance and Reliability: Fault models, Replication techniques, Consensus algorithms.	9
4	Distributed File Systems and Databases: Distributed file system design, Distributed databases, Consistency models. Security in Distributed Systems: Security threats and vulnerabilities, Authentication and authorization, Secure communication protocols.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamental concepts and challenges of distributed system and different architectural and fundamental models.	K2
CO2	Explain various communication mechanisms like RPC, RMI, and message-oriented communication	K2
CO3	Apply algorithms for clock synchronization, mutual exclusion, and leader election	K3
CO4	Describe the architecture and types of middleware, case studies to understand practical implementations of middleware	K2
CO5	Identify security threats and mechanisms in distributed systems and Implement authentication, authorization, and secure communication techniques.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3			2							
CO2	3	3	2	2								
CO3	3	3	3									
CO4	3	3	2	3	2							
CO5	3	3	3									1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Distributed Systems Architecture: A Middleware Approach	Arno Puder, Kay Römer, and Frank Pilhofer	Morgan Kaufmann.	2005
2	Distributed Systems	Maarten van Steen and Andrew S. Tanenbaum	McGraw-Hill	4e, 2024
3	The Complete Book of Middleware	Judith M. Myerson	Auerbach Publications	2002

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Distributed Systems: Principles and Paradigms	Andrew S. Tanenbaum and Maarten Van Steen	Prentice Hall	2007
2	Distributed Systems: Concepts and Designs	George Coulouris, Jean Dollimore, Tim Kindberg, and Gordon Blair	Addison-Wesley	2012
3	Distributed Systems A Complete Guide	The Art of Service	The Art of Service - Distributed Systems Publishing	2020

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_cs87/preview
2	https://nptel.ac.in/courses/106106168
3	http://acl.digimat.in/nptel/courses/video/106106168/L21.html
4	https://elearn.nptel.ac.in/shop/nptel/cloud-computing-and-distributed-systems/?v=c86ee0d9d7ed

SEMESTER S7

PRIVACY IN CPS

Course Code	PECPT743	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	–	Course Type	Theory

Course Objectives:

1. Introduction to Privacy in CPS
2. To understand Privacy Preservation Techniques in CPS
3. To Implement Privacy in CPS
4. To introduce the Emerging Trends and Future Directions in Privacy for CPS

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of Cyber-Physical Systems (CPS): Definition, characteristics ,Components and architecture of CPS Importance of privacy in CPS. Fundamentals of Privacy: Definition and dimensions of privacy, Privacy principles and frameworks, Legal and ethical considerations Privacy Threats and Vulnerabilities in CPS: Types of privacy threats in CPS, Attack vectors and vulnerabilities Case studies of privacy breaches in CPS Introduction to Privacy Enhancing Technologies (PETs): Overview of PETs, Role of PETs in CPS, Examples of PETs in practice	9
2	Data Anonymization and Masking: Techniques for data anonymization, Data masking methods, Balancing privacy and utility Encryption and Secure Communication: Fundamentals of encryption, Encryption techniques for CPS, Secure communication protocols Access Control and Authentication: Access control mechanisms, Authentication methods in CPS, Role-based access control (RBAC).Differential Privacy: Concept of differential privacy, Techniques and algorithms, Applications of differential privacy in CPS	9

3	Privacy by Design (PbD): Principles of PbD, Implementing PbD in CPS', Case studies of PbD in action Privacy Risk Assessment: Identifying privacy risks in CPS, Conducting privacy impact assessments (PIAs),Mitigating privacy risks, Privacy Policies and Compliance: Developing privacy policies for CPS, Compliance with privacy regulations (e.g., GDPR, CCPA),Monitoring and enforcement of privacy policies Practical Exercises: Hands-on implementation of privacy-preserving techniques, Collaborative project work, Presentation and peer review of projects	9
4	Advanced Privacy-preserving Technologies : Homomorphic encryption, Secure multi-party computation, Blockchain for privacy Privacy in Emerging CPS Applications: Internet of Things (IoT),Smart cities and infrastructures, Autonomous systems and robotics Privacy Challenges and Opportunities: Current challenges in CPS privacy, Research directions and opportunities, Future trends in privacy for CPS Case Studies and Real-World Applications: Analysis of privacy-preserving CPS in various industries, Lessons learned from real-world implementations, Future outlook and career opportunities in CPS privacy	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Outline of Cyber Physical system	K2
CO2	Explain Privacy preservation techniques in CPS	K2
CO3	To identify Privacy Threats and vulnerabilities in CPS	K3
CO4	Outline to Privacy Enhancing Techniques.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									
CO2	2	3	2					1				1
CO3	2	2	3		3			1				1
CO4	2	2	2			3		1				1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Privacy and Security for Cyber-Physical Systems	Yassine Maleh, Mohamed Shojafar, Ahmed A. Al-Nemrat, and Abdelouahid Derhab	CRC press	1e, 2019
2	Security and Privacy in Cyber-Physical Systems: Foundations and Applications	Houbing Song, Glenn A. Fink, and Sabina Jeschke	Wiley	1e, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Privacy Engineering: A Dataflow and Ontological Approach	Ian Oliver	Createspace Independent Publishing Platform	2014
2	Privacy in a Digital, Networked World: Technologies, Implications and Solutions	Sherali Zeadally and Mohamad Badra	Springer	1e, 2015
3	Security and Privacy in Internet of Things (IoTs): Models, Algorithms, and Implementations	Fei Hu	CRC press	1e, 2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/AQClJAif5w8?si=E6vvQ2cRYli1jnUc
2	https://youtu.be/xErbGFBnZeo?si=xBG0WrR5-IDsXv5o
3	https://youtu.be/OYsY5B9pqYU?si=rFopBPQTqngLVUIP
4	https://youtu.be/p7kYStiASLo?si=1VhlOfz4Nh0gx8bn

SEMESTER S7

USER EXPERIENCE DESIGN

Course Code	PECPT744	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PECPT415, OECPT612, OECPT614	Course Type	Theory

Course Objectives:

1. To understand the fundamentals of UX design and its importance in CPS
2. To apply user research methods to inform UX design decisions
3. To design intuitive and user-centred interfaces for CPS devices.
4. To identify security and privacy concerns in CPS UX, design challenges and opportunities.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of UX Design and its importance in CPS: Human-centered design principles, Design thinking approach, Human-Computer Interaction (HCI) Design: Principles of HCI design, Interaction design patterns, Designing interfaces for CPS devices.	9
2	User Experience (UX) Design Principles: User experience design principles: User journey mapping: Wireframing and prototyping. CPS-specific UX Design Considerations: Designing for IoT devices and sensors, UX design for real-time systems, Security and privacy considerations	9
3	Designing for Smart Systems: Designing for smart homes and buildings, UX design for industrial automation, Designing for autonomous systems. UX Design for Artificial Intelligence (AI) and Machine Learning (ML) Systems: UX design considerations for AI and ML systems, Designing for transparency and explainability, UX design for AI-powered CPS applications	9

4	Cybersecurity and UX Design: Security considerations in UX design, Designing for secure user authentication and authorization, UX design for secure data transmission and storage. User Experience (UX) Design Tools and Software: Overview of UX design tools and software, Proficiency in using UX design tools (e.g., Sketch, Figma, Adobe XD), Best practices for UX design tool selection and usage.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify the key principles of UX design and Human-Computer Interaction (HCI) Design	K2
CO2	Explain User Experience (UX) Design Principles CPS-specific UX Design Considerations for IoT Devices and sensors	K2
CO3	Understand the Design considerations for Smart Systems, UX Design for AI and ML Systems.	K3
CO4	Identify security and privacy concerns in CPS UX	K2
CO5	Identify User Experience (UX) Design Tools and Software.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										1
CO2	3	3	2	2	3							1
CO3	3	3	3	3	3							1
CO4	3	3	3									1
CO5	3	3	3	3	3							1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Designing User Experience for Cyber-Physical Systems	Ali Hessami & Soar	Springer	2020
2	The Elements of User Experience	Jesse James Garrett	New Riders	2003
3	The Design of Everyday Things	Don Norman	Basic Books	2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	User Experience Design: A Practical Guide	Nathan Shedroff and Christopher Noessel	CRC Press	2012
2	UX Design: A Beginner's Guide	Ben Hunt	Packt Publishing	2018
3	User Experience (UX) Design: Creating User-Centered Design Solutions	Russ Unger and Carolyn Chandler-	Elsevier	2012

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_ar05/preview
2	https://archive.nptel.ac.in/noc/courses/noc22/SEM1/noc22-ar02/
3	https://www.youtube.com/watch?v=GDcOKMTxAq4
4	https://www.youtube.com/watch?v=mcXaJw4o164

SEMESTER S7

DATA VISUALIZATION AND INTERPRETATION

Course Code	PECPT745	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3: 0: 0: 0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None		Theory

Course Objectives:

1. To interpret data plots and understand core data visualization concepts such as correlation, linear relationships, and log scales.
2. To gain the ability to choose appropriate visualization systems for different types of data.
3. To apply visualization techniques to improve data interpretability
4. To develop skills to create visualizations that explain complex data science concepts effectively

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to data visualization: Introduction of visual perception, visual representation of data, Gestalt principles, information overloads, Creating visual representations, visualization reference model, visual mapping, visual analytics, Design of visualization applications.	9
2	Visualization systems: Classification of visualization systems, Interaction and visualization techniques misleading, Visualization of one, two and multi-dimensional data, text and text documents.	9
3	Interactive visualization: Why interactivity is needed, Handling multiple views, Brushing and Linking, D3 instructions and assignments Data Interpretability: Causes of misinterpretation, Role of communicative visualization in sciences	9

4	Visualization in data science: methods and examples- Explaining machine learning models, Interpretability challenges and solutions, Transparency and human-machine trust, Impactful case studies: biology, healthcare, cyber security, climate science, social science.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Students should evaluate and analyze a real-world problem, assess the proposed solutions, conclude which solution is most appropriate for the problem, and then implement the chosen solution using executable code or simulate it using relevant software or supported hardware platforms.

Criteria for evaluation:

- 1. Problem Definition (K4 - 4 points)**
 - a. Clearly defines the real-world problem.
 - b. Examine and identifies relevant contextual factors (constraints, resources, objectives).
- 2. Problem Analysis (K4 - 4 points)**
 - a. Break-down and presents a well-reasoned solution approach.
 - b. Compare and justify the proposed solutions with evidence and logical reasoning.
- 3. Evaluate (K5 - 4 points)**
 - a. Thoroughly evaluate the proposed solutions.
 - b. Compares trade-offs, advantages, and disadvantages.
 - c. Considers feasibility, scalability, and practical implications.
- 4. Implementation (K5 - 4 points)**
 - a. Select the most feasible solution by implementing the proposed solutions.
 - b. Successfully translates the chosen solution into an executable code or simulation using relevant software suit or supported hardware platform.
 - c. Demonstrates proficiency in simulation / emulation practices (readability, efficiency, error handling).
- 5. Conclusion (K4- 2 points, K5 – 2 points)**
 - a. Summarizes findings and insights. State which solution is most appropriate for the problem. (K4)
 - b. Reflects critical thinking and informed decision-making. (K5)

Scoring:

1. **Accomplished (4 points):** Exceptional analysis, clear implementation, and depth of understanding.
2. **Competent (3 points):** Solid performance with minor areas for improvement.
3. **Developing (2 points):** Adequate effort but lacks depth or clarity.
4. **Minimal (1 point):** Incomplete or significantly flawed.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• 2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. • Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understanding of data visualization and key terms	K2
CO2	Design effective data visualization for visual mapping and design	K4
CO3	Demonstrate skills on creating visual representation of data	K3
CO4	Evaluate and critique interactive visualizations based on their effectiveness and usability.	K5
CO5	Examine visualizations that effectively communicate scientific data and concepts	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1								
CO2	3	3	3	2	2							
CO3	2	3	3	3	2					1		
CO4	2	3	3	2	2					2		
CO5	3	3	3	3	2					2		

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Interactive Data Visualization: Foundations, Techniques, and Applications	Ward, Grinstein Keim	Natick: A K Peters, Ltd	1/e, 2010
2	Data Visualization: A Practical Introduction	Kieran Healy	Kieran Healy	1/e, 2018
3	Data Visualization: a successful design process	Andy Krik	Packt publishing	1/e, 2012
4	Interactive Data Visualization for the Web: An Introduction to Designing with D3	Scott Murray	O'Reilly Media	2/e, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Visual Display of Quantitative Information	E. Tufte	Graphics Press.	2/e, 2001
2	Interactive Data Visualization: Foundations, Techniques and Applications.	Ward, Grinstein Keim	Natick: A K Peters, Ltd.	1/e, 2014
3	Visualization Analysis and Design (AK Peters Visualization Series)	Tamara Munzner	CRC Press	1/e, 2014

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/qdnM8Fpvdqc?si=jk_fLg4w9bVb5vqt
2	https://youtu.be/MUP2m46uw8I?si=uR3ZyXEZrTH_6DEG
3	https://elearn.nptel.ac.in/shop/iit-workshops/completed/data-visualization-with-r/?v=c86ee0d9d7ed
4	https://elearn.nptel.ac.in/shop/iit-workshops/completed/data-visualization-with-r/?v=c86ee0d9d7ed

SEMESTER S7

EMBEDDED SYSTEMS DESIGN

Course Code	PECPT751	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCCPT302, PECPT631	Course Type	Theory

Course Objectives:

1. To design and optimize single-purpose processors and explore ASICs and ASIPs.
2. To introduce FPGA and behaviour synthesis using VHDL.
3. To implement controller designs using Arduino.
4. To address power-aware system design and code efficiency.
5. To explore DSP applications and address generation units.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Embedded System, ASICs and ASIPs, design ninga Single Purpose Processors and Optimization, Introduction to FPFA, Behaviour synthesis of FPGA using VHDL	9
2	Sensors and Signals, Discretisation of signals and A/D converters, Quantization Noise, SNR and D/A converters, Arduino Uno, Serial Communication and Timer, Controller design using Arduino.	9
3	Power Aware Embedded Systems, SD and DD algorithms, Parallel Operations and VLIW, Code efficiency DSP Application and address generation unit, RMS algorithm	9
4	EDF algorithm and Resource constraint issue, Priority Inversion and Priority Inheritance Protocol, Priority Ceiling Protocol.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Ability to design and optimize embedded system.	K2
CO2	Proficiency in FPGA design and VHDL synthesis.	K2
CO3	Capability to design controllers using Arduino.	K3
CO4	Knowledge of power-aware system design and parallel operations.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2						2			
CO2	3	3	2	2					2			
CO3	3	3	2						2			
CO4	3	3	2						2			

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Embedded System Design: A Unified Hardware/ Software Introduction	Frank Vahid and Tony Givargis	John Wiley & Sons	1e, 2001
2	Digital Design and Computer Architecture	David Harris and Sarah Harris	Morgan Kaufmann	2e, 2012
3	Digital System Design with FPGA: Implementation Using Verilog and VHDL	Cem Unsalan and Bora Tar	McGraw-Hill Education	2e, 2017
4	Arduino Cookbook	Michael Margolis	O'Reilly Media	3e, 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Power-Aware Computing	Massoud Pedram	Springer	1e, 2002
2	Parallel Programming for Modern High Performance Computing Systems	Pawel Czarnul	CRC Press	1e, 2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1 to 4	https://archive.nptel.ac.in/courses/106/105/106105159/

SEMESTER S7

FAULT TOLERANCE IN NETWORKED SYSTEMS

Course Code	PECPT752	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GBEST204, PBCPT404, PECPT522	Course Type	Theory

Course Objectives:

1. To understand fundamental fault tolerance concepts and their relevance in networked systems.
2. To learn and apply redundancy, replication, and recovery techniques.
3. To develop skills in fault detection, diagnosis, and recovery strategies.
4. To explore fault tolerance in various modern networked environments.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Fault Tolerance - Definitions and key concepts, Importance in networked systems Types of Faults and Failures - Crash failures, omission failures, timing failures, and Byzantine failures, Impact on system performance and reliability Redundancy and Replication Techniques - Hardware vs. software redundancy, Data redundancy (RAID levels and their implications), Replication strategies: primary-backup, quorum-based	9

2	Fault Detection Mechanisms - Basic fault detection techniques: heartbeat, monitoring, logging, Tools and frameworks for fault detection (basic overview) Fault Diagnosis and Isolation - Techniques for identifying and isolating faults, Practical examples and simplified tools Recovery Strategies - Checkpointing and rollback basics, Simple transaction recovery mechanisms	9
3	Distributed Consensus and Coordination - Introduction to consensus algorithms: Paxos, Raft, Basic concepts of coordination services: Zookeeper, Consul Load Balancing and Failover - Principles of load balancing in fault-tolerant systems, Simple failover strategies and practical case studies Network Fault Tolerance - Basic network redundancy and routing protocols, Handling network partitioning and latency issues	9
4	Fault Tolerance in Modern Systems - Fault tolerance in cloud computing and microservices, Real-world applications and case studies Security Considerations - Impact of security breaches on fault tolerance, Designing fault-tolerant systems with basic security considerations Emerging Trends and Future Directions - Fault tolerance in IoT and edge computing, Current research and future directions Project Presentations and Review - Student presentations of fault-tolerant system projects, Review of key concepts and course wrap-up	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain fundamental concepts of fault tolerance	K2
CO2	Identify and classify various types of faults and failures and analyze the impact of faults and failures on networked systems.	K2
CO3	Understand fault recovery strategies to ensure system resilience and data integrity.	K2
CO4	Apply network fault tolerance techniques	K3
CO5	Understand current trends and technologies in fault tolerance	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	2	1							
CO2	3	2	3	2	1							
CO3	3	2	3	2	1							
CO4	3	2	3	2	1							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fault-Tolerant Systems	Israel Koren, C. Mani Krishna	Morgan Kaufmann	2/e
2	Distributed Systems: Principles and Paradigms	Andrew S. Tanenbaum, Maarten Van Steen	Pearson Education India	2/e

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Reliable and Secure Distributed Programming	Christian Cachin, Rachid Guerraoui, Luis Rodrigues	Springer	2/e
2	Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems	Martin Kleppmann	Shroff/O'Reilly	2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1 to 4	https://www.youtube.com/@6.824

SEMESTER S7

INTRUSION DETECTION AND PREVENTION

Course Code	PECPT753	CIE Marks	40
Teaching Hours/Week (L:T:P:R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the fundamental concepts, types, and methodologies of intrusion detection systems.
2. To analyze various types of network threats and vulnerabilities, assessing their impact on security.
3. To evaluate and compare detection techniques for their effectiveness in identifying intrusions.
4. To study and analyze security protocols like SSL/TLS, IPSec, and SSH in securing data in transit.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Network Security: Overview of Network Security, Types of Threats and Attacks, Security Policies and Mechanisms, Introduction to Intrusion Detection and Prevention Systems, Intrusion Detection Systems (IDS): Definition and Purpose of IDS, Types of IDS- Network-based, Host-based, and Hybrid IDS Architectures, IDS Components and Functionality, Signature-based vs. Anomaly-based Detection.	9
2	Intrusion Prevention Systems (IPS): Definition and Purpose of IPS, Differences between IDS and IPS, IPS Techniques and Mechanisms, Deployment Strategies for IPS, Challenges in IPS Implementation. Detection Techniques: Signature Detection Techniques Anomaly Detection Techniques, Statistical Methods for Anomaly Detection.	9

3	Security Information and Event Management (SIEM): Introduction to SIEM, Role of SIEM in Intrusion Detection and Prevention, Log Management and Analysis, Correlation of Events, Understanding the role of SIEM in network security. Incident Response and Management: Incident Response Lifecycle, Identification, Containment, Eradication, and Recovery.	9
4	Emerging Trends and Technologies: Cloud Security and IDS/IPS, IoT Security Challenges, Machine Learning and AI in Intrusion Detection, Blockchain for Security, Future Trends in Network Security, Network Security Protocols: SSL/TLS, IPSec, SSH.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamental concepts and principles of intrusion detection systems, including types, architectures, and methodologies.	K2
CO2	Identify and explain the key differences between Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS).	K2
CO3	Understand the fundamental concepts of SIEM and its importance in cybersecurity, components and architecture of SIEM systems.	K2
CO4	Analyze the key security challenges associated with cloud computing, including data privacy, compliance, and shared responsibility models.	K4
CO5	Understand the emerging trends and advancements in network security protocols, including the evolution of SSL/TLS	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										1
CO2	3	3	2	2	3							1
CO3	3	3	3	3	3							1
CO4	3	3	3									1
CO5	3	3	3	3	3							1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Intrusion Detection and Prevention	Richard Bejtlich	Syngress	1e, 2005
2	Network Intrusion Detection and Prevention	Andrew R. Jones	Cisco Press	1e, 2008
3	Intrusion Detection and Prevention Systems	John R. Vacca	Springer	2e, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Intrusion Detection and Prevention Systems: Concepts, Techniques, and Applications	Mohamed F. Elshal	CRC Press	1e, 2019
2	Intrusion Detection and Prevention for Network Security	Junaid Chaudhry	Apress	1e, 2018
3	Intrusion Detection and Prevention Systems: A Comprehensive Review	Springer	Springer	1e, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.swayam2.ac.in/cec20_cs09/preview
2	https://www.youtube.com/watch?v=2YGUvopGkQc
3	https://www.youtube.com/watch?v=v8VWD3KnizE
4	https://ssmt.iiit.ac.in/Cyber%20Security/Module%208_1Intrusion%20Detection%20Systems

SEMESTER S7

COGNITIVE SYSTEMS AND AI IN CPS

Course Code	PECPT754	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Program Elective 5

Course Objectives:

1. To understand Basic Concepts of Artificial Intelligence and Expert Systems.
2. To provide knowledge on Various Techniques and Tools involved in Artificial Intelligence.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: History, Definition of AI, Emulation of human cognitive process, knowledge search trade off, stored knowledge, semantic nets. An abstract view of modelling, elementary knowledge. Computational logic, analysis of compound statements using simple logic connectives, predicate logic, knowledge organization and manipulation, knowledge acquisition.	9
2	LISP and other programming languages- introduction to LISP, syntax and numerical function, LISP and PROLOG distinction, input output and local variables, Interaction and recursion, property list and arrays alternative languages, formalized symbolic logics- properties of WFRS, non-deductive inference methods. Inconsistencies and uncertainties-Truth maintenance systems, default reasoning and closed world assumption, Model and temporary logic.	9

3	Matching techniques- Need for matching, matching problem, partial matching, Fuzzy matching, RETE matching algorithm. Knowledge organization- Indexing and retrieval techniques, integration of knowledge in memory organization systems, Perception, communication and Expert systems. Overview of Linguistics, Basic passim techniques, Semantic analysis and representation structures, natural language generation and system.	9
4	Pattern recognition system- understanding speech recognition, Image transformation, low level processing, medium and high level processing, vision system architecture, Rule based system architecture, knowledge acquisition and validation, knowledge system building tools, use of AI and ES in manufacturing and design, types of learning- general learning model, performance measures, learning automate genetic algorithm, learning by induction - LEX, ID3, INDUCE systems.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the Basics about Artificial Intelligence and Expert Systems.	K1
CO2	Understand the Programming Logics in Artificial Intelligence.	K2
CO3	Understand the Knowledge about the Expert Systems.	K2
CO4	Understand The Image processing and the latest developments in Knowledge systems and Tools	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3	2	1									1
CO3	3		1	2								1
CO4	3	2	1	2	2							1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Artificial Intelligence- Modern approach, Pearson Education series in AI	Russel (Stuart)	Pearson Education series in AI	2009
2	Introduction to Artificial intelligence and Expert systems	Dan W Patterson,	Prentice Hall of India	2001
3	Introduction to Artificial intelligence	Eugene Charniak, Drew Mc Dermot	Addison Wesley Longman Inc	2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Artificial intelligence and the design of expert systems	George. F, William. A. Stubblefield,	The Benjamin Cummins	1992
2	Artificial intelligence An Engineering Approach	Robert J Schalkoff	McGraw Hill International Edition	1990

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/pKeVMlkFpRc?si=Y1yzJ3gm4ckbpyDk
2	https://youtu.be/EbNJ05EVXs0?si=6gQ4bjXY1ddIV5yv
3	https://youtu.be/XG1sxRcdQZY?si=CN8rSA1MTOqnQBge
4	https://youtu.be/Z66_2_VG_k8?si=0D_6uJdZBdTq6UCF

SEMESTER S7

BIG DATA ANALYTICS

Course Code	PECPT755	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PECPT415, PECPT635	Course Type	Theory

Course Objectives:

1. To understand the fundamentals of Big Data and its significance.
2. To explore the relationship between Big Data and related technologies like Cloud Computing and IoT.
3. To learn about data generation, acquisition, and storage mechanisms for Big Data.
4. To gain hands-on experience with Hadoop and MapReduce for data processing.
5. To explore traditional and advanced methods for Big Data analysis.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	INTRODUCTION: Dawn of the Big Data Era, Definition and Features of Big Data, Big Data Value, The Development of Big Data, Challenges of Big Data. RELATED TECHNOLOGIES: Cloud Computing - Cloud Computing Preliminaries, Relationship Between Cloud Computing and Big Data, IoT - IoT Preliminaries, Relationship Between IoT and Big Data, Data Center, Hadoop - Hadoop Preliminaries, Relationship between Hadoop and Big Data	9
2	BIG DATA GENERATION AND ACQUISITION: Big Data Generation-Enterprise Data, IoT Data, Internet Data, Biomedical Data, Data Generation from Other Fields, Big Data Acquisition - Data Collection, Data Transportation, Data Pre-processing.	9
3	BIG DATA STORAGE: Storage System for Massive Data, Distributed Storage System, Storage Mechanism for Big Data - Database Technology, Design Factors, Database Programming Model HADOOP & MAP REDUCE: Data Storage and Analysis, Comparison with Other Systems, A Brief History of Hadoop, Apache Hadoop and the Hadoop Ecosystem, A Weather Dataset, Analyzing the Data with Unix Tools,	9

	Analyzing the Data with Hadoop(Map and Reduce, Java MapReduce), Scaling Out, Hadoop Streaming, Hadoop Pipes.	
4	BIG DATA ANALYSIS: Traditional Data Analysis, Big Data Analytic Methods, Architecture for Big Data Analysis - Real-Time vs. Offline Analysis, Analysis at Different Levels, Analysis with Different Complexity, Tools for Big Data Mining and Analysis.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Students should evaluate and analyze a real-world problem, assess the proposed solutions, conclude which solution is most appropriate for the problem, and then implement the chosen solution using executable code or simulate it using relevant software or supported hardware platforms.

Criteria for evaluation:

- 1. Problem Definition (K4 - 4 points)**
 - a. Clearly defines the real-world problem.
 - b. Examine and identifies relevant contextual factors (constraints, resources, objectives).
- 2. Problem Analysis (K4 - 4 points)**
 - a. Break-down and presents a well-reasoned solution approach.
 - b. Compare and justify the proposed solutions with evidence and logical reasoning.
- 3. Evaluate (K5 - 4 points)**
 - a. Thoroughly evaluate the proposed solutions.
 - b. Compares trade-offs, advantages, and disadvantages.
 - c. Considers feasibility, scalability, and practical implications.
- 4. Implementation (K5 - 4 points)**
 - a. Select the most feasible solution by implementing the proposed solutions.
 - b. Successfully translates the chosen solution into an executable code or simulation using relevant software suit or supported hardware platform.
 - c. Demonstrates proficiency in simulation / emulation practices (readability, efficiency, error handling).

5. Conclusion (K4- 2 points, K5 – 2 points)

- a. Summarizes findings and insights. State which solution is most appropriate for the problem. **(K4)**
- b. Reflects critical thinking and informed decision-making. **(K5)**

Scoring:

1. **Accomplished (4 points):** Exceptional analysis, clear implementation, and depth of understanding.
2. **Competent (3 points):** Solid performance with minor areas for improvement.
3. **Developing (2 points):** Adequate effort but lacks depth or clarity.
4. **Minimal (1 point):** Incomplete or significantly flawed.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. • Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Ability to define and describe the key features and value of Big Data.	K2
CO2	Understanding of the relationship between Big Data and technologies like Cloud Computing and IoT.	K3
CO3	Appraise Proficiency of data generation, acquisition, and pre-processing techniques.	K5
CO4	Analyse storage systems and mechanisms for handling massive data.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									
CO2	3	2	3	2	2							
CO3	3	2	3	2	2							
CO4	3	2		2								

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Big Data: Related Technologies, Challenges and Future Prospects	Min Chen, Shiwen Mao, Yin Zhang, Victor C.M. Leung	Springer	2014
2	Hadoop- The Definitive Guide	Tom White	O'reilly	2e
3	Big Data Analytics with R and Hadoop	Vignesh Prajapati	PACKT Publishing	November 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Big Data: Principles and Best Practices of Scalable Real-Time Data Systems	Nathan Marz and James Warren	Manning Publications	1/e, 2015
2	Understanding Big data	Chris Eaton, Dirk Deroos	McGraw Hill	2012

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc20_cs92/preview
2	https://nptel.ac.in/courses/106104189
3	https://www.youtube.com/watch?v=9QxZhapbo0o

SEMESTER S7

EMBEDDED SYSTEMS AND MICROCONTROLLERS

Course Code	OECPT721	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3: 0: 0: 0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	-	Course Type	Theory

Course Objectives:

1. To be able to develop simple programs for 8051 using assembly language programming
2. To interface 8051 microcontroller with peripheral devices.
3. To understand the architecture and design concept of embedded systems
4. To be able to design embedded systems based on Arduino.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	8051 microcontroller: Difference between microprocessor and microcontroller Architecture, 8051 pin diagram; Architecture, I/O Port structure, Register organization - special function registers, Memory organization, Instruction set, Addressing modes Simple Assembly language programs: Arithmetic (Addition, Subtraction, Multiplication & Division), Transfer a block of data from one internal memory location to another.	9
2	8051 microcontroller Programming and Peripherals. Timers/Counters- Serial Communication, Interrupt structure-programming. Interfacing of peripherals – LED (ALP and embedded C programming). OLED, LCD, ADC, DAC, sensors, simple Switch and keyboard interfacing, 7 segment LED (embedded C programming).	9
3	Embedded Systems Overview Introduction to Embedded Systems: Definition, Features, Simple Example of Embedded Systems, Applications of embedded systems-Consumer	9

	electronics, Robotics, Automobiles Embedded System Architecture: HW - Processor, Controller, SoC, Memory, Peripherals; SW - Application, Middleware, OS, Device Drivers, Tool chain- Assembler, Interpreter, Compiler, Linker, Loader, Debugger Embedded system design process: Requirement Analysis, Specification Development, HW & SW Co-Design and Development, Module Integration, Testing.	
4	Embedded System Board Study (Arduino Uno) Arduino Uno Board: Board Study (Board level Block schematic) - Chip (Features only – Architecture not needed), GPIO, Memory, Programming Interface Programming: Arduino IDE, Sample Code (LED, Switch, DC motor, Stepper motor control), Temperature monitoring system using LM35 Temperature sensor & Seven Segment display Embedded Operating system basic concepts: Functional layers in a computer system OS terminology, Kernel Functions (Overview only), Types of Kernels (Monolithic kernel & Microkernel), Tasks/ Processes.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the internal architecture of 8051 Microcontroller	K2
CO2	Develop simple programs for 8051 using assembly language programming	K3
CO3	Interface 8051 microcontroller with peripheral devices using ALP/Embedded C	K3
CO4	Interpret the architecture and design concept of embedded systems	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2							2			
CO2	3	2	3						2		2	
CO3	3	2	3		2				2		2	
CO4	3	2	2	2					2			

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The 8051 Microcontroller and Embedded Systems: Using Assembly and C	Muhammad Ali Mazidi	Pearson	2e, 2007
2	Embedded Systems: An Integrated Approach	Lyla B Das	Pearson	1e, 2012
3	Beginning Arduino	Michael McRoberts	Apress	1e, 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Embedded Systems Architecture, programming and Design	Raj Kamal	Tata McGraw-Hill	3e, 2013
2	Embedded Systems Architecture, A Comprehensive Guide for Engineers and Programmers	Tammy Noergaard	Newnes – Elsevier	2e, 2012
3	Arduino: The complete guide to Arduino for beginners, including projects, tips, tricks and programming	James Arthur,	Ingram	1e, 2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_ee32/preview
2	https://onlinecourses.nptel.ac.in/noc21_ee32/preview
3	https://archive.nptel.ac.in/courses/106/105/106105193/
4	https://nptel.ac.in/courses/106105193

SEMESTER S7
INTRODUCTION TO ROBOTICS

Course Code	OECPT722	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3: 0: 0: 0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	-	Course Type	Theory

Course Objectives:

1. To understand the basic concepts and components of robotic systems.
2. To learn principles of robot kinematics, dynamics, and control.
3. To gain practical experience with robot programming and sensor integration.
4. To explore various applications of robotics in industry and research.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of Robotics : Introduction to Robotics - Overview of robotics and its applications - Historical development and current trends - Basic components and classifications of robots Robot Anatomy and Components - Sensors: types, functions, and integration - Actuators: motors, servos, and their control - Control systems and feedback mechanisms Robot Kinematics - Coordinate systems and transformations - Forward and inverse kinematics Robot Dynamics - Introduction to dynamic modeling: Newton-Euler and Lagrangian methods	9

2	Control Systems and Programming : Control Systems Fundamentals - Control theory basics: PID control, state-space models - Implementing control algorithms in robotics Programming Robots - Introduction to robot programming languages and environments - Basic programming constructs and techniques Perception and Sensor Integration - Types of sensors: vision, rangefinders, IMUs - Sensor fusion and data interpretation	9
3	Motion Planning and System Integration : Motion Planning - Path planning algorithms: A*, RRT, PRM - Motion planning in dynamic environments Robotic Systems Integration - Integration of hardware and software components - Communication protocols and data exchange Robotics in Industry - Applications in manufacturing, healthcare, and other fields - Case studies and industry guest speakers - Current trends and future directions in robotics	9
4	Advanced Concepts in Robotics : Human-robot interaction: Techniques and applications - Autonomous robots: Navigation and decision-making - Robotics and AI: Integration and future trends Emerging Technologies - Soft robotics and bio-inspired robots Collaborative robots (cobots) and their applications - Swarm robotics and collective behavior	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify key applications of robotics in various fields such as manufacturing, healthcare, and research	K1
CO2	Apply Robot Kinematics and Dynamics	K2
CO3	Understand and apply basic control theory concepts, including state-space models	K3
CO4	Identify the modern applications and emerging technologies of robotics	K1

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1										
CO2	3	3	2	2								
CO3	3	2	2	2								
CO4	3	1										

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Robotics: Mechanics and Control	John J. Craig	Pearson	3/e
2	Robotics: Modelling, Planning and Control	Bruno Siciliano, Giuseppe Oriolo, Lorenzo Sciavicco, Luigi Villani	Springer	1/e
3	Foundations of Robotics: A Multidisciplinary Approach with Python and ROS	Damith Herath, David St-Onge	Springer	1/e
4	Introduction to Autonomous Robots: Mechanisms, Sensors, Actuators, and Algorithms	Christopher Heckman (Author), Nikolaus Correll	MIT Press	2022

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Robot Modeling and Control	Mark W. Spong, Seth Hutchinson, M. Vidyasagar	Wiley	1/e
2	Robotics and Control	R Mittle, I Nagrath	McGraw Hill	2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/112107289
2	https://nptel.ac.in/courses/112107289
3	https://nptel.ac.in/courses/112107289
4	https://nptel.ac.in/courses/112107289

SEMESTER S7

FUNDAMENTALS OF CYBER PHYSICAL SYSTEMS

Course Code	OECPT723	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3: 0: 0: 0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	-	Course Type	Theory

Course Objectives:

1. To understand the characteristics, requirements, and architecture of cyber physical systems.
2. To grasp the fundamentals of microprocessor and microcontroller architecture, focusing on Digital Signal Processors.
3. To master the core principles and mathematical foundations of dynamical systems.
4. To comprehend cyber-physical system software, emphasizing real-time operating systems and job scheduling.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Cyber-Physical Systems: Cyber-Physical Systems (CPS) in the real world, Basic principles of design and validation of CPS, Industry 4.0 AutoSAR, IIOT implications, Building Automation, Medical CPS.	9
2	Platform Components for Cyber Physical Systems: CPS HW platforms - Processors, Sensors, Actuators, CPS Network - WirelessHart, CAN, Automotive Ethernet, Scheduling Real Time CPS tasks: Table-driven and Event driven schedulers, Hybrid schedulers	9
3	Principles of Dynamical Systems: Dynamical Systems and Stability, Controller Design Techniques, Performance under Packet drop and Noise	9
4	CPS implementation issues: From features to automotive software components, Mapping software components to ECUs CPS Performance Analysis: Effect of scheduling, bus latency, sense and actuation faults on control performance, network congestion, Building real-time networks for CPS.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the concept of Cyber-Physical Systems (CPS) and their applications in the real world.	K2
CO2	Apply scheduling techniques to manage and optimize real-time CPS tasks effectively.	K3
CO3	Examine the performance of dynamical systems in the presence of packet drop and noise.	K4
CO4	Explain the functionality and applications of CPS network protocols such as WirelessHART, CAN, and Automotive Ethernet.	K3
CO5	Analyze the effects of network congestion on CPS performance	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	1							
CO2	2	3	3	2	2							
CO3	2	3	3	3	2							
CO4	2	2	2	2	3							
CO5	2	3	2	3	3							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Applied cyber-physical systems.	Suh, Sang C., U. John Tanik, John N. Carbone, and Abdullah Eroglu	Springer New York	1e, 2014
2	Principles of cyber-physical systems	Alur, Rajeev.	MIT Press,	1e, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	ARM System Developers Guide: Designing optimizing System Software	Andrew M Sloss, Dominic Symes, Chris Wright,	Morgan Kaufmann Publishers	1e, 2004
2	Cyber-Physical Systems: A Model-Based Approach	Taha, Walid M. Taha, Abd-Elhamid M.	Springer	1e, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc23_cs62/preview
2	https://nptel.ac.in/courses/106105241
3	https://archive.nptel.ac.in/courses/106/105/106105241/
4	https://youtu.be/OYsY5B9pqYU?si=kZigZpaQ6oAX638Y

SEMESTER S7

ADVANCED HUMAN MACHINE INTERFACES

Course Code	OECPT724	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Web Technologies, Software Engineering	Course Type	Open Elective 2

Course Objectives:

1. To learn the foundation of human machine interaction.
2. To understand the importance of human psychology in designing good interfaces.
3. To be aware of mobile interaction design and its usage in day – to – day activities.
4. To understand various design technologies to meet user requirements.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	FOUNDATIONS OF HMI: The Human: History of User Interface Designing, I/O channels, Hardware, Software and Operating environment Reasoning and problem solving . The computer: Devices, Memory, processing and networks. Interaction: Models, frameworks, Ergonomics, styles, elements, interactivity, Paradigms.	9
2	DESIGN & SOFTWARE PROCESS: Mistakes performed while designing a computer system, Human interaction with computers, importance of human characteristics human consideration, Human interaction speeds .Interactive Design basics, process, scenarios, navigation, Iteration and prototyping. HMI in software process: software life cycle, usability engineering, Prototyping in practice, design rationale. Design rules: principles, standards, guidelines, rules. Recognize the goals, Goal directed design process. Evaluation Techniques: Universal Design.	9

3	GRAPHICAL USER INTERFACE: The graphical User Interface: Popularity of graphics, the concept of direct manipulation, graphical systems, Characteristics. Web user Interface: Interface popularity, characteristics. The merging of graphical Business systems and the Web. Principles of user interface design	9
4	INTERFACE DESIGN FOR MOBILE DEVICES: Mobile Ecosystem: Platforms, Application frameworks: Types of Mobile Applications: Widgets, Applications, Games, Mobile Information Architecture, Mobile 2.0, Mobile Design: Elements of Mobile Design INTERACTION STYLES AND COMMUNICATION: Windows: Characteristics, Components, Presentation styles, Types of Windows, Management, operations. Text messages: Words, Sentences, messages and text words, Text for web pages. Icons, Multimedia and colors	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify User Interface (UI) design principles	K2
CO2	Analysis of effective user friendly interfaces.	K4
CO3	Analyze UI design and justify.	K4
CO4	Develop application for social and technical task.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	3	2									
CO2	2	1	2		3							
CO3	3	1	2		3							
CO4	2				3							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human Computer Interaction	Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale	Pearson Education	2004
2	About Face3: Essentials of Interaction design	Alan Cooper, Robert Reimann, David Cronin	Wiley publication.	3/e , 2007
3	Mobile Design and Developmentl	Brian Fling	O'Reilly Media Inc	2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Interaction Design: Beyond Human Computer Interaction	Rogers Sharp Preece	Wiley	4/e, 2015
2	The Handbook of Human Machine Interaction	Guy A. Boy	Ashgate publishing Ltd.	1/e, 2011
3	Human Machine Interaction	Kalbnde, Kanade, Iyer, lGalitz's	Wiley Publications.	1/e, 2007

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/uFYuHHglC6U?si=R-lzDLpIpsc5FSvU
2	https://youtu.be/Ln_LP7c23WM?si=h1pfqxQDctBKp0GZ
3	https://youtu.be/uFYuHHglC6U?si=gUyV9tyuY8H8UpxB
4	https://youtu.be/Spq7M-Y6gpo?si=U9DbgAPKIDzPwH7v

SEMESTER 8

CYBER PHYSICAL SYSTEMS

SEMESTER S8

ADVANCED EMBEDDED SYSTEMS

Course Code	PECPT861	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs 30 Min.
Prerequisites (if any)	Embedded system	Course Type	Theory

Course Objectives:

1. Review of embedded system technology.
2. To understand Concept of Hardware software tradeoff.
3. To study Real Time Operating System concept.
4. To understand the current trends in embedded system.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Typical Embedded System: Core of the Embedded System, Memory, Sensors and Actuators, Communication Interface, Embedded Firmware, Other System Components. Characteristics and Quality Attributes of Embedded Systems.	9
2	Hardware Software Co-Design and Program Modelling: Fundamental Issues in Hardware Software Co-Design, Computational Models in Embedded Design, Introduction to Unified Modelling Language, Hardware Software Trade-offs.	9
3	Real-Time Operating System (RTOS) based Embedded System Design: Operating System Basics, Types of OS, Tasks, Process and Threads, Multiprocessing and Multitasking, Task Scheduling, Threads, Processes and Scheduling: Putting them altogether, Task Communication, Task Synchronization, Device Drivers, How to Choose an RTOS.	9

4	The Embedded System Development Environment: The Integrated Development Environment (IDE), Types of Files Generated on Cross-compilation, Disassembler / Decompiler, Simulators, Trends in the Embedded Industry: Processor Trends in Embedded System, Embedded OS Trends, Development Language Trends, Open Standards, Frameworks and Alliances, Bottlenecks.	9
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Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To understand various components in embedded system	K2
CO2	To understand various Computational Models in Embedded Design and Introduction to Unified Modelling Language	K1
CO3	To understand Real-Time Operating System (RTOS) based Embedded System Design	K2
CO4	To understand the Embedded System Development Environment and Trends in the Embedded Industry.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										
CO2	3	2	1		3							1
CO3	3	2			3							1
CO4	2	2	3		2							1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Embedded Systems	Shibu K V	Tata McGraw Hill Education Private Limited	2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Embedded Systems – A contemporary Design Tool	James K Peckol	John Wiley	2008.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/uFhDGagZzjs?si=RTSteySgTE5OM8vq
2	https://youtu.be/TP1_F3IVjBc?si=rImi8BorgBRD_O86
3	https://youtu.be/T54qJMqpim8?si=5jxFrK7y6AZ8kQ1V
4	https://youtu.be/0xgvINDxXJI?si=fzaJrjUiBIwkRwnt

SEMESTER S8

ADVANCED NETWORKED CONTROL APPLICATIONS

Course Code	PECPT862	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3: 0: 0: 0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. To provide students with a thorough understanding of the basic principles and concepts of networked control systems (NCS).
2. To develop the ability to model, analyze, and design networked control systems, including switched feedback control and boundary control for damped wave equations.
3. To teach students how to integrate various components of NCS, including sensors, actuators, and communication networks, and address implementation challenges.
4. To introduce concepts of robust control design, including the synthesis of strictly positive real (SPR) transfer functions and ensuring system stability and robustness

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of Networked Control Systems: Introduction- Advantages and Applications of Control over Network, Brief History of Research Field of NCS, NCS Categories and NCS Components - NCS Components, Information Acquisition in a Network, Control of Actuators over a Network, Communication, NCS Challenges and Solutions - Integration of Components and Distribution of Intelligence	9
2	Overview of Agent-based Control and Management for NCS: Introduction, From Electricity to Connectivity: Why Agent-based Control and Management for Networked Systems, Hosting Mechanism and System Architecture for ABC, Design Principle for Networked Control Systems: Local Simple, Remote Complex (LSRC), Modular Construction and Learning Algorithms of Neuro-fuzzy Networks for LSRC Implementation, Issues in Software, Middleware, and Hardware Platforms, Real-world Applications.	9

3	Switched Feedback Control for Wireless Networked Systems: Mathematical Modeling of NCS as a Switched System, Optimal Output Feedback Control - Gain Tuning of Output Feedback Parameter, Stability Investigation. Networked Boundary Control of Damped Wave Equations: A Brief Introduction to the Smith Predictor, Boundary Control of Damped Wave Equations with Large Delays, Stability and Robustness Analysis, Fractional Order Case – Problem Formulation, Robustness of Boundary Stabilization, Compensation of Large Delays in Boundary Measurement.	9
4	Design of Robust Strictly Positive Real Transfer Functions: Definitions and Notation, Some Properties of SPR (WSPR) Regions, Characterization of SPR (WSPR) Regions, Robust SPR Synthesis: Intersection of WSPR Regions, Applications to Robust SPR Synthesis for Low-order Systems - The Third-order SPR Synthesis, The Fourth-order SPR Synthesis	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic concepts, advantages, challenges and solutions in integrating components of NCS	K2
CO2	Develop mathematical models, analyze stability, and design controllers for networked control systems, including handling challenges such as delays and packet loss.	K3
CO3	Utilize sensors, controllers, and actuators in a networked environment, addressing issues related to communication, latency, and reliability.	K3
CO4	Apply robust control design techniques, including the synthesis of strictly positive real (SPR) transfer functions, to ensure system stability and robustness.	K3
CO5	Apply networked control systems in various industrial applications, demonstrating the ability to implement solutions in real-world scenarios	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	2	1						
CO2	3	3	3	3	2	2						
CO3	3	3	3									
CO4	3	3	3									
CO5	3	2	3	3	2	2						

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Networked Control Systems Theory and Applications	Fei-Yue Wang, Derong Liu	Springer	1/e, 2008
2	Networked Control Systems: Theory, Applications and Analysis	Shiwen Tong, Dianwei Qian	Nova	2021

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Networked Control Systems with Their Application in Industry	Geng Liang, Wen Li, Guoping Liu	Nova Science Pub Inc	1/e, 2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1 - 4	https://youtu.be/FVmQbfsOkGc

SEMESTER S8

ADVANCED CYBERSECURITY STRATEGIES FOR CPS

Course Code	PECPT863	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PECPT743/53, PECPT633, PECPT 523, PECPT413	Course Type	Theory

Course Objectives:

1. To Understand and apply advanced cybersecurity concepts and strategies specific to Cyber-Physical Systems (CPS).
2. To develop skills to analyze and assess risks and vulnerabilities in CPS environments.
3. To Implement and evaluate defense mechanisms and security measures to protect CPS effectively.
4. To Explore and assess emerging trends and technologies in CPS security to address contemporary challenges.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Advanced Risk Management Techniques - Advanced risk assessment methodologies and frameworks - Integrating risk management into the CPS lifecycle Advanced Threat Modeling - Detailed threat modeling for sophisticated CPS architectures - Tools and techniques for dynamic threat modeling Integrative Security Strategies - Combining risk management and threat modeling strategies - Developing a cohesive security strategy for complex CPS environments	9

2	Advanced Intrusion Detection Systems - In-depth IDS technologies and architectures for CPS - Machine learning and AI in IDS for CPS - Configuring and evaluating advanced IDS systems Intrusion Prevention and Mitigation Strategies - Advanced intrusion prevention techniques - Real-time threat mitigation and response - Effective intrusion prevention in high-stakes CPS environments	9
3	Integration and Performance Evaluation - Integrating IDS and IPS into CPS - Performance evaluation and tuning of IDS/IPS systems - Best practices for IDS/IPS integration and performance optimization Secure Firmware and Software Update Mechanisms - Advanced techniques for secure firmware updates - Strategies for ensuring update integrity and authenticity - Implementing secure update mechanisms in CPS	9
4	Vulnerability Assessment and Management - Tools and techniques for assessing vulnerabilities in CPS - Managing and mitigating discovered vulnerabilities - Conducting vulnerability assessments and applying remediation strategies Incident Response for Firmware and Vulnerabilities - Incident response strategies specific to firmware and vulnerability issues - Forensic techniques for firmware-related incidents - Incident response and forensic analysis for firmware vulnerabilities. Emerging Trends and Technologies - Latest advancements in CPS security technologies - Impact of emerging technologies such as quantum computing and blockchain on CPS security	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. - Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand advanced cybersecurity concepts and strategies specific to CPS.	K2
CO2	Analyze and assess risks associated with CPS.	K4
CO3	Understand and apply robust defense mechanisms for CPS.	K2
CO4	Explore emerging trends and technologies in CPS security.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	1							
CO2	3	3	2	3	1							
CO3	3	2	3	2	1							
CO4	3	1	1	1	1							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Security: Principles and Practice	William Stallings, Lawrie Brown	Pearson	4/e, 2020
2	Cybersecurity and Privacy in Cyber Physical Systems	Yassine Maleh, Mohammad Shojafar <i>et al.</i>	Springer	1/e, 2019
3	Security in Cyber-Physical Systems: Foundations and Applications	Wenye Wang, Yuguang Fang	Wiley	1/e, 2017
4	Handbook of SCADA/ Control Systems Security	Robert Radvanovsky, Jacob Brodsky	CRC Press	2/e, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Art of Cyberwarfare: An Investigator's Guide to Espionage, Ransomware, and Organized Cybercrime	Jon DiMaggio	No Starch Press	1/e, 2022
2	Blockchain and Cryptography: From Theory to Applications	Mohsen Attaran, Angappa Gunasekaran	Springer	1/e, 2020
3	Security and Resilience in Intelligent Data-Centric Systems and Communication Networks	Massimo Ficco, Francesco Palmieri	Academic Press	1/e, 2017
4	Cyber-Physical Security: Protecting Critical Infrastructure at the State and Local Level	Richard A. Caralli, James F. Stevens, Lisa R. Young	CRC Press	1/e, 2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-4	https://www.youtube.com/watch?v=O6NKY2oE99M
	https://www.youtube.com/watch?v=T-spfvrmBl
	https://www.youtube.com/watch?v=G-JWhKrMDk4
	https://www.youtube.com/watch?v=EAQYhWWJbBM

SEMESTER S8

ADVANCED INTERACTION TECHNIQUES AND USER RESEARCH

Course Code	PECPT864	CIE Marks	40
Teaching Hours/Week (L:T:P:R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	—	Course Type	Theory

Course Objectives:

1. To Understand the principles of user-centered design and the challenges associated with input and interaction techniques.
2. To Explore the concept of Tangible User Interfaces (TUIs), focusing on the use of physical objects as input/output devices.
3. To learn spatial interactions in Augmented Reality (AR) and Virtual Reality (VR) environments to enhance immersion in gaming experiences.
4. To Investigate and design inclusive mobile user interfaces, considering common UI patterns, navigation, gestures, and accessibility for users with disabilities.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Historical Development of Major Advances in HCI: evolution of Human-Computer Interaction (HCI) from early interfaces (such as command-line interfaces) to modern graphical user interfaces (GUIs). Recent Developments and Challenges in Input and Interaction Techniques: Emerging input methods (touch, gesture, voice, etc.) and their impact on user experience, challenges related to designing intuitive and efficient interaction techniques.	9
2	Tangible and Embodied User Interactions: tangible interfaces (physical objects as input/output devices) and their applications. Embodied interactions that consider the user's body and movement. Mobile Interactions: Mobile HCI, including design principles for small screens, touch-based interactions, and mobile app usability, challenges specific to mobile devices.	9

3	Augmented and Virtual Reality (AR/VR): AR and VR interfaces, including spatial interactions, 3D visualization, and user experience design, challenges related to AR/VR hardware and software. Game User Interfaces: Game UI/UX design, including HUDs (heads-up displays), menus, controls and feedback mechanisms, player engagement and immersion.	9
4	Mobile Interaction Patterns: Common mobile UI patterns (e.g., navigation, gestures, touch targets), challenges specific to mobile devices. Interaction Techniques for Special User Groups: Investigate designing inclusive interfaces for users with disabilities (e.g., visual impairments, motor limitations).	9

**Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand principles of user-centered design and Challenges in Input and Interaction Techniques.	K2
CO2	Understand the concept of TUIs, which involve physical objects as input/output devices.	K2
CO3	Understand spatial interactions in AR and VR environments and enhance immersion in gaming experiences.	K2
CO4	Understand common mobile UI patterns, including navigation, gestures, and touch targets, designing inclusive interfaces for users with disabilities.	K2
CO5	Investigate designing inclusive interfaces for users with disabilities including Visual Impairments ,Motor Limitations	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										
CO2	3	3										
CO3	3	3										
CO4	3	3	3	3	3							
CO5	3	3	3	3	3							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Interaction Techniques and User Research: Emerging Trends and Methods	Ahmed Seffah, Homa Javahery	IGI Global	1e, 2018
2	Interaction Design: Beyond Human-Computer Interaction.	Yvonne Rogers, Helen Sharp, Jenny Preece	Wiley	5e, 2019
3	User Experience (UX) Design: Creating User-Centered Design Solutions	Russ Unger, Carolyn Chandler	Morgan Kaufmann	2e, 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human-Computer Interaction: An Empirical Study	Andrew Sears, Julie A. Jacko	CRC Press	3e, 2019
2	User Research Methods: Practical Strategies for User Experience Research	Nate Bolt, Tony Tulathimutte	Morgan Kaufmann	1e, 2015
3	Human-Computer Interaction: An Introduction to Design	Dix, Finlay, Abowd, Beale	Pearson	4e, 2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1 - 4	https://nptel.ac.in/courses/107103083
	http://acl.digimat.in/nptel/courses/video/106103220/L04.html
	https://nptel.ac.in/courses/121106013
	https://onlinecourses.nptel.ac.in/noc19_hs40/preview

SEMESTER S8

SYSTEMS ENGINEERING AND INTEGRATION

Course Code	PECPT865	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PECPT415, PECPT 635	Course Type	Theory

Course Objectives:

1. To understand the fundamental principles of systems engineering
2. To learn methodologies for designing and integrating complex systems.
3. To explore the challenges and solutions in CPS engineering.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Systems Engineering: Definition and Scope, Systems Engineering Life Cycle, Key Concepts and Principles Systems Engineering Processes: Requirements Engineering, Functional Analysis and Allocation, System Architecture and Design System Modeling and Simulation: Introduction to System Modeling, Tools and Techniques for Modeling	9
2	System Integration Concepts : Definition and Importance of Integration, Levels of System Integration, Integration Strategies and Approaches Integration Processes and Techniques : Interface Management, Integration Testing ,Verification and Validation Challenges in System Integration: Interoperability Issues, Managing Complexity, Risk Management in Integration	9
3	Design and Integration of Cyber-Physical Systems: System Design Methodologies: Model-Based Systems Engineering (MBSE), System Modeling and Simulation Integration of CPS: Sensor and Actuator Integration,Real-Time Systems and Embedded Computing, Networking and Communication Protocols Challenges in CPS Integration: Interoperability and Standards, Security and Privacy ,Scalability and Performance	9

4	Advanced Topics in Systems Engineering: Systems of Systems (SoS): Definition and Characteristics, SoS Engineering Challenges, SoS Architectures and Design Cyber-Physical Systems (CPS): Introduction to CPS, CPS Architecture and Integration, Security and Privacy in CPS Emerging Trends in Systems Engineering: Model-Based Systems Engineering (MBSE), Agile Systems Engineering, Digital Twins	9
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Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Students should evaluate and analyze a real-world problem, assess the proposed solutions, conclude which solution is most appropriate for the problem, and then implement the chosen solution using executable code or simulate it using relevant software or supported hardware platforms.

Criteria for evaluation:

1. **Problem Definition (K4 - 4 points)**
 - a. Clearly defines the real-world problem.
 - b. Examine and identifies relevant contextual factors (constraints, resources, objectives).
2. **Problem Analysis (K4 - 4 points)**
 - a. Break-down and presents a well-reasoned solution approach.
 - b. Compare and justify the proposed solutions with evidence and logical reasoning.
3. **Evaluate (K5 - 4 points)**
 - a. Thoroughly evaluate the proposed solutions.
 - b. Compares trade-offs, advantages, and disadvantages.
 - c. Considers feasibility, scalability, and practical implications.
4. **Implementation (K5 - 4 points)**
 - a. Select the most feasible solution by implementing the proposed solutions.
 - b. Successfully translates the chosen solution into an executable code or simulation using relevant software suit or supported hardware platform.
 - c. Demonstrates proficiency in simulation / emulation practices (readability, efficiency, error handling).

5. **Conclusion (K4- 2 points, K5 – 2 points)**

- a. Summarizes findings and insights. State which solution is most appropriate for the problem. **(K4)**
- b. Reflects critical thinking and informed decision-making. **(K5)**

Scoring:

1. **Accomplished (4 points):** Exceptional analysis, clear implementation, and depth of understanding.
2. **Competent (3 points):** Solid performance with minor areas for improvement.
3. **Developing (2 points):** Adequate effort but lacks depth or clarity.
4. **Minimal (1 point):** Incomplete or significantly flawed.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply systems engineering principles to the design and integration of CPS.	K3
CO2	Proficiency in using methodologies and tools for CPS design and integration.	K3
CO3	Evaluate the challenges and solutions in CPS engineering.	K5
CO4	Analyze, develop, and manage integrated CPS.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2		2							
CO2	3	2	3	2	2							
CO3	3	2	3	2	2							
CO4	3	2		2								

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Systems Engineering	Kossiakoff, A., Sweet, W.N., Seymour, S.J., and Biemer S.M.	John Wiley Sons Inc., New Jersey,	2/e, 2011
2	Systems Engineering and Analysis	Benjamin S. Blanchard and Wolter J. Fabrycky	Prentice Hall	5/e, 2010
3	Systems of Systems Engineering: Principles and Applications	Mo Jamshidi	CRC Press,	2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cyber-Physical Systems: Foundations, Principles and Applications	Houbing Song, Danda B. Rawat, Sabina Jeschke, Christian Brecher	Academic Press	1/e, 2016
2	The Engineering Design of Systems: Models and Methods	Dennis M. Buede, William D. Miller	Wiley	3/e, 2016
3	Model-Based Systems Engineering with OPM and SysML	Dov Dori	Springer	1/e, 2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-4	https://onlinecourses.nptel.ac.in/noc21_mg39/preview

SEMESTER S8

SYSTEM INTEGRATION AND DESIGN

Course Code	OECPT821	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3: 0: 0: 0	ESE Marks	60
Credits	3	Exam Hours	2.30
Prerequisites (if any)	-	Course Type	Theory

Course Objectives:

1. To identify the need and importance of system integration in modern IT environments.
2. To understand and analyze existing business processes relevant to integration
3. To apply architectural design concepts to a case study
4. To understand the importance of detailed program design in successful integration projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to System Integration: Definition, Need of System integration, System integration vs. data integration, System integration challenges, System integration types and examples, Ways to connect systems, How to approach system integration, Deployment options for integrated systems,	9
2	Preparation for the Integration Architecture Design: Integration Architecture Design Process, Business Need Analysis, Functional and Non-Functional Requirements Analysis, Business Processes Analysis,	9
3	Architecture design: Elements of an Architecture Design, Creating an Architecture Design, Hardware And Software Specification, Applying The Concepts At Tune Source	9
4	Program Design: Introduction, Moving from Logical to Physical Process Models, Designing Programs, Structure Chart, Program Specification.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic concepts and definitions of system integration including definitions, types, and key concepts.	K2
CO2	Apply various methods and approaches to connect different systems and deploy integrated systems effectively.	K3
CO3	Analyze business needs, functional and non-functional requirements, and existing business processes to inform integration architecture design.	K4
CO4	Examine and document existing business processes to align with integration goals.	K4
CO5	Develop comprehensive integration architecture designs, including hardware and software specifications, and detail programs and structure charts for system integration.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	2	1							
CO2	2	3	2	3	2							
CO3	3	3	3	2	2							
CO4	2	2	2	3	2					1		
CO5	3	3	3	3	3					2	2	1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	System Analysis and design	Dennis, Wixom, Roth	John Wiley & Sons, Inc.	5/e, 2012
2	Designing IT systems Integration	Marcin Ziemek	Deimos Software	1/e, 2023

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	System Integration	Jeffrey O Grady	CRC Press	1/e, 2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/107106009
2	https://youtu.be/D_j8j6T0zRU?si=tJ22litIpXWFLS3Z
3	https://onlinecourses.nptel.ac.in/noc22_ee58/preview
4	https://nptel.ac.in/courses/107106009

SEMESTER S8

KINEMATICS AND DYNAMICS OF ROBOT

Course Code	OECPT822	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	–	Course Type	Theory

Course Objectives:

1. To understand the fundamentals of robot kinematics and dynamics.
2. To understand the robot kinematics with velocity and force domain.
3. To understand the trajectory planning of robot manipulators.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	ROBOT KINEMATICS Representing Position and orientation – Homogeneous matrices – Denavit Hardenberg representation - Describing a Robot Arm - Forward Kinematics - Inverse Kinematics, Case studies on describing the real life robot manipulators using D-H representation.	9
2	VELOCITY AND FORCE RELATIONSHIPS Velocities in manipulators, Forward and inverse velocity kinematics - Manipulator Jacobian and Transforming Velocities between Coordinate Frames - Static forces in manipulators - Jacobians in the force domain - Cartesian transformation of velocities and static forces.	9
3	ROBOT DYNAMICS Acceleration of a rigid body - Equations of Motion – Forward and inverse dynamics - Iterative Newton-Euler dynamic formulation – Closed form dynamic equations – Lagrangian formulation of manipulator dynamics – Formulating manipulator dynamics in Cartesian coordinates.	9

4	TRAJECTORY PLANNING Joint space schemes – Cartesian space schemes – Path generation in run time – Description of paths in a robot programming language – Planning path when using the dynamic model – Collision-free path planning.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (K)
CO1	Develop the kinematic equations for any given robot manipulator using Denavit Hartenberg representation.	K2
CO2	Analyze the robot kinematic equations for the given force and/ or velocity domains.	K4
CO3	Formulate the robot dynamic equations for the given manipulator.	K3
CO4	Identify the trajectory of the given robot manipulator using the joint space or Cartesian space schemes.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1										
CO2		3	2		2							
CO3	2	1	2		2							
CO4	2	1	2		2							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Craig, Introduction to Robotics Mechanics and Control,	John J. Craig,	Pearson Education International,	2005
2	Robotics: Fundamental Concepts and Analysis,	Ghosal, A.	Oxford University Press,	2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Kinematics and Dynamics of Mechanical Systems	Kevin Russell, Qiong Shen, Raj S. Sodhi	CRC Press	2019
2	Theory of Applied Robotics: Kinematics, Dynamics, and Control	Reza N. Jazar	Springer	2010
3	Robotics Technology and Flexible Automation	S. R. Deb, Sankha Deb	McGraw Hill Education (India) Private Limited	2/e, 2010

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/ZHC7oavdrSY?si=mtrqP0a2ZP7cpmCk
2	https://youtu.be/ge1j8zMxVX8?si=CS-hwEndx_gK8us5
3	https://youtu.be/74d4vauVcWQ?si=mbHKiiTRT4Zko-VF
4	https://youtu.be/zSvCAW-mowg?si=QHDcn2sUz-JfrcMI

SEMESTER S8

SECURITY AND PRIVACY IN CYBER-PHYSICAL SYSTEMS

Course Code	OECP823	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	–	Course Type	Theory

Course Objectives:

1. To identify and Analyze Security Threats and Privacy Concerns
2. To apply Techniques for Securing Data and Ensuring Privacy
3. To evaluate and Implement Security Measures
4. To understand and Utilize Emerging Technologies

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Privacy in CPS - Privacy principles and concerns - Data collection, handling, and storage in CPS - Privacy regulations Data Protection Techniques - Data encryption and decryption methods - Authentication and authorization mechanisms - Access control models Security Fundamentals for CPS - Basic concepts of cybersecurity - Threats and vulnerabilities specific to CPS - Security goals: confidentiality, integrity, availability	9
2	Secure Communication in CPS - Secure communication protocols - Cryptographic algorithms and key management - Wireless and network security considerations Integrity and Availability in CPS - Techniques for ensuring data integrity - Fault tolerance and redundancy - Protection against denial of service attacks Threat Modeling and Risk Assessment - Identifying and assessing risks in CPS - Threat modeling techniques - Risk mitigation and management strategies	9

3	Safety and Security Integration - Safety-critical systems and their security implications - Techniques for integrating safety and security Emerging Technologies and Security Implications - Impact of AI and machine learning on CPS security - Blockchain and decentralized security models - Challenges related to IoT and edge computing Security Standards and Compliance - Overview of security standards - Compliance requirements and best practices - Implementing and auditing security policies	9
4	Practical Applications and Case Studies - Analysis of real-world CPS security incidents - Hands-on projects and simulations - Group presentations on case studies Emerging Threats and Countermeasures - Recent trends in CPS threats and attacks - Advanced countermeasures and defense techniques - Adapting security strategies to evolving threats Future Trends and Innovations - Future directions in CPS security and privacy - Role of emerging technologies in shaping the future - Predictions and challenges for next-generation CPS	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. - Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify and Analyze Security Threats and Privacy Concerns	K2
CO2	Apply Techniques for Securing Data and Ensuring Privacy	K3
CO3	Evaluate and Implement Security Measures	K3
CO4	Understand and Utilize Emerging Technologies	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	1							
CO2	3	3	2	3	1							
CO3	3	2	3	2	1							
CO4	3	1	1	1	1							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Securing Cyber-Physical Systems	Al-Sakib Khan Pathan	CRC Press	1/e
2	Cybersecurity and Privacy in Cyber Physical Systems	Yassine Maleh, Mohammad Shojafar, Ashraf Darwish, Abdelkrim Haqiq	CRC Press	1/e
3	Computer Security: Principles and Practice	William Stallings, Lawrie Brown	Pearson	4/e

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Privacy Engineering: A Dataflow and Ontological Approach	Ian Oliver	Createspace Independent Publishing Platform	2014
2	Privacy in a Digital, Networked World: Technologies, Implications and Solutions	Sherali Zeadally and Mohamad Badra	Springer	1e, 2015
3	Security and Privacy in Internet of Things (IoTs): Models, Algorithms, and Implementations	Fei Hu	CRC press	1e, 2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/AQClJAif5w8?si=E6vvQ2cRYli1jnUc
2	https://youtu.be/xErbGFBnZeo?si=xBG0WrR5-IDsXv5o
3	https://youtu.be/OYsY5B9pqYU?si=rFopBPQTqngLVUIP
4	https://youtu.be/p7kYStiASLo?si=1VhlOfz4Nh0gx8bn

SEMESTER S8

AUGMENTED AND VIRTUAL REALITY FOR CPS

Course Code	OECPT824	CIE Marks	40
Teaching Hours/Week (L:T:P:R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the fundamental concepts and components of Virtual Reality (VR) systems.
2. To explain the principles of digital entertainment and development tools in VR, along with hardware and software components in AR/VR systems.
3. To analyze the benefits and challenges of using AR/VR in Cyber-Physical Systems (CPS) applications.
4. To identify and evaluate potential security threats and vulnerabilities in AR/VR CPS applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Virtual Reality (VR): Fundamental concepts and components of VR. Input and output interfaces in VR: trackers, sensors, digital gloves, movement capture, video-based input, and 3D menus. Visual, auditory, and haptic output devices. AR/VR in CPS: Overview of AR/VR technologies, Definition and characteristics of CPS, Applications of AR/VR in CPS.	9

2	Interactive Techniques in VR: Body tracking, hand gestures, and 3D manipulation, Development tools and frameworks for VR. Application of VR in Digital Entertainment: VR technology in film and TV production. Physical exercises and games using VR. Demonstrations of digital entertainment through VR. AR/VR Fundamentals: AR/VR hardware and software components, Types of AR/VR (Marker-based, Markerless, Immersive, Non-Immersive), AR/VR development tools and platforms.	9
3	Augmented and Mixed Reality: Taxonomy and features of augmented reality (AR). Differences between AR and VR. Challenges in AR systems, Visualization techniques for AR. Enhancing interactivity in AR environments. CPS Components and Architecture: Introduction to CPS components (sensors, actuators, control systems), CPS architecture and design principles, Integration of AR/VR with CPS components.	9
4	AR/VR Applications in CPS: Industrial automation and control, Healthcare and medical systems, Smart homes and buildings, Transportation systems. Security and Privacy in AR/VR CPS: Security threats and vulnerabilities in AR/VR CPS, Privacy concerns and data protection, Secure design and development practices.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamental concepts and components of Virtual Reality (VR) and describe the components of a VR system.	K2
CO2	Explain the principles of digital entertainment in VR, the role of development tools and frameworks in VR development., hardware and software components in AR/VR systems.	K2
CO3	Explain the role of sensors, actuators, and control systems in CPS, Identify and describe the components of CPS.	K2
CO4	Illustrate the benefits and challenges of using AR/VR in CPS applications and the effectiveness of AR/VR solutions in CPS applications.	K2
CO5	Identify and explain potential security threats and vulnerabilities in AR/VR CPS and Evaluate the risks associated with these threats and vulnerabilities.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3	3	3		3							
CO3	3											
CO4	3	3	3	3	2							
CO5	3	3	3	3								

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Securing Cyber-Physical Systems	Al-Sakib Khan Pathan	CRC Press	1/e
2	Cybersecurity and Privacy in Cyber Physical Systems	Yassine Maleh, Mohammad Shojafar, Ashraf Darwish, Abdelkrim Haqiq	CRC Press	1/e
3	Computer Security: Principles and Practice	William Stallings, Lawrie Brown	Pearson	4/e

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Augmented and Virtual Reality in Cyber Physical Systems: A Guide for Engineers	Hassan Karimi	Springer	1e, 2022
2	Handbook of Augmented and Virtual Reality	S. S. Iyengar	Springer	1e, 2020
3	Augmented and Virtual Reality: A Beginner's Guide	David A. Cook	Apress	2020

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=zLMgdYI82IE
2	https://www.youtube.com/watch?v=MGuSTAqlZ9Q
3	https://www.youtube.com/watch?v=DCQYBHz7RDs
4	https://www.youtube.com/watch?v=Nq3mPFgpREE